

BASE24 Release 6.0 Version 9

Implementation Guide



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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one document version to another are indicated with the symbol “|” in the outside margin.

Release 6.0 Version 9 (M. Ahrens, D. Barton, S. Eschliman, R. McCall, P. McMeekin, C. Meyers, C. Rink, E. Rock, R. Schieuer, L. Watson, J. Wishart)
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1: Introduction

The *BASE24 Release 6.0 Version 9 Implementation Guide* is the primary reference document for information needed to upgrade existing BASE24 Release 6.0 Version 8 systems to BASE24 Release 6.0 Version 9.

This section provides an overview of the planning requirements for BASE24 users migrating to Release 6.0 Version 9. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for Release 6.0 Version 9, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to Release 6.0 Version 9. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the *BASE24 Release 6.0 Version 9 Implementation Guide* to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to Release 6.0 Version 9.

BASE24 Release 6.0 products include the following:

- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value-Added Products:
 - BASE24-card
 - BASE24-customer service
 - BASE24-infoBase
 - BASE24-inventory

Who should use this guide?

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to Release 6.0 Version 9. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to Release 6.0 Version 9, users must be on Release 6.0 Version 8. Refer to the *BASE24 Release 6.0 Version 8 Implementation Guide* to assist in planning the tasks necessary to get to Release 6.0 Version 9.

Using this guide, you can do the following:

- Determine what ACI services are available to assist in the migration to Release 6.0 Version 9.
- Develop a project plan to migrate to Release 6.0 Version 9.
- Determine staffing and training needs.
- Gain a detailed understanding of the Release 6.0 Version 9 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 Release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on an XPNET 3.0 or 3.1 system.

FIX

FIX 2.28 (or newer) is required to apply any new BASE24 fix files produced after 19 October 2006. FIX 2.28, like the previous version of FIX, will allow application of fix files to a specified file name in a concatenated list of fix files. This option can be combined with the FIRSTSCN and LASTSCN options to apply only fixes to the specified file name in a specified SCN range. As of 19 October 2006, FIX 2.28 (or the most current version of FIX) was automatically shipped with each software order. Please replace your existing FIX object with the most current version.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to Release 6.0 Version 9. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for Release 6.0 Version 9 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's Release 6.0 Version 8 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of Release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from release 5.3 to the latest version of Release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of Release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from Release 6.0 Version 1 to Release 6.0 Version 2. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from Release 6.0 Version 2 to Release 6.0 Version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from Release 6.0 Version 3 to Release 6.0 Version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from Release 6.0 Version 4 to Release 6.0 Version 5. No changes are made to these programs other than necessary fixes.

The *UC09 subvolumes (e.g., AT60UC09, BA60UC09, PS60UC09) contain the conversion program for users migrating from Release 6.0 Version 5 to Release 6.0 Version 6. No changes are made to these programs other than necessary fixes.

The *UC0A subvolumes (e.g., AT60UC0A, BA60UC0A, PS60UC0A) contain the conversion program for users migrating from Release 6.0 Version 6 to Release 6.0 Version 7. No changes are made to these programs other than necessary fixes.

The *UC0B subvolumes (e.g., AT60UC0B, BA60UC0B, PS60UC0B) contain the conversion program for users migrating from Release 6.0 Version 7 to Release 6.0 Version 8. No changes are made to these programs other than necessary fixes.

The *UC0C subvolumes (e.g., AT60UC0C, BA60UC0C, PS60UC0C) contain the conversion program for users migrating from Release 6.0 Version 8 to Release 6.0 Version 9.

If a release 5.3 user wants to convert to the latest Version of Release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 Version 1 user wants to convert to Release 6.0 Version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 Version 2 user wants to convert to Release 6.0 Version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 Version 3 user wants to convert to Release 6.0 Version 4, the conversion programs on *UC07 should be run.

If a Release 6.0 Version 4 user wants to convert to Release 6.0 Version 5, the conversion programs on *UC08 should be run.

If a Release 6.0 Version 5 user wants to convert to Release 6.0 Version 6, the conversion programs on *UC09 should be run.

If a Release 6.0 Version 6 user wants to convert to Release 6.0 Version 7, the conversion programs on *UC0A should be run.

If a Release 6.0 Version 7 user wants to convert to Release 6.0 Version 8, the conversion programs on *UC0B should be run.

If a Release 6.0 Version 8 user wants to convert to Release 6.0 Version 9, the conversion programs on *UC0C should be run.

If a Release 6.0 user wants to convert from Version 1 to Version 9, multiple conversion programs must be run starting with *UC05. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user needs to run the conversion programs on *UC06, *UC07, *UC08, *UC09, *UC0A, and on *UC0B. Finally, the user should run the conversion programs on *UC0C.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the Release 6.0 Version 9 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 Release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 Release 6.0 network. This document is included as a Microsoft Word document and is shipped with the Release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to Release 6.0 Version 9. This document is included as a HP NSK TGAL document and is shipped with the Release 6.0 Version 9 software. This document is named AAREAD0A and is located on the BA60UC0A subvolume.

HELP24 and HELP24 InfoLink

HELP24 and HELP24 InfoLink are excellent sources of assistance and information. The following access options appear on the support page of the ACI Worldwide Inc. Internet website (<http://www.aciworldwide.com/support>).

- A current listing of HELP24 phones worldwide.
- An option to send an e-mail message to HELP24.
- An option to sign up for HELP24 InfoLink.
- An option to access HELP24 InfoLink once you have received a user ID and password.

Once you can log on to HELP24 InfoLink, you can do the following:

- Access the HELP24 database to open a new Clarify case.
- Access installation and fix release procedures online.
- Access all BASE24, BASE24-es, and NET24-XPNET documentation online.
- Search the site for information about BASE24.

The HELP24 eSupport Users Guide contains procedures for opening a new Clarify case, accessing the Clarify database, and accessing documentation on-line. Once you log on to HELP24, the path to the HELP24 eSupport Users Guide is Support Center> InfoLink > HELP24 eSupport Users Guide. Once you log on to HELP24, the path to the installation procedures is Support Center> InfoLink > Product Info > BASE24> BASE24 6.0 Install Info and Guides. The menu displayed gives you three options:

- Pre-Install Checklists- for BASE24 6.0 and ITS
- Install Guides—Procedures for installing a software release or Version for the first time.
- Retail Install Guides—Procedures for installing Retail 3.1 or Check Auth 3.2.

Once you log on to HELP24, the path to the Fix Information is Support Center> InfoLink > Fix Information > BASE24

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with Release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility within existing code that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the *BASE24-atm Transaction Processing Manual*, *BASE24-pos Transaction Processing Manual*, or the *BASE24 ITS Kernel Configuration Manual* for additional information about this utility.

Online (Inline) Conversion

A software routine within existing code that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

2: Functionality

This section contains an overview of the BASE24 Release 6.0 Version 9 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the Release 6.0 Version 9 offerings. These enhancements affect all BASE24 users of a specific product or module. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that require a new separate licensable module. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value Added Products

BASE24 users should familiarize themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall Release 6.0 Version 9 offering.

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BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in Release 6.0 Version 9, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Changes for all BASE24 Release 6.0 Version 9 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

PCI Event and Trace Message

All non-essential sensitive cardholder information is being removed from event and trace messages to simplify compliance with the PCI Data Security Standard.

Prerequisites: None

Support Additional Flexibility in EMV Processing

This enhancement allows customers to configure multiple EMV Issuer Script MAC lengths. This flexibility is needed in those situations where a customer has card portfolios of multiple lengths. Currently, the MAC length is configured at the institution level which is not adequate. In addition, this enhancement provides support for the "Go Online Next Transaction" functionality of the ARPC Response Code/Card Status Update field and allow for the configuration of the "Offline Counter Updates" field.

Prerequisites: EMV Extensions

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Optional Enhancements

There are no BASE24 Base Release 6.0 Version 9 enhancements that are optional.

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BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the Release 6.0 Version 9 offerings. All BASE24-atm users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- None

Add Support for AMEX Format in EMV IAD

This enhancement adds support for the AMEX format in EMV IAD. This enhancement does not, however, support AMEX offline PIN Change scripts. A future change in TSS is necessary to support that functionality.

Prerequisites: None

Diebold Agilis 2.2 Upgrade

Diebold released its Agilis v2.2 software which incorporates a number of new features. The BASE24 Diebold 10XX/478X Device Handler was enhanced to support several of these new Agilis v2.2 features and ensure all customers can use the device handler with these models of Diebold ATMs.

The specific changes made to the device handler are:

- Support for Enhanced Status messages on the card reader.
- Separate existing 'Feed Failure' messages (message # 58, 59, 60, 61) into specific failure messages for no bills, cassette failure, and pick unit failure.
- Support of new Op Codes '070' thru '078' for state 'SC', including any associated status messages.
- Support for Enhanced Status messages on the IDM.
- The ability to print the bottom of the check on the customer's receipt.
- Include additional receipt and depositor print data on the customer receipt.
- The ability to display the image from both sides of the check (face and endorse sides).

Prerequisites: AT-DA150 Diebold 10XX/478X Device Handler

MasterCard Chargeback TLF Enhancement

MasterCard introduced new Chargeback requirements that can be complied with using the information on the Track2. Specifically, MasterCard mandated that acquirers provide the Track2 service code (or at least the first digit) under certain conditions for a second presentment. In 2007, ACI applied a fix for case #431460 which meant that the service code is no longer suppressed with the rest of the Track2 discretionary data. However, BASE24-atm doesn't log the Track2 data to the TLF, so the service code is not included in the log file.

An enhancement was made to allow the creation and logging of a new token to the TLF, consisting of the non-sensitive Track 2 information of the card originating the transaction.

Prerequisites: AT-AE000 ATM AUTH.

Multiple-Currency with Like Denominations (Diebold)

An issue was reported with the previously released Multi-Currency Dispense enhancement whereby if 2 currencies were configured and the same denomination was present for each currency, the terminal selected the first note of the denomination, regardless of the currency code.

An enhancement was made to the Multi-currency Dispense related device handler modules to ensure the proper dispensing of same bill denominations for multiple currencies (for example the dispense of both 20 Peso bills and 20 Dollar bills at the same device).

Prerequisites: AT-DA302 Diebold 10xx/478x Multiple Currency Dispense

NCR 5xxx Optional Receipt Support with Mobile Top-up

The purpose of this project was to correct a conflict between mobile top-up and optional receipts. During mobile top-up processing, operation code (op-code) position 6 is used to identify the Telecommunications service provider (Telco) when more than eight pre-defined Telcos are being used. Prior to this enhancement, the device handler overwrote position 6 of the op-code buffer if optional receipt processing was invoked. This enhancement will now allow the op-code buffer of a mobile top-up transaction to remain intact when optional receipt processing is invoked.

Prerequisites: MC-PP004 NCR 5xxx Mobile Top-up Extension

NCR Status Code and Model Number Update

Support was also added for various new M-status codes and associated event messages. This required adding the new NCR ATM model numbers to the NCR 5XXX Device Handler's global product class table, building unique event messages for the various new status codes the device handler can now receive in status messages and adding support for several new Device Identifier Graphics (DIGs).

Prerequisites: AT-DA031 NCR 5XXX Device Handler

PCI Track2 Context Data Removal

Includes Triton, Tidel, and Diebold CSP

A committee of major EFT networks, bankcard associations and software vendors was formed in May 2006 to discuss ways of protecting the safety and soundness of the various networks and associations, their mutual members and consumers. The primary focus of the committee was to discuss the Payment Card Industry Data Security Standard (PCI DSS) requirements and how they relate to PIN debit activity.

The PCI DSS requirement which prohibits the storage of sensitive authorization data subsequent to the original authorization was of particular interest to the Committee. The solution proposed by the committee, which affects transaction handlers on an industry-wide basis, impacts the online message structure and data element (DE) requirements for a few specific message types. The net effect is to eliminate the need for merchants and acquiring processors to store full track data for transactions following the original authorization.

The working group's recommendation is that in place of full track data on ATM and POS reversals and POS pre-authorization completion messages, acquirers should only send the card number (DE 002 – PAN) and the PAN Entry Mode (DE 022). This initiative is applicable to 0200 completion, 0220 forced completion, and 0420 reversal messages.

In accordance with the committee's recommendation, the purpose of this enhancement was to analyze the Triton, Tidel & Diebold CSP Device Handler to see if Track 2 data was being retained after a transaction is processed. When Track 2 data was found to exist, the Device Handlers were enhanced to eliminate the retaining of Track 2 data from the TDF file after a transaction has been processed.

Prerequisites: AT-DA280 Triton Device Handler, AT-DA270 Tidel Device Handler, AT-DA027 Diebold CashSource Plus Device Handler.

Support Additional Flexibility in EMV Processing

This enhancement allows customers greater flexibility in managing EMV Cards. The following changes were made in support of this requirement:

- Support for multiple EMV Issuer Script MAC lengths. This flexibility is needed in those situations where a customer has card portfolios supporting different MAC lengths. Prior to this enhancement, the MAC length was configured at the logical network level which was not adequate. A change was made to add an EMV Issuer Script MAC Length value to the CPF to be used in creating issuer scripts. An override EMV Issuer Script MAC Length value was also added to the CAF, to allow for reissued EMV cards requiring a different MAC length.
- Support for the configuration of the “Update Counters” bits in the ARPC Response Code data element (M/Chip 4 cards) and the Card Status Update (CSU) data element (CCD cards), used in the generation of the Issuer Authentication Data. A change was made to add configuration parameters to the CPF to be used in setting these bits.
- Support for the setting of the “Go Online On Next Transaction” bit in the ARPC Response Code data element (M/Chip 4 cards) and the Card Status Update (CSU) data element (CCD cards), used in the generation of the Issuer Authentication Data. A change was made to set this bit whenever an issuer script is generated, so that the issuer receives the results of the issuer script processing with the next EMV transaction performed on the card.
- Support for multiple EMV Issuer Script Template values. This flexibility is needed in those situations where a customer has card portfolios supporting different template tags. Prior to this enhancement, the template tag was configured at the logical network level which was not adequate. A change was made to add an EMV Issuer Script Template value to the CPF to be used in creating issuer scripts. An override EMV Issuer Script Template value was also added to the CAF, to allow for reissued EMV cards requiring a different template tag.

Prerequisites: AT-CC998 ATM Application EMV Extension

Tidel Display Balances on Screen Support

In addition to the standard ATM transaction set, the Tidel ATM is now capable of supporting the display of account balances on the ATM screen. The Extended Features Optional Group is used to accommodate the balances in the ATM message.

Prerequisites: AT-DA270 Tidel Device Handler

Triton EMV Offline PIN Synchronization

In addition to the standard ATM transaction set, the Triton Device Handler will now support EMV offline/online PIN change and EMV offline PIN unblock transactions. These transactions are used to keep the online PIN and the offline EMV PIN in sync when the online PIN is changed, and to reset the bad PIN tries counter on the EMV card.

Prerequisites: AT-DA280 Triton Device Handler, AT-AE400 Offline PIN Management, AT-CC997 Triton EMV.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Triton Mini-Statement

In addition to the standard ATM transaction set, the Triton Device Handler will now support mini-statement transaction functionality. The Extended Transaction Code String, Miscellaneous Field ID “I” is used to accommodate the new transaction code.

BASE24-atm Triton customers will need to enter the Management Functions Settings in the Terminal Configuration to turn on mini-statement transactions under Select Transaction Types.

Prerequisites: AT-DA280 Triton Device Handler

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Optional Enhancements

The following BASE24-atm Release 6.0 Version 9 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

NCR Standard APTRA Edge NDC Support

This enhancement supports NCR's new APTRA Edge Standard NDC specification as a new NCR device extension. It is currently applicable to customers located in the United States only.

Previously, customers wanting to use NCR's APTRA Edge terminal software would commit to costly customization, since much of the APTRA Edge application configuration is determined at the time of the terminal software build. NCR's APTRA Edge Standard NDC Message Format software eliminates much of this customization work by giving customers the ability to remotely modify their standardized APTRA Edge configuration, via XML instructional data downloads to the terminal, thus overriding the standard configuration built into the application software.

At this time, the NCR Standard APTRA Edge NDC extension provides support for standard financial transactions, standard terminal management, surcharging, non-currency dispense, PIN change and Open Account Relationship.

The APTRA Edge Device Handler support has been developed as an extension to the existing NCR 5XXX Device Handler. This will permit customers to operate both APTRA Edge standard NCD terminals and standard NCD+ terminals via a common NCR 5XXX Device Handler.

Prerequisites: AT-DA031 NCR 5XXX Device Handler, AT-DA041 NCR Standard APTRA Edge NDC Extension

The above enhancement is a separate licensable module: AT-DA041 NCR Standard APTRA Edge NDC Extension. Please contact your Customer Manager for further information.

Tidel AKDS Support

This enhancement adds Automated Key Distribution System (AKDS) functionality for use with Tidel Devices. AKDS allows a new ATM Master Key to be downloaded to the ATM without requiring security personnel to visit the ATM and manually enter the new keys. The BASE24-atm Tidel Device Handler was modified to process AKDS messages for Tidel EasyPoint dialup ATMs. The new AKDS extension downloads a secure Master Key from BASE24 to the ATM. In order to download a secure Master Key from BASE24 to the ATM, BASE24 and the EPP device must first authenticate each other and exchange their public keys and signatures. Once this is accomplished, the appropriate Master Key can be downloaded to the device.

Prerequisites: AT-DA270 Tidel Device Handler

The above enhancement is a separate licensable module.

Tidel Mobile Top-Up

This enhancement consists of a new Mobile Top-Up extension and changes to the Tidel Device Handler to support the existence of the extension.

The new Tidel Mobile Top-Up Extension enables cardholders to replenish or top up their mobile telephone accounts at an ATM. The replenishment transaction is a non-currency dispense transaction.

A telecommunications service provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account.

The BASE24-atm product authorizes the funds portion of the transaction.

Prerequisites: AT-DA270 Tidel Device Handler, MC-PP019 Tidel Mobile Top-up Extension

The above enhancement is a separate licensable module: MC-PP019 Tidel Mobile Top-up Extension. The interface to the telecommunications service provider is not a standard product feature and must be licensed as a CSM. Please contact your Customer Manager for further information.

Triton AKDS Support

The Triton Device Handler has been enhanced to support the Triton Automated Key Distribution System (AKDS) enhancement. In order to download a new PIN Master Key and/or a MAC Master Key to a Triton ATM, AKDS will be used. The BASE24-atm Triton Device Handler has been modified to process AKDS messages for both dialup and leased line ATMs. The new AKDS extension will download a secure Master Key from BASE24 to the ATM. In order to download a secure Master Key from BASE24 to the ATM, BASE24 and the EPP device must first authenticate each other and exchange their public key certificates. Once this is accomplished, the appropriate Master Key can be downloaded to the device.

Prerequisites: AT-DA280 Triton Device Handler

The above enhancement is a separate licensable module.

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BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the Release 6.0 Version 9 offerings. All BASE24-pos users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- None

Consolidate Variable in SPDH/RTAU Procedure

This enhancement consolidates the number of variables that are passed to certain procedure in the SPDH, which provides increased capacity to define variables required for custom modifications. The maximum number of variables that can be passed to a procedure is 32. At least one procedure was passing 31 variables. To provide for future development of CSMs, the variables have been consolidated into a single structure.

Prerequisites: PS-DP140 Standard POS Device Handler (SPDH)

DUPKT Support with Message Data Encryption

This project is to enhance the BASE24-pos Data Encryption module to support DUKPT as part of either the full or configurable message data encryption options added previously in BASE24 Release 6.0 Version 8.

Prerequisites: PS-DP140 Standard POS Device Handler (SPDH) and PS-DP410 POS Data Encryption

Hypercom Enhancements

This enhancement supports the downloading of initialization parameter data specifically in support of Hypercom's Optimum series devices which support the SPOS32 application, EMV terminal capability information as configured through Pathway, and support for Issuer Script Results as part of a reversal message.

Prerequisites: PS-DP390 BASE24-pos Hypercom Device Handler

Institution Level Security with OLT for CRT Auth

This enhancement provides institutional level security for OLT access in CRT Authorization. This ensures that only those operators with permission are capable of generating transactions for specified retailers and their respective merchant processors or institutions. Support is provided by restricting access by retailer FIID on the Online Transaction (OLT) screen. The BASE24 user will be allowed to enter transactions only for those retailers defined in BASE24 and which match the institution ID as specified for that user's profile.

Prerequisites: BP-DP130 POS CRT Authorization Device Handler

PCI CVD2 Suppression in CRT Auth

This enhancement is to ensure that cardholder sensitive data such as CVD2 is not stored in any file utilized by the POS CRT Authorization or POS CRT Administrative applications. Such data should not be stored or viewable to operators accessing these applications. Authorization processing has been modified to ensure that the Track 2 discretionary data and CVD2 data is not included in a record written to the PRF. Additionally, the OLT screen processing has been modified to ensure that CVD2 data is not logged to the TQF, and is not included in the PSTM unless the message type is a request.

Prerequisites: BP-DP130 POS CRT Authorization Device Handler

PCI Track2 Context Data Removal for POS Device Handlers

The purpose of this enhancement is to eliminate the retention of Track 2 context data from the PTDF/PTD file after a transaction has been processed. This enhancement addresses the Visa II POS, APACS and Hypercom Device Handlers. This enhancement further helps to protect consumer data in acquirer and issuer transactions.

Prerequisites: PS-DP988 APACS Device Handler, PS-DP390 Hypercom Device Handler, or PS-DP420 Visa II Device Handler.

POS Limit Checking on Responses

In Release 6.0 Version 7, an ATM project called Simultaneous Transactions Fraud Enhancement was released. The project added CAF limit checking on response messages that were approved by the host. This allowed another chance to catch a transaction that was over the daily limit if two transactions happened so close together that each would pass the check in the outgoing message before a response was received on the first message. The purpose of this new enhancement is to add similar functionality to POS authorization processing. The changes will be in-line and delivered to all customers. An LCONF param, POS-RESP-LMTS-CHK, will allow customers to turn the functionality for POS.

Prerequisites: BASE24-pos

SPDH Additional EMV Data Elements

This enhancement adds a new field to the SPDH message to identify the EMV Terminal Capability. While this information is included on the terminal record, this enhancement allows all the same data to be included in the transaction message as an assurance for network compliance. The transaction data sent by the device would override the data in the terminal record.

Prerequisites: PS-DP140 Standard POS Device Handler (SPDH) and PS-CC976 SPDH EMV Extension

SPDH Support for Network Mandates

Visa's October 2007 mandates require that those supporting Healthcare Auto-Substantiation have the ability to support four new amount types:

- 4U = Amount prescription/Rx
- 4V = Amount vision/optical
- 4W= Amount clinic/other qualified medical
- 4X = Amount dental

The Enhancement also includes support for MasterCard's new Card Account Level codes, and Visa's new form factor to identify device entry for contactless payments.

Prerequisites: PS-DP140 Standard POS Device Handler (SPDH)

Support Additional Flexibility in EMV Processing

This enhancement allows customers greater flexibility in managing EMV Cards. The following changes were made in support of this requirement:

- Support for multiple EMV Issuer Script MAC lengths. This flexibility is needed in those situations where a customer has card portfolios supporting different MAC lengths. Prior to this enhancement, the MAC length was configured at the logical network level which was not adequate. A change was made to add an EMV Issuer Script MAC Length value to the CPF to be used in creating issuer scripts. An override EMV Issuer Script MAC Length value was also added to the CAF, to allow for reissued EMV cards requiring a different MAC length.
- Support for the configuration of the “Update Counters” bits in the ARPC Response Code data element (M/Chip 4 cards) and the Card Status Update (CSU) data element (CCD cards), used in the generation of the Issuer Authentication Data. A change was made to add configuration parameters to the CPF to be used in setting these bits.
- Support for the setting of the “Go Online On Next Transaction” bit in the ARPC Response Code data element (M/Chip 4 cards) and the Card Status Update (CSU) data element (CCD cards), used in the generation of the Issuer Authentication Data. A change was made to set this bit whenever an issuer script is generated, so that the issuer receives the results of the issuer script processing with the next EMV transaction performed on the card.
- Support for multiple EMV Issuer Script Template values. This flexibility is needed in those situations where a customer has card portfolios supporting different template tags. Prior to this enhancement, the template tag was configured at the logical network level which was not adequate. A change was made to add an EMV Issuer Script Template value to the CPF to be used in creating issuer scripts. An override EMV Issuer Script Template value was also added to the CAF, to allow for reissued EMV cards requiring a different template tag.
- This enhancement includes the introduction of support for delayed authorizations, which is a function specific to POS processing. When an EMV card requests an online authorization, but the terminal cannot send the transaction to the acquirer, it is likely that the card will decline the transaction. In some cases, the merchant may want to approve the transaction, e.g. if the cardholder is known to the merchant. When the terminal is able to communicate with the acquirer again, the transaction may be sent online as a “delayed authorization”, to determine whether the issuer is prepared to accept the transaction.

Prerequisites: AT-CC986 POS Application EMV Extension

Visa Contactless MSD with CVN 17

This enhancement supports Visa's Contactless CVN 17 requirement for card issuing, and for terminal acquirers using the Standard POS Device Handler. Visa's 2.x contactless CVN 17 involves transactions initiated with EMV compatible applications, and which processes contactless magnetic stripe data differently than that of dynamic card verification (dCVV). With CVN 17, the cryptogram is nearly identical to EMV cards that customarily include smart chip applications used in the international markets.

Prerequisites: For acquirers, the prerequisite includes PS-DP 140 Standard POS Device Handler (SPDH), PS-CC950 Contactless Chip Card – Magnetic Stripe Format, and PS-CC976 SPDH EMV Extension

For issuers that wish to support the CVN17 authentication in BASE24, both the POS Application EMV Extension and respective TSS and HSM EMV extension products must be licensed.

Visa II Contactless Support

This enhancement enables the BASE24-pos Visa II (VVDH) Device Handler to support both Contactless Magnetic Stripe, and Visa's 2.x requirement for Contactless Magnetic Stripe Data (MSD) with CVN 17 transactions. The device handler was also enhanced to populate the Point of Service Data Token (token ID C4) from data included in the message as received from the terminal.

Prerequisites: Visa II Device Handler. POS Application EMV Extension and a respective TSS and HSM EMV Extension will be required to authenticate transactions involving Visa 2.x CVN 17 requirement.

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Optional Enhancements

There are no optional enhancements to BASE24-pos.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 9.

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General Enhancements

There are no general enhancements to Interchange Interfaces.

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Optional Enhancements

There are no optional enhancements to Interchange Interfaces.

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BASE24 Value Added Products

All BASE24 value added product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Value Added Products, refer to Section 10.

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General Enhancements

There are no general enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-infoBase
- BASE24-inventory Voucher Manager

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Optional Enhancements

There are no optional enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-infoBase
- BASE24-inventory Voucher Manager

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3: Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- AS Tech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 Release 6.0 Version 9. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to Release 6.0 Version 9. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 Release 6.0 requires a minimum of HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files. Customers choosing to implement BASE24-es Transaction Security Services (known as TSS) with BASE24 Release 6.0 must review the BASE24-es Hardware and Software Requirements for the BASE24-es release they will deploy.

5. UPGRADE TO XPNET 3.0 or 3.1

Customers should note that XPNET 3.0 is the minimum requirement for BASE24 Release 6.0.

6. PROJECT SCOPE MEETING

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 9.

7. MIGRATION PLANNING

Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 9.

9. DEFINE REQUIREMENTS DEFINITION

The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new Version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in Release 6.0 Version 9 or if they are specific to the customer's environment and will need to be upgraded to the new Version.

10. DEVELOP PROJECT PLAN

The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new Version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with Release 6.0 Version 9 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 Release 6.0 Version 9 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions and outlines can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 Release 6.0 Version 9 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in Release 6.0 Version 9, a host program will need to be developed to interpret the new token data that can be extracted with Release 6.0 Version 9.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in Release 6.0 Version 9 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 Version 9 software will need to be installed at the customer site. Licensed Release 6.0 Version 9 software modules will be provided to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested or as directed by the interchange.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system should be performed as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to Release 6.0 Version 9. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer's project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

Publications

The BASE24 documentation set has been updated to include the Release 6.0 Version 9 enhancements. The updated BASE24 documentation is available in three formats.

- HELP24 eSupport InfoLink. All files for the manuals in the BASE24 6.0 documentation set are located on the HELP24 eSupport InfoLink database. You have the option of copying the files for individual manuals from the HELP24 eSupport InfoLink database to your PC.

In addition, the HELP24 eSupport InfoLink database contains documentation that is added or revised between ZIP files or CD-ROMs.

- Software Management Electronic Distribution. ZIP or ZIP Executable files that contain the same contents as the CD-ROM are available through the ACI electronic distribution system.
- CD-ROM. The CD-ROM contains all PDF files for the manuals in the documentation set.

The BASE24 6.0 documentation set contains the most current version of each document. If a BASE24 manual has not been updated for Release 6.0, the manual for the prior release appears on the Release 6.0 menu with a note identifying the prior release.

Documentation for XPNET 3.0 and XPNET 3.1 is available in a separate set titled NET24-XPNET 3.1 that is available in the same formats as the BASE24 documentation.

BASE24 customers can download files from the HELP24 eSupport InfoLink database or order the documentation ZIP files or CD-ROM from their ACI Customer Manager.

Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 Version 9.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

The course descriptions, outlines and current education schedule can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

HELP24

Please use ACI Worldwide's eSupport site to report problems or ask questions about BASE24 Release 6.0 Version 9. eSupport is an Internet Web based application that interfaces with the HELP24 Clarify database. Using eSupport, you can search our KnowledgeBase Solutions for information and resolutions to problems you are experiencing. You can also directly open Clarify cases. An eSupport option, InfoLink, is your link to a wealth of information that can help you in your day to day business. It allows you on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. Contact your local Help24 office for an eSupport logon.

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows us to troubleshoot issues in a secure, online environment. Compared to the dial-up environment we have used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows you to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option that provides the ability to transfer files from your PC to ours, or vice versa. This feature can be used in the event the file(s) are too large to email.

For more information, please refer to the WebEx User's Guide located under Product Info on Infolink. Contact your local HELP24 office to schedule your WebEx demonstration.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks is estimated and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to Release 6.0 Version 9 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate Release 6.0 Version 9 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to Release 6.0 Version 9. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in Release 6.0 Version 9. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

4: BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading to Release 6.0 Version 9.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

General enhancements describe the Base modules changes that affect all BASE24 users using a particular product or module, along with the associated conversion and implementation requirements that all BASE24 users must address.

Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement a new separate licensable module, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 10 to identify the changes that BASE24 users need to consider when upgrading to Release 6.0 Version 9.

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Standard Information Format

A standard format is used in Sections 4 through 10 to highlight and describe the various changes, conversion requirements, and implementation requirements BASE24 users need to consider when upgrading to Release 6.0 Version 9. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, only those implementing the enhancement).

CUSTMACS Changes

Describes CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to Release 6.0 Version 9.

- General Changes
- PCI – Event and Trace Messages
- Support Additional Flexibility in EMV Processing

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described in section 4.

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General Changes

- Changes for all BASE24 Release 6.0 Version 9 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

A complete list of the merged files can be found in the SCN file BA60UD0C.SCNMERGE.

- The Cardholder Authorization File (CAF) has been modified to include additional fields on screen 13 for the Support Additional Flexibility in EMV Processing enhancement. The Cardholder Authorization File screen, requester and server have been modified to process all of the new fields. For further information, refer to the section on the Support Additional Flexibility in EMV Processing enhancement.
- The Card Prefix File (CPF) has been modified to include additional fields on screen 11, 12 and 13 for the Support Additional Flexibility in EMV Processing enhancement. The Card Prefix File screen, requester and server have been modified to process all of the new fields. For further information, refer to the sections on the Support Additional Flexibility in EMV Processing and the Add Support for AMEX format in EMV IAD enhancements.
- The Log Message Configuration File (LMCF) has been modified to include the new Directive, PAN Digits, and file maintenance fields. In addition, the size of the Message Number field has been increased to 5 bytes. The Log Message Configuration File screen, requester and server and SKELB have been modified to process these new fields. For further information, refer to the section on PCI – Event and Trace Messages.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCAF, DDLFCPF, DDLFLMCF
- Inline changes to BA60AFT.RQCAFS, RQCPFS, SCRNCASF, SCRNCASF, SVCAFS, SVCPFS

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Cardholder Authorization File (CAF)

For customers converting from Release 5.3 to Release 6.0 Version 9, the CAF conversion program must be run to convert the CAF. Running the conversion program is required for any BASE24 customer. Users must create the new CAF file using the BADDLFUP file.

No separate conversion program was created to move the CAF from Version 8 to Version 9. Customers must do a full file refresh using the new CAF layout. This will initialize the new CAF fields.

The CAF conversion program on the BA60UC04 subvolume can be used to convert the CAF records from the Release 5.3 format to the Release 6.0 Version 9 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVCAFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cafconv.t,name $cafm,nowait/ cnvcafm -  
force -list -listreplace
```

Release 5.3 users should update the CNVCAFR obey file on the BA60UC04 subvolume with the appropriate information. The CNVCAFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any BASE24 customer. Users must create the new CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the Release 5.3 format to the Release 6.0 Version 9 format. The CPF conversion program on the BA60UC0C subvolume can be used to convert the CPF records from the Release 6.0 Version 8 format to the Release 6.0 Version 9 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm -  
force -list -listreplace
```

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 Version 8 users should update the CNVCPFR obey file on the BA60UC0C subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0C on the BA60UC0C subvolume for Release 6.0 Version 8 customers.

Log Message Configuration File (LMCF)

The LMCF conversion program must be run to convert the LMCF. Running the conversion program is required for any BASE24 customer.

The LMCF conversion program on the BA10LMCF subvolume is used to convert the LMCF records to SKELB Version 29 (XPNET 3.0) or Version 09 (XPNET 3.1) format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVLMCFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#lmfcv.t,name $lmcfm,nowait/ cnvlmcfm -  
force -list -listreplace
```

Users should update the CNVLMCFR obey file on the BA10LMCF subvolume with the appropriate information. The CNVLMCFR file should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREADME file on the BA10LMCF subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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PCI – Event and Trace Messages

This section describes the implementation requirements associated with the PCI – Event and Trace Messages enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI – Event and Trace Messages enhancement.

This enhancement enables the customer to mask or otherwise alter BASE24 event and trace messages (collectively referred to from this point as *event messages*) to conform to the Payment Card Industry (PCI) Data Security Standard. The following PCI Data Security Standard storage and display requirements affect the handling of event messages in BASE24:

- Sensitive authentication data may not be stored subsequent to authorization, even if encrypted. Sensitive authentication data is defined as the full magnetic stripe, the CVC2/CVV2/CID, and the PIN/PIN block.
- The personal account number (PAN) must be masked when displayed. At most, the first six and last four digits may be displayed.
- The cardholder name, service code and expiration date must be protected if stored in conjunction with the PAN.

When a BASE24 process logs an event message, what it actually does is send a string of formatted text to the Event Management System (EMS) collector process, which in turn stores the formatted event message text in the current event log file. The EMSperus process is used to read an event message from the event log file and display it on a BASE24 operator's screen. Because event messages are stored and displayed in this manner, they are affected by the PCI requirements noted above.

This enhancement makes use of the Log Message Configuration File (LMCF) to alter event messages that do not meet PCI standards. The LMCF, accessible from a BASE24 Pathway screen, is a facility that enables the BASE24 operator to change the routing code, severity, or text of an event message without changing the program that logs the event. The values entered on the LMCF screen override the values specified in a BASE24 program. When a BASE24 program logs an event message, the SKELB library procedures reference the LMCF records in memory to determine whether the event message requires change. If an LMCF record that matches the event message is located, SKELB changes the event as specified before sending it to the EMS collector process. Otherwise, the event message is sent to the EMS collector without change.

A new field, Directive, has been added to the LMCF screen. The SKELB event logging procedures have been modified to act upon two values that may be entered in the Directive field – *MASK* and *MAX*. Use of these directives can eliminate sensitive authentication data and cardholder information from virtually any event message before it is sent to the EMS collector process. These directives enable BASE24 operators to

- mask selected parameters in an event message.
- mask all parameters in an event message.
- truncate an event message to a specified length.

Online Maintenance File (OMF) support has been added to the LMCF screen. This provides a detailed audit trail of every LMCF record that is added, updated or deleted.

An LMCFPCI file is provided to BASE24 customers. This file contains default LMCF records for every BASE24-base, BASE24-atm, BASE24-pos, BASE24-fhm and BASE24 switch interface event message that contains sensitive authentication data or cardholder information. Each LMCFPCI record contains a MASK or MAX directive appropriate for that event message. A conversion program is provided to merge records from the LMCFPCI with existing records in the customer's current LMCF.

A limited number of utility modules have been modified to remove sensitive authentication data and cardholder information from event messages by altering the source code itself. It was necessary to alter these event messages at the code level rather than use LMCF directives because, by their very nature, the utility procedures are called by multiple application modules. By changing the event messages within the utility procedures, it is not necessary to create multiple LMCF records for the same events, each with the product module ID of the calling application. The risk of omission is reduced, particularly if a future, as yet unknown, application module calls one of these utility procedures.

A small number of application modules have been modified by changing some event message numbers to unique values. This enables operators to mask or truncate those event messages using LMCF directives.

Software Impacts

All XPNET customers:

- SKELB changes - SCR 3440
 - XPNET 3.0 requires Version 29 or greater of SKELB
"REL3^VER0^SUTL29^20081201"
 - XPNET 3.1 requires Version 10 or greater of SKELB
"REL3^VER1^SUTL10^20081201"
- LMCF screen, requester and server changes - SCR 3440

All BASE24 customers:

- New files BA10LMCF.AAREADME, BALMCFM, BALMCFM, BALMCFMM, CNVLMCFM, CNVLMCFR, CNVLMCFS, CONTENTS, LMCFFUP, LMCFHIST and LMCFPCI
- Inline changes to BA60SRC.BAUTILS and HISWUTLS
- Inline changes to BA60RPT.OMFTBL
- Inline changes to BA60DDL.DDLFOMF
- Updated event (log) message manual BA60LOGM.BABAUTIL

All BASE24 Multiple Currency or Multiple Currency Dispense customers:

- Inline changes to BA60UMCU.BAMCUTLS
- Updated event (log) message manual BA60LOGM.BABAMCUT

All BASE24-atm customers:

- Inline changes to AT60SRC.ATUTILS
- Inline changes to AT60AUTH.AUTHLIBS

All BASE24-atm Multiple Currency or Multiple Currency Dispense customers:

- Inline changes to AT60UMCU.ATMCUTLS
- Updated event (log) message manual BA60LOGM.ATATMCUT

All BASE24-atm Multiple Currency Dispense customers:

- Inline changes to AT60UMCD.ATMCDHLS
- Updated event (log) message manual BA60LOGM.ATATMCDH

All BASE24-pos Multiple Currency customers:

- Inline changes to PS60UMCU.PSMCUTLS
- Updated event (log) message manual BA60LOGM.PSPSMCUT

All BASE24-pos Stored Value customers:

- Inline changes to PS60SV.DORMESCS

All BASE24 Alaska Option ISO Interface customers:

- Inline changes to SW60AOI.AOIS
- Updated event (log) message manual SW60AOI.AOILOGM

All BASE24 BIC ANSI Interface customers:

- Inline changes to SW60BIC.BICS
- Updated event (log) message manual SW60BIC.BICLOGM

All BASE24 Banknet Interface customers:

- Inline changes to SW60BNET.BNETLIBS
- Updated event (log) message manual SW60BNET.BNETLOGM

All BASE24 Discover ISO Interface customers:

- Inline changes to SW60DSCV.DSCVS
- Updated event (log) message manual SW60DSCV.DSCVLOGM

All BASE24 FDR ISO Interface customers:

- Inline changes to SW60FDRI.FDRIS
- Updated event (log) message manual SW60FDRI.FDRILOGM

All BASE24 NBGC Interface customers:

- Inline changes to SW60NBGC.NBGCS
- Updated event (log) message manual SW60NBGC.NBGCLOGM

All BASE24 NYCE ISO Interface customers:

- Inline changes to SW60NYCI.NYCIS
- Updated event (log) message manual SW60NYCI.NYCILOGM

All BASE24 Shazam ISO Interface customers:

- Inline changes to SW60SHZM.SHZMS
- Updated event (log) message manual SW60SHZM.SHZMLOGM

All BASE24 VisaNet Interface customers:

- Inline changes to SW60VISA.VISALIBS
- Updated event (log) message manual SW60VISA.VISALOGM

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNPCIE.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

To implement this enhancement, the current LMCF assign must point to the new LMCF created by the LMCF conversion program.

LCONF Param Changes

This section describes the param records that are required to be added to the LCONF to support this enhancement. Details of the params are provided in the *BASE24 Logical Network Configuration File Manual*. Samples of the params are included in the LCONFBA file located on the BA60MISC subvolume.

PCI-DISPOSITION

This param instructs the SKELB library what action it should take when a Guardian error occurs on the LMCF during the initialization or warmboot of an XPNET satellite process. If this parameter is present and has the value "A", SKELB will cause the process to abend. If this parameter is not present or does not have the value "A", SKELB will log an event message warning that it cannot perform PCI masking/truncation, and the process will continue as normal.

Database Configuration

Log Message Configuration File (LMCF)

The following new fields have been added to the LMCF Screen 1:

The following field specifies whether the SKELB event logging procedures should mask or truncate the event message that corresponds to this LMCF record.

Screen Name:	DIRECTIVE
Screen:	LMCF Screen 1
DDL Name:	LMCFPCI.REC.PCI.DIRECTIVE

Valid values are:

- MASK [*parameter number*], [*parameter number*], ...[*parameter number*]

where [*parameter number*] is an optional value that specifies the ordinal number of each parameter in the event message format template to be masked. Multiple parameter numbers are comma-separated. For example, MASK 1, 3 means mask the 1st and 3rd parameters of the event message.

If a parameter number is specified, the parameter is masked only if it uses one of the following formatting characters. If no parameter numbers are specified, every parameter in the event message that uses one of the following formatting characters is masked.

\ *	Variable-length string of any length
\ ?	Variable-length string of any length
\\	Fixed-length string of any length

If a parameter is 13-19 characters in length, it is masked according to the AFT-PAN-DIGITS LCONF parameter or the PAN DIGITS LMCF field, replacing specified digits with asterisks. It is assumed that a parameter of this length is a card number or account number. If a masked parameter is any other length, every character is replaced with asterisks.

- *MAX length*

where *length* is a required value that specifies the maximum number of characters of the event message's text that will be logged. The remainder of the event message text is truncated. Valid values for *length* are 0-342.

- The default value is blanks (no masking, no truncation).

The following field is a three-digit value that specifies how to mask a PAN if it is displayed in the event message that corresponds to this LMCF record. If this field contains valid digit values, it overrides the masking specified in the AFT-PAN-DIGITS LCONF parameter. If this field is blank or contains the value "ULF" (Use LCONF File), AFT-PAN-DIGITS is used. The default value is blanks.

Screen Name: PAN DIGITS
Screen: LMCF Screen 1
DDL Name: LMCFPCI.REC.PCI.LCONF-PAN-DIGITS

Descriptions of the digits are as follows:

<u>Digit Number</u>	<u>Description</u>
1	The maximum number of left most digits to be displayed unmasked. Valid values are 0-6 or blank (default).
2	The minimum number of digits to be masked. Valid values are 0-9 or blank (default).
3	The number of rightmost digits to be displayed unmasked. Valid values are 0-4 or blank (default).

NOTE: These new fields will be visible on the LMCF screen only if the LMCF assign in the LCONF points to the new-format LMCF. If the LMCF assign points to the old-format LMCF, the old LMCF screen will be displayed and no event message masking or truncation will be performed.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Servers

The new LMCF-VSN server must be defined in the PATHCONF file as SERVER-LMCF-VSN. Its PROGRAM parameter should be set to the path name of the SVLMCV server object file included with SCR 3440.

The new LMCF-PCI server must be defined in the PATHCONF file as SERVER-LMCF-PCI. Its PROGRAM parameter should be set to the path name of the SVLMCP server object file included with SCR 3440.

These servers can be added to the system online through the PATHCOM utility. After adding the servers to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined servers.

The existing LMCF server SERVER-LMCF should not be removed from the PATHCONF nor should it be altered.

Requesters

The following requesters must be SCUP COPY'd from the POBJ file included with SCR 3440 to the customer's POBJ.

- T3270-LMCF
- T3270-LMCF-NO-PCI
- T3270-LMCF-PCI
- T6520-LMCF
- T6520-LMCF-NO-PCI
- T6520-LMCF-PCI

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The SKELB library file used by the BASE24 XPNET satellite processes should be replaced with the new SKELB library file included with SCR 3440.

Partial implementation of this enhancement is possible. However, ACI recommends that customers configure their networks for full implementation. The following chart shows the effects of different implementation options.

<u>Implementation</u>	<u>Effect</u>
New LMCF, new requesters/servers, new SKELB	Full PCI enabled (recommended)
Old LMCF, new requesters/servers, old SKELB	Functional but PCI disabled
Old LMCF, new requesters/servers, new SKELB	Functional but PCI disabled
Old LMCF, old requester/server, new SKELB	Functional but PCI disabled
Old LMCF, old requester/server, old SKELB	Functional but PCI disabled
New LMCF, old requester/server, old SKELB	Fails because the new LMCF record format is unknown to old SKELB
New LMCF, old requester/server, new SKELB	PCI enabled but not configured since the LMCF records cannot be modified (the new LMCF record format is unknown to the old requester/server)

File Conversions

The LMCF conversion program must be run to convert the Log Message Configuration File (LMCF). The LMCF conversion program will convert the customer's current LMCF to a new record format and combine the records in ACI's default LMCFPCI file with existing records in the customer's LMCF. If the same record exists in both the customer's LMCF and ACI's LMCFPCI, the customer's LMCF record will not be replaced with the LMCFPCI record. Rather, the customer's LMCF record will be retained and the conflict will be reported in the conversion report file. The customer can use the information in the report to resolve the conflict manually.

Details of this conversion program can be found in the General Changes earlier in Section 4.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

Operators should be aware that portions of some event messages will be masked with asterisks or truncated as a result of implementing this enhancement. Events that are masked will end with the note ****PCI MASKED****. Events that are truncated will end with the note ****PCI TRUNCATED****.

Disk Impacts

The LMCF requires additional disk space to accommodate the default records provided by ACI for this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1

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Support Additional Flexibility in EMV Processing

This section describes the implementation requirements associated with the Support Additional Flexibility in EMV Processing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Support Additional Flexibility in EMV Processing enhancement.

Section 5 contains BASE24-atm specific requirements associated with the Support Additional Flexibility in EMV Processing enhancement.

Section 6 contains BASE24-pos specific requirements associated with the Support Additional Flexibility in EMV Processing enhancement.

The Support Additional Flexibility in EMV Processing enhancement allows customers greater flexibility in managing EMV cards. The following requirements have been addressed in this project:

- Support for multiple EMV Issuer Script MAC lengths.
- Support for the configuration of the “Update Counters” bits in the ARPC Response Code data element (M/Chip 4 cards) and the Card Status Update (CSU) data element (CCD cards), used in the generation of the Issuer Authentication Data.
- Support for the setting of the “Go Online On Next Transaction” bit in the ARPC Response Code data element (M/Chip 4 cards) and the Card Status Update (CSU) data element (CCD cards), used in the generation of the Issuer Authentication Data.
- Support for multiple EMV Issuer Script Template values.

The Support Additional Flexibility in EMV Processing enhancement also provides support for “delayed authorizations”. This function is specific to BASE24-pos and is described in Section 6 of this document.

Script MAC Length Configuration

This flexibility is needed in those situations where a customer has card portfolios supporting different MAC lengths. Prior to this enhancement, the MAC length was configured at the logical network level, which is not adequate.

BASE24 has been enhanced to allow issuers to configure the MAC length required at the BIN level on the Card Prefix File (CPF). Valid values are 0 (specifying that the existing LCONF value should be used), 4, 6, and 8 (specifying 4 bytes, 6 bytes, 8

bytes respectively). An override EMV Issuer Script MAC Length value has also been added to the Cardholder Authorization File (CAF), to allow for reissued EMV cards requiring a different MAC length. Valid values are 0 (specifying that the CPF value should be used), 4, 6, and 8 (specifying 4 bytes, 6 bytes, 8 bytes respectively).

The value(s) configured for the length of the Script MAC on the CPF and CAF must not be greater than the corresponding value in the LCONF script parameters.

BASE24 has been enhanced to use the configured value(s) when formatting issuer script data to be returned to the card. Note that script data also includes a number of “length” bytes which have to be “adjusted” based on the MAC length configured.

The enhanced functionality does not apply to Common Core Definition (CCD) cards, for which a MAC length of 4 bytes is mandated.

Update Counters Configuration

BASE24 has been enhanced to include new configuration parameters at the BIN level on the Card Prefix File (CPF). These parameters allow issuers to specify how the card should process its offline counters when an EMV response message is received from the issuer. There is one parameter for approved financial transactions, one parameter for approved non-financial transactions, and one parameter for non-approved transactions (because issuers may want to take a different action depending on the type and outcome of the transaction). The values supported for each parameter are the same, and the valid values are:

- 0 = Do not update offline counters;
- 1 = Set counters to upper offline limits;
- 2 = Reset counters to zero;
- 3 = Add transaction to counter.

BASE24 has been enhanced to use the appropriate value when formatting the data used in generating the Authorization Response Cryptogram (ARPC) in the Issuer Authentication Data.

It is not possible to configure the “Update Counters” bit settings by an individual card.

The enhanced functionality applies only to M/Chip 4 and Common Core Definition (CCD) cards, as these are the only EMV applications supported by BASE24 that define these bit settings.

Set Go Online On Next Transaction

BASE24 has been enhanced to specify that the next transaction performed on the card should be sent online, when an EMV response message containing an issuer script is sent. BASE24 has been enhanced to use the value when formatting the data

used in generating the Authorization Response Cryptogram (ARPC) in the Issuer Authentication Data. Whenever an issuer script is generated, this logic will ensure that the issuer receives the results of the issuer script processing with the next EMV transaction performed on the card.

If BASE24 is responsible for ARQC verification/ARPC generation, but issuer scripts are created by a back-end Host system, then the “go online on next transaction” bit will NOT be set in approved transactions, because the ARPC will be generated before BASE24 is aware that a script is being sent. Similarly, if BASE24 is responsible for ARQC verification/ARPC generation, and creates issuer scripts based on a notification from a back-end Host system, then the “go online on next transaction” bit will NOT be set in approved transactions, because the ARPC will be generated before the Host notifies BASE24 that a script is required.

The enhanced functionality applies only to M/Chip 4 and Common Core Definition (CCD) cards, as these are the only EMV applications supported by BASE24 that define this bit setting.

Script Template Configuration

This flexibility is needed in those situations where a customer has card portfolios supporting different template tags. Prior to this enhancement, the template tag was configured at the logical network level, which is not adequate.

BASE24 has been enhanced to allow issuers to configure the script template required at the BIN level on the Card Prefix File (CPF). Valid values are 0 (specifying that the existing LCONF value should be used), 1 and 2 (specifying ‘71’ and ‘72’ respectively). An override EMV Issuer Script Template value has also been added to the Cardholder Authorization File (CAF), to allow for reissued EMV cards requiring a different template tag. Valid values are 0 (specifying that the CPF value should be used), 1 and 2 (specifying ‘71’ and ‘72’ respectively).

It is not possible to configure different script template tags per prefix/card for different types of script, e.g. tag ‘71’ for a CARD BLOCK script and tag ‘72’ for a PUT DATA script.

BASE24 has been enhanced to use the configured value(s) when formatting issuer script data to be returned to the card.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCAF, DDLFCPF, DDLGCAFX, DDLGISO.
- Inline changes to BA60AFT.SCRNCAF, SCRNCPPF, RQCAFS, RQCPFS, SVCAFS, SVCPPF, SECTBL.
- Inline changes to BA60REFR.REFRG, REFRS.
- Inline changes to BA60EXT.OMFXS;
- Inline changes to BA60UC04.CNVCPFS;
- New conversion program (BA60UC0C.CNVCPFS) to convert the CPF to the required format.

All BASE24-fhm customers:

- Inline changes to FH60DDL.DDLFHSTM.
- Inline changes to FH60CAF.FHMCASF.
- Inline changes to FH60RPT.FHMRPTS.

For full details on the ATM software impacts for this enhancement see Software Impacts in Section 5 (ATM) of this document.

For full details on the POS software impacts for this enhancement see Software Impacts in Section 6 (POS) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNFLEX.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Card Prefix File (CPF)

The following new field specifies the template tag to be used when an issuer script is sent to the card:

Screen Name: SCRIPT TEMPLATE TAG
Screen: CPF Screen 11
DDL Name: SCRIPT-TPLT-TAG
Valid Values are: 0 (Use LCONF value), 1 (71), 2 (72)
Default: 0 (Use LCONF value)

The following new field specifies the MAC length to be used when an issuer script is sent to the card:

Screen Name: SCRIPT MAC LENGTH
Screen: CPF Screen 11
DDL Name: SCRIPT-MAC-LGTH
Valid Values are: 0 (Use LCONF value), 4, 6, 8
Default: 0 (Use LCONF value)

The following new field specifies how the card should update the offline counters when processing a declined response message from the issuer:

Screen Name: UPDATE COUNTERS – DECLINED TRANSACTION
Screen: CPF Screen 11
DDL Name: UPDATE-CNTRS.DCLN-TXN
Valid Values are: 0 (Do not update offline counters)
 1 (Set counters to upper offline limits)
 2 (Reset counters to zero)
 3 (Add transaction to counter)
Default: 0 (Do not update offline counters)

The following new field specifies how the card should update the offline counters when processing an approved financial response message from the issuer:

Screen Name: UPDATE COUNTERS – APPROVED FINANCIAL TRANSACTION
Screen: CPF Screen 11
DDL Name: UPDATE-CNTRS.APPRV-FNCL-TXN
Valid Values are: 0 (Do not update offline counters)
1 (Set counters to upper offline limits)
2 (Reset counters to zero)
3 (Add transaction to counter)
Default: 0 (Do not update offline counters)

The following new field specifies how the card should update the offline counters when processing an approved non-financial response message from the issuer:

Screen Name: UPDATE COUNTERS – APPROVED NON-FINANCIAL TRANSACTION
Screen: CPF Screen 11
DDL Name: UPDATE-CNTRS.APPRV-NON-FNCL-TXN
Valid Values are: 0 (Do not update offline counters)
1 (Set counters to upper offline limits)
2 (Reset counters to zero)
3 (Add transaction to counter)
Default: 0 (Do not update offline counters)

To make room for these fields on CPF screen 11, parameters specific to ATM EMV transaction processing have been moved to CPF screen 12, and one parameter specific to POS EMV transaction processing has been moved to CPF screen 13.

A full description of these fields can be found in the *BASE24-atm EMV Support Manual* and the *BASE24-pos EMV Support Manual*.

Cardholder Authorization File (CAF)

The following new field specifies the template tag to be used when an issuer script is sent to the card:

Screen Name: SCRIPT TEMPLATE TAG
Screen: CAF Screen 13
DDL Name: SCRIPT-TPLT-TAG
Valid Values are: 0 (Use CPF value), 1 (71), 2 (72)
Default: 0 (Use CPF value)

The following new field specifies the MAC length to be used when an issuer script is sent to the card:

Screen Name: SCRIPT MAC LENGTH
Screen: CAF Screen 13
DDL Name: SCRIPT-MAC-LGTH
Valid Values are: 0 (Use CPF value), 4, 6, 8
Default: 0 (Use CPF value)

A full description of these fields can be found in the *BASE24-atm EMV Support Manual* and the *BASE24-pos EMV Support Manual*.

For full details on the database configuration impacts for ATM EMV processing see Database Configuration in Section 5 (ATM) of this document.

For full details on the database configuration impacts for POS EMV processing see Database Configuration in Section 6 (POS) of this document.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry has been added for the new CPF Screen 12.

PATHCONF Changes

Not applicable

CRT Access Security Changes

The Security File must be updated to allow operators the appropriate access to the new CPF Screen 12.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the General Changes earlier in Section 4.

Cardholder Authorization File (CAF)

For customers converting from Release 5.3 to Release 6.0 Version 8, the CAF conversion program must be run to convert the CAF. Details of this conversion program can be found in the General Changes earlier in Section 4.

No separate conversion program was created to move the CAF from Version 8 to Version 9. Customers must do a full file refresh using the new CAF layout. This will initialize the new CAF fields to appropriate values.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process the new field in records containing a CPF or CAF record image.

Refresh Process

Host code changes are required for all BASE24 users that utilize the Release 6.0 CAF Refresh file message format to provide the new fields in the CAF EMV segment.

From Host Maintenance Process

Host code changes are required for all BASE24 users that utilize FHM to provide the new fields in the CAF EMV segment.

Card Management Systems

Card management systems must be capable of providing the additional card data required by BASE24.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with Release 6.0 Version 9, along with the changes, conversion requirements, and implementation requirements.

There are no optional enhancements for the base modules.

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5: BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from Release 6.0 Version 8.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to Release 6.0 Version 9.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to Release 6.0 Version 9.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Add Support for AMEX format in EMV IAD
- Diebold Agilis 2.2 Upgrade
- MasterCard Chargeback TLF Enhancement
- Multi-Currency Support With Like Denominations (Diebold)
- NCR 5XXX Optional Receipt Support with Mobil Top-Up
- NCR Status Code and Model Number Update
- PCI - Track 2 Context Data Removal for Diebold CSP
- PCI - Track 2 Context Data Removal for Tidel
- PCI - Track 2 Context Data Removal for Triton
- Support Additional Flexibility in EMV Processing
- Tidel Display Balances on Screen Support
- Triton EMV Offline PIN Synchronization
- Triton Mini-Statement Support

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes for all BASE24 Release 6.0 Version 9 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The Terminal Data File (TDF) has been modified to include several fields added to the addl-data section of the TDF for Triton AKDS support. For further information regarding these new fields see the Triton AKDS Support enhancement later in Section 5.
- Visa Contactless MSD with CVN 17 Support. This enhancement introduces a new value in the ATC CHECK TYPE field in the EMV segment of the CPF, to allow the Application Transaction Counter (ATC) to be checked against the ATC LIMIT field in the Base segment of the CPF. The new value allows ATC limit checking to be performed even when the EMV segment is not present in the appropriate CAF record. The BASE24-atm Authorization EMV extension module has been enhanced to support the new value.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.RQCPFS, SCRNCPPF and SVCPPFS
- Inline changes to BA60DDL.DDLFCPPF

CUSTOMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not Applicable

Database Configuration

Not Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Terminal Data File (TDF)

The TDF conversion program must be run to convert the TDF.

The ATD conversion program on the AT60UC04 subvolume can be used to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 Version 9 format. This program will also split the Release 5.3 TDF into a Release 6.0 Version 9 ATD and TDF file, depending on the device type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. Information on compiling and running this program are found in the following section for the ATM Terminal Data File (ATD).

The TDF conversion program on the AT60UC0C subvolume can be used to convert the TDF records from the Release 6.0 Version 8 format to the Release 6.0 Version 9 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVTDFS file. Running this conversion program is required for any BASE24 customer utilizing the TDF in transaction processing. Users must create the new TDF file using the ATDDLUP file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#tdfcnv.t,name $tdfm,nowait/ cnvtdfm  
-force -list -listreplace
```

Release 6.0 Version 8 users should update the CNVTDFR obey file on the AT60UC0C subvolume with the appropriate information. The CNVTDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD0C on the BA60UC0C subvolume for Release 6.0 Version 8 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

BASE24-atm
General Enhancements
General Changes

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Add Support for AMEX format in EMV IAD

This section describes the implementation requirements associated with the Add Support for AMEX format in EMV IAD enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Add Support for AMEX format in EMV IAD enhancement.

Note: This enhancement does not support AMEX offline PIN Change scripts. A change in TSS is necessary to support that functionality.

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24 EMV customers:

- Inline changes to BA60IEMV.CAMUTILS and EMVG.

All BASE24-atm EMV customers:

- Inline changes to AT60IATH.AUTHEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNEIAD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

A new value has been added to CPF screen 11 EMV IAD FORMAT and ALTERNATE IAD FORMAT. The value of 5 will represent the AMEX IAD format.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Diebold Agilis 2.2 Upgrade

This section describes the implementation requirements associated with the Diebold Agilis 2.2 Upgrade enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a brief functional overview of the Diebold Agilis 2.2 Upgrade enhancement.

The current BASE24 Diebold 10XX/478X Device Handler supports much, if not most, of Diebold Agilis pre-v2.2 functionality. This enhancement adds to that existing functionality by providing support for many of the newer features of Agilis v2.2.

The following changes are included:

Enhanced Status Message Support on the Card Reader

Support has been added for enhanced status messages on the card reader, device type/number “CR01”. This includes support for all new related status codes, for both Solicited and Unsolicited status messages, including those initiated by Diebold’s new Enhanced Anti-Fishing (EAF) option for certain of their magnetic card readers. The following card reader enhanced status messages are supported:

- CR01:3D:58:60. This status message will be transmitted to the device handler as CR01:3E:58:60. It is produced when a card becomes jammed in the card reader indicating a possible fraud attempt. A skimming attack is assumed, therefore the device handler will raise an appropriate event message, then shutdown the ATM.
- CR01:3F:80:01. This status message indicates the possible removal of a foreign device such as a skimmer from the card reader. The device handler will raise an appropriate event message. If the terminal is out of service, the device handler will place it in service.
- CR01:3F:80:02. This status message indicates the possible addition of a foreign device such as a skimmer to the card reader. The device handler will raise an appropriate event message. If the terminal is in service, the device handler will place it out of service.

Support for Separate Feed Failure Messages for No Bills, Cassette Failure, and Pick Unit Failure Status Messages.

The BASE24 Diebold 10XX/478X Device Handler has been modified to raise distinct event messages for 'no bills', 'cassette failure', and 'pick unit failure' errors. Previously, these errors had been grouped under single 'Feed Failure' event messages '58', '59', '60', and '61'. Each of these event messages has been split into distinct event messages, depending upon the specific error, and based upon data values contained in the status message.

Support for New Operation Codes '070' thru '078' for State 'SC ', and Associated Status Messages.

The (ENTIBOLD) Diebold entry program now supports both old and new sets of Operation Codes associated with the Enhanced Smart Card state 'SC '. Support previously existed for the legacy Operation Codes (010, 011, 090, 095, 097, 099). However, Diebold has recently developed a set of new Operation Codes (070 thru 078) for this Smart Card state. These new Operation Codes provide for enhanced functionality. In addition, the BASE24 Diebold 10XX/478X Device Handler has also been updated to support processing of new EMV-specific status codes associated with these new Operation Codes, including raising appropriate event messages as required.

Adding support for the new Enhanced Smart Card state 'SC ' Operation Codes required moving support for the old Operation Codes to a separate Diebold entry program object. Customers who still require support for the legacy Operation Codes will use this new Diebold entry program object (ENTIBLD2). Customers needing support for the new Operation Codes will continue to use the familiar 'ENTIBOLD' object. Both Diebold entry program objects support all other Diebold entry program functionality. A new AAREADME documentation file has been added to the Diebold entry program subvol (AT20IBLD), and should be consulted for more information regarding the current structure of the Diebold entry program.

Diebold's Legacy Smart Card Op Codes:

<u>Operation</u>	<u>Description</u>
010	Authorization Request Cryptogram
011	ICC Transaction Termination
090	Set Tag and Values
091-094	Reserved
095	Update Buffer with ICC Data
097	Select Application / Initiate Application
098	Reserved
099	ICC Session Initialization

Diebold's New Smart Card Op Codes:

<u>Operation</u>	<u>Description</u>
070	Begin ICC Initialization
071	Complete ICC Initialization
072	Begin Application Selection and Initialization
073	Complete Application Selection and Initialization
074	Automatic Language Selection
075	Reselect Application
076	Set ICC Transaction Data
077	Generate ARQC
078	Terminate Transaction

Support for Enhanced Status Messages on the IDM.

Support has been added for both Solicited and Unsolicited status messages on the IDM (Intelligent Depository Module), device type "DP01". The following IDM enhanced status messages are supported:

- DP01:3C:80:00. Document recognition failed. The device handler will raise an appropriate event message, then resume normal processing.
- DP01:3C:81:11. Store and Forward database is corrupt or inoperative. The device handler will raise an appropriate event message, then shutdown the ATM
- DP01:3F:00:40. Suspicious magnetics detected on document. The device handler will raise an appropriate event message, then resume normal processing.

- DP01:3F:42:7C. The depository gate is jammed open. Note that this status is actually transmitted to the device handler as “DP01:3E:42:7C. The device handler will raise an appropriate event message, then resume normal processing.
- DP01:3F: 81:01. RSS closed the SAF deposit batch. The device handler will raise an appropriate event message detailing buffer data (batch number, check total, and cash total), then resume normal processing.
- DP01:61:00:00. An unexpected status message indicating the deposit bin has been closed. The device handler will raise an appropriate event message, then resume normal processing. Note that this is an Unsolicited Status message only.
- DP01:61:00:01. An unexpected status message indicating the deposit bin has been opened. The device handler will raise an appropriate event message, then resume normal processing. Note that this is an Unsolicited Status message only.

Support for Printing the Endorse Side of the Check on Customer Receipts.

The BASE24 Diebold 10XX/478X Device Handler now includes support for printing the endorse side of the check or document on customer receipts. This is accomplished using the control sequence, “ESC r nnn”, where ‘nnn’ is the length of the check image in millimeters. The “ESC r nnn” control sequence requires an ATM equipped with an IDM having dual scanners.

<u>Control Sequence</u>	<u>ASCII Hex</u>	<u>Description</u>
ESC R nnn	1B 52 nnn	Prints scanned image of check front
ESC r nnn	1B 72 nnn	Prints scanned image of check back

Note that, like ESC R nnn, control sequence ESC r nnn is applicable for receipt printers only, nnn is the length of the check image in millimeters, and both require a terminal equipped with an IDM with dual scanners.

Support for Displaying Both Face and Endorse Sides of the Check.

Support has been added for a new control sequence that will allow displaying both face and endorse sides of the check or document at the ATM. Note that displaying the back, or endorse side of a check requires an IDM with dual scanners. Customers wishing to implement this functionality will need to make the appropriate changes to their customer screens. The control sequence governs which side of the document or check, face or endorse, is presented. Location of the displayed image is determined by the position of the control sequence within the screen frame.

ESC @ nnn is a screen control sequence used to display either side of the check or other document. The 'nnn' parameter is a 3-digit check Icon number used to determine which document view is displayed, as well as its size. The range of the Icon number is 000 – 009. The particular document view (face or endorse) is displayed at the current cursor position. Note that if the document to be displayed is other than a check, the view presented is unpredictable. This is because the terminal keys in on the check's MICR data line to determine the correct orientation. Consequently, documents without MICR data may not orient correctly.

The format for the control sequence is as follows:

ESC @ nnn, where 'nnn' is one of the following . . .

- 000 Face Side, 640 x 240 pixels
- 001 Endorse Side, 640 x 240 pixels
- 002 Face Side, 320 x 120 pixels
- 003 Endorse Side, 320 x 120 pixels

Changes are required for screen(s) only. Configuration file (CNFGIX) changes have been left to the customer to make since they are highly subjective.

Software Impacts

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- Changes to AT20IBLD.CNFGDDL, ENTFORMS, ENTIBLDG, ENTIBLDS, ENTSTATS, IBOLDMM, SCRNFOM, AT601000.C1000G, C1000S, AT60AFT.SVRPCTS
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X EMV Device Handler customers:

- Changes to AT60I100.C100EMVS
-

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold ATMs must be loaded with Diebold Agilis v2.2 ATM software.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional Event Messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

- TP-820746-001E, *Agilis 91x User's Guide*
- TP-820843-0010, *Agilis XFS for Opteva, Version 3.8.0, Programmer's Reference Guide*
- TP-821152-001B, *Agilis 91x Terminal Programming Manual*
- TP-821153-001C, *Agilis 91x Status Reference Manual*
- TP-821232-001B, *Agilis 91x EMV Programmer Guide*, 2007

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MasterCard Chargeback TLF Enhancement

This section describes the implementation requirements associated with the MasterCard Chargeback TLF enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the MasterCard Chargeback TLF enhancement.

The BASE24-atm Authorization process has been modified to append a new token, the Track 2 Token (token ID “S6”), to each transaction before writing the transaction to the Transaction Log File (TLF). This token will contain the Track 2 data from the transaction, with sensitive data suppressed as required for PCI compliance. Refer to the BASE24 Tokens Manual for the format of this token. The token is provided for use in offline processing, to assist customers in complying with MasterCard’s 2007 Chargeback requirements. MasterCard wants acquirers to provide the Track 2 service code (or at least the first digit) under certain conditions for a second presentment.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.COBTKN
- Inline changes to BA60DDL.DDLBATKN
- Inline changes to BA60SRC.BATKNCVS and BATKNID

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG and AUTHLIBS
- Updated event (log) message manual BA60LOGM.AT5070

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNMCCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

To ensure that the Track 2 token to be logged to the BASE24-atm Transaction Log File (TLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If this token is to be extracted with the TLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

The TLF may require additional disk space to contain the Track 2 token. If the BASE24 network acts as an issuer for transactions received from an interchange, the ILF may also require additional space. The token will not impact the ILF on transactions on which BASE24 is the acquirer only, because the BASE24-atm authorization process creates the token on the response side, just before logging.

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the *BASE24 Tokens Manual* provides information on assessing the impacts of tokens on log file disk space requirements.

BASE24-atm
General Enhancements
MasterCard Chargeback TLF Enhancement

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Multi-currency with Like Denominations (Diebold)

This section describes the implementation requirements associated with the Multi-Currency with Like Denominations (Diebold) enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Multi-Currency with Like Denominations (Diebold) enhancement.

Software Impacts

All BASE24-atm Diebold 1000 customers:

- Inline changes to AT601000.C1000D and C1000S.

All BASE24-atm Diebold 1000 customers Mobile Top Up customers:

- Inline changes to AT60U100.ATU100MM, C100MCUM and C100MCUS.
- New file AT60U100.U100NAMS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNMCDF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

A new file, U100NAMS, has been added to the AT60U100 subvolume. This file allows currency codes to be defined for each cassette. These values are used in conjunction with the currency value that has been defined in the C100NAMS file to ensure the correct note is selected when 2 or more currencies are present with the same note value. Instructions are included in the new U100NAMS file to help with setting the correct values.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc

NCR 5XXX Optional Receipt Support with Mobile Top-Up

This section describes the implementation requirements associated with the NCR 5XXX Optional Receipt Support with Mobile Top-Up.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR 5XXX Optional Receipt Support with Mobile Top-Up.

This enhancement corrects an issue when Optional Receipts and Mobile Top-Up are used together.

Software Impacts

All BASE24-atm NCR 5XXX customers:

- Inline changes to AT60N50.N50S.

All BASE24-atm NCR 5XXX Mobile Top-Up customers:

- A typographical error was corrected in AT60TN50.TN50NAMS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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NCR Status Code and Model Number Update

This section describes the implementation requirements associated with the NCR Status Code and Model Number Update enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a brief functional overview of the NCR Status Code and Model Number Update enhancement.

This enhancement updates the Base24 NCR 5XXX Device Handler to support many of NCR's newest ATM models, and related M-status codes. Support has also been added for associated event messages.

Support has been added for the following NCR ATM model numbers:

5879	5362	5871	6622	6624	6626	6628
6631	6632	6634	6638	6625	6618	6636

Support has been added for the following device-associated M-status codes:

USB Smart Dip Card Reader

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
D	42	Fatal	Not Authorized
	96	Fatal	F Tamper Switch Triggered
	97	Fatal	R Tamper Switch Triggered
	99	Warning	USB Recovered (Smart Dip Card Reader)

USB EPP Keyboard

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
L	6	Routine	Security Model Reset Successful
	7	Fatal	Security Model Reset Unsuccessful

USB EPP Encryptor

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
L	73	Routine	Invalid Key Code Table
	77	Routine	Signature Verify Failed
	78	Routine	Data Not Found
	80	Fatal	USB Comms Error
	81	Fatal	Not Authorized
	82	Fatal	Master Key Deleted
	83	Suspend	USBEP2 PIN Pad Key Stuck
	84	Fatal	Tamper Detected
	85	Routine	Firmware Update Failed
	86	Fatal	Self Test Failed
	87	Routine	Hardware or Firmware Error

Integrated Motorized Card Reader/Writer

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
D	42	Fatal	Not Authorized
	96	Warning	Device Communications Inoperative
	97	Warning	Device Communications Operative
	98	Good	Device Reconnection (state lost)
	99	Good	Device Reconnection (new device)

80 MM Receipt

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
G	17	Fatal	Transport Jam
	18	Fatal	Capture Jam
	21	Warning	No Receipt During Control
	23	Warning	Capture Bin Full
	24	Fatal	Power Supply Voltage Error
	25	Fatal	Control Board Error
	26	Fatal	Head Overheated (recoverable)
	27	Fatal	Head Burnt Out (non-recoverable)
	28	Warning	Paper Low
	29	Fatal	Exit Jam On Capture
	30	Warning	Knife Needs Replaced

80 MM Receipt (Cont)

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
G	31	Fatal	Receipt Transport Not Connected
	32	Fatal	Fatal on Media Not Cleared.

80 MM Journal

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
H	19	Fatal	Paper Not Winding
	20	Fatal	Mechanism Jammed
	24	Fatal	Power Supply Voltage Error
	25	Fatal	Control Board Error
	26	Fatal	Head Overheated (recoverable)
	27	Fatal	Head Burnt Out (non-recoverable)
	28	Fatal	Paper Low

USB Envelope Dispenser

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
\	56	Fatal	Fish Solenoid Jammed
	57	Routine	Envelope Left
	80	Fatal	Dispenser Command Rejected
	81	Fatal	Dispenser Command Failed
	82	Fatal	Not Authorized

USB Envelope Depository

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
F	51	Suspend	Shutter Moved
	56	Fatal	Fish Solenoid Jammed
	57	Routine	Envelope Left
	58	Routine	No Envelope Entered
	80	Fatal	Depository Command Rejected
	81	Fatal	Depository Command Failed
	82	Fatal	Not Authorized

CPM4

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
q	25	Warning	MICR Reader Error
	26	Warning	Image Processing Error (front)
	27	Warning	Image Processing Error (back)
	46	Fatal	Printer Inoperative
	49	Fatal	Initialization Fault
	50	Fatal	Document Too Long
	51	Fatal	Document Too Thick
	52	Fatal	Document Too Narrow
	53	Fatal	Jam in Stacking Bin 1
	54	Fatal	Stacking Bin 1 Fault
	55	Fatal	Jam in Stacking Bin 3
	56	Fatal	Stacking Bin 3 Fault
	57	Fatal	Double Feed Detect
	58	Fatal	Shutter Communications Fault
	59	Fatal	Shutter Fault.

USB Coin Dispenser

<u>DIG</u>	<u>M- status</u>	<u>Severity</u>	<u>Description</u>
Y	15	Fatal	Download Error
	16	Good	Not Enough Coins
	17	Good	Not Empty

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

- Changes to AT60N50.N50G, AT60N50.N50S
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX SSB Device Handler customers:

- Changes to AT60DN50.N50SSDS

All BASE24-atm NCR 5XXX Wincor BNA/CPM Device Handler customers:

- Changes to AT60CWN5.N50WCPMS, AT60DWN5.N50WBNAS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNNSC

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional Event Messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

- NCR: *"Deltas 17"* (not dated)
- NCR: *"New Model Numbers"* (not dated)

PCI - Track 2 Context Data Removal for Diebold CSP

This section describes the implementation requirements associated with the PCI - Track 2 Context Data Removal for Diebold CSP enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI - Track 2 Context Data Removal for Diebold CSP enhancement.

When the track discretionary data is suppressed, the primary account number (PAN), expiration date and service code are retained. The rest of the track data is replaced with ASCII zeroes.

The BASE24-atm Diebold CSP Device Handler module has been enhanced to suppress Track 2 discretionary data in the internal STM held in memory, when the transaction is no longer in flight. The enhancement also ensures that Track 2 discretionary data does not reside in the temporary context areas of the Terminal Data File (TDF) when the transaction has been completed or cancelled.

Software Impacts

All BASE24-atm Diebold CSP Device Handler customers:

- Inline changes to AT60DCSP.DCSPS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host systems may no longer receive full Track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternately, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35 (Track 2 Data). Hosts must be capable of processing response, reversal, and advice messages without the discretionary data or using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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PCI - Track 2 Context Data Removal for Tidel

This section describes the implementation requirements associated with the PCI - Track 2 Context Data Removal for Tidel enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI - Track 2 Context Data Removal for Tidel enhancement.

When the Track 2 discretionary data is suppressed, the primary account number (PAN), expiration date and service code are retained. The rest of the track data is replaced with ASCII zeroes.

The BASE24-atm Tidel Device Handler module has been enhanced to suppress Track 2 discretionary data in the internal STM held in memory, when the transaction is no longer in flight. The enhancement also ensures that Track 2 discretionary data does not reside in the temporary context areas of the Terminal Data File (TDF) when the transaction has been completed or cancelled.

Software Impacts

All BASE24-atm Tidel Device Handler customers:

- Inline changes to AT60TIDL.TIDELS
- Minor inline changes to AT60TIDL.TIDELG

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host systems may no longer receive full Track 2 data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternately, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35 (Track 2 Data). Hosts must be capable of processing response, reversal, and advice messages without the discretionary data or using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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PCI - Track 2 Context Data Removal for Triton

This section describes the implementation requirements associated with the PCI - Track 2 Context Data Removal for Triton enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI - Track 2 Context Data Removal for Triton enhancement.

When the Track 2 discretionary data is suppressed, the primary account number (PAN), expiration date and service code are retained. The rest of the Track 2 data is replaced with ASCII zeroes.

The BASE24-atm Triton Device Handler module has been enhanced to suppress Track 2 discretionary data in the internal STM held in memory, when the transaction is no longer in flight. The enhancement also ensures that Track 2 discretionary data does not reside in the temporary context areas of the Terminal Data File (TDF) when the transaction has been completed or cancelled.

Software Impacts

All BASE24-atm Triton Device Handler customers:

- Inline changes to AT60TRIT.TRITONS
- Minor inline changes to AT60TRIT.TRITONG

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host systems may no longer receive full Track 2 data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternately, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35 (Track 2 Data). Hosts must be capable of processing response, reversal, and advice messages without the discretionary data or using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Support Additional Flexibility in EMV Processing

This section describes the implementation requirements associated with the Support Additional Flexibility in EMV Processing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Support Additional Flexibility in EMV Processing enhancement.

For additional detailed information on this enhancement see Section 4 (Base).

Software Impacts

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24-atm EMV customers:

- Inline changes to AT60IATH.AUTHEMVG, AUTHEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNFLEX.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

A new CPF screen (screen 12) has been added to include parameters specific to ATM EMV transaction processing.

For full details on the database configuration impacts for this enhancement see Database Configuration in Section 4 (Base) of this document.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

For full details on the file conversion impacts for this enhancement see File Conversions in Section 4 (Base) of this document.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Tidel Display Balances on Screen Support

This section describes the implementation requirements associated with Tidel Display Balances on Screen Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Tidel Display Balances on Screen Support enhancement.

The BASE24-atm Tidel Device Handler module has been enhanced to support the displaying of balances on the ATM screen.

The Extended Features Optional Group is used to accommodate the balances in the ATM message.

The cardholder is now presented with an additional choice, to perform a balance inquiry at the screen in addition to the standard balance inquiry which will produce a receipt containing the balance information.

Due to a limitation of the Tidel Software, the ATM cannot display negative balances, or balances larger than 8 characters. If a negative balance, or a large balance, is present, the device handler will not send the balances to the ATM, and the transaction will be denied. In this circumstance the cardholder may still perform a standard balance inquiry and receive balance information on the receipt.

Software Impacts

All BASE24-atm Tidel Device Handler customers:

- Inline changes to AT60TIDL.RQTTIDLS, SCRNTIDL, TIDELG, TIDELS, TIDLDDLS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNTIBL.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Terminal Definition File (TDF)

Field Name: CUSTOMER BALANCE INFORMATION
Screen: TDF Screen 8
DDL Name: DEV-INFO-TIDEL.CUST-BAL-INFO

Note that this field already exists; it is now being displayed on the Tidel TDF screen for the first time.

Indicates which balances will be presented to the customer. The Device Handler normally uses the CUST-BAL-INFO field in the STM to determine which balance information to format. However, if a balance inquiry request is made and the STM indicates no balances are displayed, the entry in this field will be used. Valid values are:

0 = Use both amounts (new default value)
1 = Use the value in the AMT-2 field
2 = Use the value in the AMT-3 field

Amount 2 reflects the credit balance for transactions impacting a credit account, and it is the ledger balance for non-credit transactions. Amount 3 reflects the available balance for transactions that impact non-credit accounts, and it is the available credit for transactions that impact credit accounts.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

ATMS must have the Balance on screen feature enabled to be able to support the Tidel Display Balances on Screen Support enhancement.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-atm Terminal Definition File (TDF)

For Tidel Device Handler customers, the TDF conversion program can be run to set the default value for the cust-bal-info field to a "0" (use both amounts).

Details of the conversion program can be found in the General Changes earlier in Section 5.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- Tidel: *ACCOMSPC Version 08-10-2007. ATM Communication Specification, Part #201-3653-017, Revision Q.*

Triton EMV Offline PIN Synchronization

This section describes the implementation requirements associated with the Triton EMV Offline PIN Synchronization enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton EMV Offline PIN Synchronization enhancement.

The BASE24-atm Triton Device Handler module has been enhanced to support EMV offline PIN synchronization. In addition to the standard ATM transaction set, the Triton Device Handler is now capable of supporting EMV offline/online PIN change and EMV offline PIN unblock transactions. These transactions are used to keep the online PIN and the offline EMV PIN (the PIN kept on the card) in synchronization when the online PIN is changed, and to reset the bad PIN tries counter on the EMV card.

The Triton Device Handler has been enhanced to allow a request message to contain a transaction code of '01' (PIN Change) without a new PIN (field identifier '@'). This is how the Triton device identifies an EMV PIN Unblock transaction. The transaction code in the BASE24-atm internal message is set to '82' (EMV PIN Unblock) if the transaction code contains '01' (PIN Change), but the message does not contain a new PIN (field identifier '@'). Logic has also been added to allow an EMV PIN Unblock transaction to be reversed.

For some interchanges, e.g. VisaNet, there is a requirement to set the Authorization Response Code (EMV tag '8A') to '85' in approved EMV responses to offline PIN management requests. When formatting an EMV response message, the Triton Device Handler retrieves the EMV Status Token, regardless of whether the EMV Response Token is present. If the transaction has been approved, and the ISO-RC-IND field in the EMV Status Token is set to '9' (indicating that the ISO response code received from the interchange should be returned to the terminal as the Authorization Response Code), then tag '8A' in the response message is set to '85' (ASCII hexadecimal '3835'), instead of '00' (ASCII hexadecimal '3030').

The offline PIN and the bad PIN tries counter on the EMV card are updated by an issuer script returned to the Triton ATM by BASE24. If the EMV card is unable to process the script, then the terminal must send a reversal message to the Triton Device Handler, to enable BASE24-atm to reverse the PIN Change transaction. The outcome of issuer script processing is included in an EMV data element called the Issuer Script Results. The Triton Device Handler adds the EMV Issuer Script Results Token (Id = 'BJ') when a reversal message containing the Issuer Script Results is received from the terminal. To do this, the Issuer Script Results (TSEMVID = 'DF05')

is extracted from the encoded EMV untagged-data block (field identifier 'uh'), before being appended to the BASE24-atm internal reversal (0420) message.

The Triton specification defines 3 new reversal reason codes ('08', '09', '10') specific to offline PIN management processing. These codes are all mapped to the default BASE24-atm reversal code of '10' (hardware error), because the set of valid BASE24-atm reversal codes does not include a more appropriate value.

Software Impacts

All BASE24-atm Triton Device Handler customers:

- Inline changes to AT60TRIT.TRITONG and TRITONS.

All BASE24-atm Triton Device Handler EMV customers:

- Inline changes to AT60ITRT.TRITEMVG and TRITEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNTRPS.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

This enhancement uses the EMV Issuer Script Results Token (Token ID = 'BJ') to carry information regarding the processing of an issuer script by the EMV card.

If this token is to be logged to the BASE24-atm Transaction Log File (TLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If this token is to be extracted with the TLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Triton devices must be able to provide the Host with request messages that contain a transaction code of '01' (PIN Change) without a new PIN (field identifier '@'), to allow BASE24 to identify EMV PIN Unblock transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

This enhancement uses the EMV Issuer Script Results Token (Token ID = 'BJ') to carry information regarding the processing of an issuer script by the EMV card. If this token is to be included in TLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-atm users must consider this additional data in host programs that process this data.

For PIN change processing, the system that authorizes the online PIN Change transaction, i.e. BASE24 or an issuer Host system, must also generate the 'PIN Change' script command. (Note that the original PIN must be verified before the PIN Change transaction can be authorized.) Similarly, for EMV PIN unblock processing, the system that authorizes the transaction, i.e. BASE24 or an issuer Host system, must also generate the 'PIN Unblock' script command. (Note that the original PIN must be verified before the PIN Change/Unblock script command can be generated.) This may require enhancements to Host systems.

User/Operational Impacts

Disk Impacts

The TLF and ILF may require additional disk space if any of the tokens used by this enhancement are included in the transaction records logged to these files.

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the *BASE24 Tokens Manual* provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

- Triton Systems: *Triton Terminal Communications Protocol and Message Format Specification*, Version 5.35

Triton Mini-Statement Support

This section describes the implementation requirements associated with Triton Mini-Statement Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton Mini-Statement Support enhancement.

The BASE24-atm Triton Device Handler module has been enhanced to support mini-statement transaction functionality. For the Triton Device Handler, data compression is not supported (by the device) for the statement print transaction, and only one request and response will be supported between the ATM, Base24 and the Host. Multiple pages cannot be supported.

The Extended Features Optional Group is used to accommodate the balances in the ATM message.

Software Impacts

All BASE24-atm Triton Device Handler customers:

- Inline changes to AT60TRIT. TRITDDLs, TRITNAMS, TRITONG and TRITONS
- Updated event (log) message manual BA60LOGM.ATTRIT
- Changes to AT60TRIT.TRITMAN

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNTRMS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Terminal Definition File (TDF)

BASE24-atm Triton customers must update their Triton TDF records to allow a Statement Print transaction. This field is already supported and is configured on Screen 3 of the TDF. There are no AFT changes or file conversions required for this enhancement.

```
BASE24-ATM  ATM TERMINAL DATA      TES7  BK30  08/10/18  22:52  03 OF 22
TERM ID:  S74RTRITOMA      FIID: BK30  REGION:  001  BRANCH:  001  LN: TES7
```

SPECIAL TRANSACTION CLASS

DEP WITH CASHBACK	0	(NOT ALLOWED)
CASH CHECK	0	(NOT ALLOWED)
STATEMENT PRINT	0	(NOT ALLOWED)
PIN CHANGE	0	(NOT ALLOWED)
NON CRNCY DISP	0	(NOT ALLOWED)
NON CRNCY DISP CC	0	(NOT ALLOWED)
LOAD VALUE	0	(NOT ALLOWED)

0 = NOT ALLOWED 1 = ALLOWED INTRACOUNTY 2 = ALLOWED INTRASTATE
3 = ALLOWED INTERSTATE 4 = ALLOWED INTERNATIONALLY

TOKEN RETRIEVAL OPTION: 1 (TDF)

```
RECORD LAST CHANGED: 08/10/17 12:46 BY USER: 0036 , 00000501 CHANGE
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Customers must enter the Management Function Settings in Terminal Configuration to turn on mini-statement transactions under Select Transaction types, in order to be able to support the Triton Mini-Statement Support enhancement.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-atm Terminal Definition File (TDF)

For Triton Device Handler customers, the TDF conversion program must be run to set the default value for the prntr-cap field in the device dependent data.

Details of the conversion program can be found in the General Changes earlier in Section 5.

Host/Co-Network Impacts

The Host needs to format statement data appropriately for the mini-statements. Data compression techniques cannot be used, and only a single page of data can be sent to be printed on the receipt. The data is expected to conform to the number of columns that can be printed at on a Triton ATM's receipt printer, and must be able to fit on the maximum number of lines that are available for printing on the ATM's receipt. If the data is not formatted in this way, warning message will be raised by the Triton Device Handler via EMS, but the statement will continue to print.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- Triton: *Triton Terminal Communications Protocol and Message Format Specification*, Version 5.35

Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with Release 6.0 Version 9, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- NCR Standard APTRA Edge NDC Support
- Tidel AKDS
- Tidel Mobile Top-Up
- Triton AKDS

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding Release 6.0 Version 9 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding Release 6.0 Version 9 changes, conversion requirements, and implementation requirements for future reference.

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NCR Standard APTRA Edge NDC Support

This section describes the implementation requirements associated with the NCR Standard APTRA Edge NDC Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a brief functional overview of the NCR Standard APTRA Edge NDC Support enhancement.

This enhancement updates the Base24-atm NCR 5XXX Device Handler to support NCR's Standard APTRA Edge NDC Message Format interface. This support has been implemented as a separately bindable extension module. This new extension module is responsible for processing transaction request messages originating from an NCR ATM running NCR's Standard APTRA Edge interface software.

ATM Download:

Unlike NCR's NDC+ terminal software, Standard APTRA Edge NDC does not use host downloaded states or screens. Instead, proprietary Standard APTRA Edge states and screens are included in the Edge terminal software build. It should be noted that while the Standard APTRA Edge interface software does not use host downloaded states or screens, it can accept them in a download and, if received, are simply discarded.

Proprietary Operation Codes:

The Standard APTRA Edge interface also has its own proprietary Op Codes. Mapping these Standard Edge interface Op Codes to the appropriate internal BASE24-atm NCR 5XXX Device Handler Op Codes is one of the functions of the new Standard APTRA Edge bindable extension module.

Remotely Configuring the Standard APTRA Edge NDC interface ATM:

Functionality of the NCR Standard APTRA Edge NDC terminal software is largely preconfigured through the NCR ATM software build and resident data files. However, the Standard APTRA Edge ATM is also remotely configurable via XML data language downloads. By downloading to the terminal specific XML commands, which overlay and over-ride existing configuration, customers can remotely customize their Standard APTRA Edge ATMs. This can be accomplished as part of a regular ATM load, using Screen Data Load Messages and the existing CNFG5070 configuration file. Using existing DCT functionality, regular Customer Screens can be 'painted' with XML commands to customize the Edge terminal's configuration. It is the presence of

XML data that causes the Edge terminal to not discard these 'Screen' load messages as it would regular Customer Screens.

Downloading modifications to the Data Dictionary and/or XML file(s) is accomplished using the standard NDC+ Screen / Keyboard Data Load Message (Message Class '3'), but with a slightly modified format. The general format for the Screen / Keyboard Data Load Message carrying XML data instructions is as follows.

<u>Field</u>	<u>No. of Chars</u>	<u>M / O</u>	<u>Description</u>
a	Var	M	Protocol-dependent Header
b	1	M	Message Class = '3' (Data Command)
c	1	M	Response Flag (reserved for future use)
fs	1	M	Field Separator
d	3	M	Logical Unit Number (LUNO)
fs	1	M	Field Separator
e	3	O	Message Sequence Number (reserved for future use)
fs	1	M	Field Separator
f	1	M	Message Sub-Class = '1' (Customization Data)
g	1	M	Message Identifier = '1' (Screen and/or Keyboard Data)
fs	1	M	Field Separator
h	3	*	Screen Number (Record Number)
I	Var	**	Screen Data Field (XML Data)

* For XML data records, the Screen Number is actually the number of the particular load record, and is user defined. It exists only to permit the user to distinguish between various records within the configuration file.

** For XML data records, the Screen Data Field will contain the actual XML instruction(s). Note that this field is defined as variable in length. NCR has defined the maximum size of the entire Screen / Keyboard Data Load Message to be 2000 bytes. However, the actual maximum record size will be significantly less than 2000 bytes as it is limited by the size of the DCT Screen input palette, and because the configuration instructions are written as nested XML code.

The following is an example of an instructional update of the Data Dictionary as it might be constructed on a Customer Screen entry screen of the DCT.


```
<CONFIGURATION_DATA reboot="y">  
  <DD_UPDATE>  
    <GROUP>CONSUMERFLOW</GROUP>  
    <NAME>ENVELOPEDEPOSIT</NAME>  
    <VALUE>YES,1,8</VALUE>  
  </DD_UPDATE>  
</CONFIGURATION_DATA>
```

In this example, the Envelope Deposit transaction is affected. This instruction enables the Envelop Deposit transaction at the ATM (the 'YES' part of the instruction), and places the transaction selection at the 8th position of the first menu (the "1,8" part of the instruction).

Note that the downloaded instruction is transmitted to the ATM as a nested block of XML instructional data, starting with <CONFIGURATION_DATA> and ending with </CONFIGURATION_DATA>. The example above also includes the [reboot="y"] clause. This clause must be included whenever a reboot of the ATM is required. Some, but not all, Data Dictionary updates require rebooting. In fact, updates to "CONSUMERFLOW" such as the one in the example above do not require rebooting. The reboot="y" command is included in the example above only to demonstrate the use of the command.

The actual update instructions for the Data Dictionary are included within the <DD_UPDATE> </DD_UPDATE> parameters. This example contains a single update instruction, but additional Data Dictionary update commands can be included within a single load message, by inserting them between the <CONFIGURATION_DATA> </CONFIGURATION_DATA> parameters. However, the entire update command must be contained within a single load message, and every load message must contain the <CONFIGURATION_DATA> </CONFIGURATION_DATA> parameters.

Here is how the preceding example instruction would look, ready to be downloaded, formatted within the configuration file's Load Message record (spaces appear for readability only):

```
9 3 0 fs 000 fs 000 fs 11 fs 001 <CONFIGURATION_DATA  
reboot="y"><DD_UPDATE><GROUP>CONSUMERFLOW</GROUP>  
<NAME>ENVELOPEDEPOSIT</NAME><VALUE>YES,1,8</VALUE>  
</DD_UPDATE></CONFIGURATION_DATA>
```

Customers should consult the *NCR Aptra Edge Remote Configuration Guide, release 02.03.xx, January 11, 2006*, for more specific information on formatting instructional data.

In addition to the Data Dictionary, there are several XML files that may be updated as well. These include the following (default locations are indicated):

- ButtonToKEYMAP.xml – c:\Webatm\WebPages\scripts
- Coupon.xml – c:\Webatm
- Journal.xml – c:\Webatm
- Receipt.xml – c:\Webatm
- SoundEng.xml – c:\Webatm
- SoundSpa.xml – c:\Webatm

Just like Data Dictionary updates, XML file updates must be included within the <CONFIGURATION_DATA> </CONFIGURATION_DATA> block. Specific instructions are further nested within <XML_UPDATE> </XML_UPDATE> blocks. The initial <XML_UPDATE> command must include the complete path and filename of the XML file being modified. The filename clause is followed by another <XML_Udate> </XML_Udate> block, and then a nested <Update> </Update>, <Insert> </Insert>, or <Delete> </Delete> block, depending on the intent of the instruction.

The following is an example of an instructional update of the ButtonToKeyMap XML file as it might be constructed on the Customer Screen entry screen of the DCT.

```
<CONFIGURATION_DATA reboot="y">
  <XML_UPDATE FILENAME="C:\WebATM\WebPages\scripts\ButtonToKeyMap.xml">
    <XMLUpdate>
      <Update>
        <VALUE>YES,1,8</VALUE>
      </Update>
      <NodeKey>
        <Path>Document/FDKKey</Path>
      </NodeKey>
      <XML>
        <FDKKeyMap Button="1" FDKKey="1"/>
        <FDKKeyMap Button="2" FDKKey="2"/>
        <FDKKeyMap Button="3" FDKKey="3"/>
        <FDKKeyMap Button="4" FDKKey="4"/>
        <FDKKeyMap Button="5" FDKKey="5"/>
        <FDKKeyMap Button="6" FDKKey="6"/>
        <FDKKeyMap Button="7" FDKKey="7"/>
        <FDKKeyMap Button="8" FDKKey="8"/>
      </XML>
    </XMLUpdate>
  </ XML_UPDATE>
</CONFIGURATION_DATA>
```

This example command affects the FDKKey Settings as defined within the specified XML file.

Note that a single Screen / Keyboard Data Load Message may contain multiple Data Dictionary update commands, multiple XML file update commands or a combination of both Data Dictionary and XML file update commands.

Again, customers should consult the *NCR Aptra Edge Remote Configuration Guide, release 02.03.xx, January 11, 2006*, for more specific information on formatting instructional data.

Note that some modifications to the Data Dictionary will require an ATM reboot, while all modifications to XML files require rebooting of the ATM.

Interfacing with both NDC+ and Standard APTRA Edge NDC interface ATMs:

It is anticipated that customers would want the flexibility to process transactions originating from both NDC+ and Standard Aptra Edge terminals using a single NCR 5XXX Device Handler object. Since the basic message format is the same for both Standard APTRA Edge and NDC+ terminals (NDC+ message format), this is possible as long as the device handler knows from which type of terminal a message originates. The BASE24-atm NCR 5XXX Device Handler is able to distinguish between terminals by using the Standard APTRA Edge Software ID of "V351".

Currently Supported Processing:

The Release 6.0 Version 9 NCR Standard APTRA Edge NDC Device Handler was built on NCR's Standard APTRA Edge interface version 2.4, and supports the following function set:

- Non-Currency Dispense
- Open Account Relationship (Multiple Account Select)
- PIN Change
- Standard Set of Financial Transactions
- Standard Terminal Management
- Surcharging

The Release 6.0 Version 9 NCR Standard APTRA Edge NDC Device Handler does not currently support:

- Bunch Note Acceptor (BNA)
- Check Processing Module (CPM)
- Europay, MasterCard, Visa (EMV) processing

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

- Changes to AT60N50.N50D, N50G, N50S
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX Standard APTRA Edge NDC Device Handler customers:

- New AT60EN50 subvol.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNEGE

CUSTOMACS Changes

The following Make flag has been added to the CUSTOMACS file (Note that this CUSTOMACS change was made in Version 8). Set the flag to true to bind the Standard APTRA Edge NDC module to the NCR 5XXX Device Handler.

atm_dh_n50_edge_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

NCR ATMs must be loaded with NCR's Standard APTRA Edge NDC message format interface software.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional Event Messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

- *NCR Aptra Edge Remote Configuration Guide*, rel 02.03.xx, January 11, 2006
- *NCR Aptra Edge Standard NDC Message Format Guidelines*, rev H, December 12, 2005
- *NDC+ Programmer's Reference Manual*, rel 7.0, August 2002

Tidel AKDS Support

This section describes the implementation requirements associated with Tidel AKDS Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Tidel AKDS Support enhancement.

The BASE24-atm Tidel Device Handler has been enhanced to support the Tidel Automated Key Distribution System (AKDS) enhancement. A new bindable extension subvolume, AT60ATDL, has been developed to support the Tidel AKDS Support enhancement.

This enhancement enables customers to perform AKDS downloads from the Tidel Device Handler to Tidel ATMs configured to support signature-based AKDS.

The Tidel Device Handler has been modified to process AKDS messages for dialup and leased-line ATMs. Tidel allows only double-length Master keys to be downloaded to the ATM via AKDS.

AKDS commands can be performed via NCPCOM, or at the ATM, using the Remote Key Load options presented in the Managers screens.

Tidel is owned by NCR, therefore the EPPs in Tidel ATMs are manufactured by NCR. The Certificate Authority (CA) to be used will be the NCR Trusted Center.

Prerequisite: Transaction Security Services (TSS) Version 07.4 must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS, FLGSDOC
- Changes to BA60DDL.DDLACNST
- Changes to BA60SRC.BASEFM, BASEMM, BASRCM, SECUTLLM
- Changes to BA60MISC.BAMISCTD

All BASE24-atm customers:

- Inline changes to AT60DDL.DDLFTDF
- Changes to AT60SRC.ATDXG, ATMFM, ATMMM
- Inline changes to AT60AFT.RQATDS, SVATDTBL and SVATDTS
- Changes to BA60MISC.LCONFAT

All BASE24-atm Tidel Device Handler customers:

- Inline changes to AT60TIDL.TIDELD, TIDELG, TIDELM, TIDELS, TIDLDDL
- Updated event (log) message manual BA60LOGM.ATTIDL
- Changes to AT60TIDL.TIDLMAN

All BASE24-atm Tidel Device Handler customers requiring the new AKDS functionality:

- A new extension subvolume, AT60ATIDL, with all of the files required to support the Tidel AKDS functionality.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNTIAK.

CUSTOMACS Changes

The following Make flag has been added to the CUSTOMACS file. Set the flag to true to bind the Tidel AKDS module to the Tidel Device Handler.

atm_dh_tidel_akd_on flag

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to support this enhancement.

Note that the first of these params already exists for the NCR 5XXX Device Handler, and is simply being used by the Tidel AKDS module too. This param, therefore, already exists in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-AKDS-TSS-TIMER

This param is used by the Tidel Device Handler. It specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

The following params are new for the Tidel Device Handler. They have been added to the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-TIDL-AKDS-SERIAL-NUM-VRFY

This param is used by the Tidel Device Handler to specify whether Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number), or "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

ATM-DH-TIDL-AKDS-HOST-LOAD-TIMER

This param is used by the Tidel Device Handler to specify the number of seconds the ATM waits for a Host RKM Command message to be sent during an AKDS load, before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 60 is used.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

Terminal Definition File (TDF)

The field "Hardware" on screen 1 of the ATD has been updated to include a possible value of "8" for the Tidel ATMs. During AKDS processing, the Tidel Device Handler will set this value to "8".

The Key Signer field on screen 22 already exists, but must be set to an appropriate value for Tidel AKDS. This field is used to indicate the name to be used in the HOST field of the TSS Host Public Key Security Datasource (HSPKD). If multiple Host Public Keys are required for an individual vendor's ATMs, the Key Signer field can be used to assign a separate HOST name for each of those Public Keys.

This field allows up to 16 alphanumeric characters of free-format text to be entered. Embedded spaces are allowed. The default value is "HOST".

The EPP Format field on screen 22 already exists. This field is used to indicate whether the EPP, and therefore the ATM, supports Signature- or Certificate-based AKDS processing. During AKDS processing, the Tidel Device Handler will set this value to indicate that signature-based AKDS is being performed

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS.

Terminal Software

The Tidel ATM must be running Tidel's 4.22.00 Software.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the TDF screen 22 must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) Version 07.4 must be installed prior to implementation of this enhancement.

Device Definition

If a TCP/IP connection is used, no changes are necessary. If the ATM is to be connected via a dialup connection, using the VISA protocol, changes will be necessary to the device definition in XPNET. Changes for the VISA dialup protocol are noted here. The device defined for the ATM's station must be set up to use the ACISTD variant of the VISA protocol. This ensures that the connection to the ATM is not dropped after each step of the AKDS message sequence.

An example of such a definition follows:

> info device d74r^dial, detail

```

DEVICE:  \K9.P74R^NODE.D74R^DIAL
TYPE      255          ACLENGTH      2
RECVLEN   2000         XMITLEN       2000
APPLID
PROFILE
TRADDRESS OFF          TTRANSPARENT OFF
DEFRSP    OFF          TRLEN        0
SELECTION  NORMAL      NRSSCP       OFF
INPUTFORMAT
OUTPUTFORMAT
POLLFORMAT
SELECTFORMAT
DESTRROUTE
XNCONF
RPCONF
DEVICE:  \K9.P74R^NODE.D74R^DIAL
WILDCHAR  "?"          NUM DEST ENTRIES:  1

ENTRY     DESTINATION   OFFSET   MATCH CHARS
1         P74R^TIDEL    3        "A01"
DEVICE:  \K9.P74R^NODE.D74R^DIAL
WILDCHAR  "?"          NUM PROT ENTRIES:  1

ENTRY     PROTOCOL      OFFSET   MATCH CHARS
1         ACISTD        0        "?"

```

File Conversions

BASE24-atm Terminal Definition File (TDF)

For users wishing to take the Tidel AKDS enhancement, either the TDF conversion program must be run to convert the TDF, or the customer may modify the fields manually via PATHWAY for affected ATMS.

Details of the conversion program can be found in the General Changes earlier in Section 5.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-atm users that use OMF Extract to accept and process information from the TDF record.

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, the Enable PKI field on the Security Device Configuration window of the ACI Desktop must be checked. If this field is not checked for a security device, it is not used for PKI functions. This enables you to consolidate all PKI functions in just one or a few devices.

Initial Registration and Certification for the BASE24 System

Initial setup procedures must be carried out for Transaction Security Services before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain a public and private key pair, a signature of the EPP's public key, the NCR's (Certificate Authority's) public key, and a unique serial number. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security File (HSPKD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair.

The public key must be sent to the Certificate Authority (NCR Trusted Center) to be verified and signed. The Certificate Authority must return BASE24's public key signature, as well as the Certificate Authority's own public key. These values must be entered into the Transaction Security Services database.

Details of the registration and certification process for the BASE24 system can be found in the *Transaction Security Services Processing Guide*, for TSS Version 07.4, in appendix B: Public Key cryptography. Follow the details under the NCR 5XXX Terminal Implementation procedures.

Tidel Device Handler – NCS/NCPCOM Commands

The following 9500 command is supported in the Tidel Device Handler. It is entered at the BASE24 AFT NCS screen or through NCPCOM.

LOADALLKEYS

This 9500 command causes the device handler to initiate a Load All Keys sequence.

This command allows the authentication and public key exchange, the download of the Master Key, and a load of the working MAC and/or PIN key to be processed in a single command.

Once the device handler receives the 9500LOADALLKEYS message, it will set a flag in the TDF record indicating that an AKDS load is required. Nothing more happens until the ATM performs a financial transaction, at which time the device handler adds a new field to the response message to the ATM, indicating that an AKDS load must be initiated. It is then that the ATM will initiate the AKDS sequence. The device handler, EPP and Transaction Security Service (TSS) exchange a series of messages to authenticate each other and exchange the public keys. These public keys are used for encryption and mutual authentication in all future master key downloads. Then a series of messages follows in order to generate the Master Key, and download it to the EPP.

The load of the new MAC and/or PIN working key is the last step of the Master key download.

This command can be performed as often as is necessary to preserve the integrity of the ATM's keys.

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LOADALLKEYS <terminal ID>"

<process name>: The symbolic name assigned to the Tidel Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the Terminal Data file (TDF).

Supported Keys

Only the ATM Master key can be downloaded to an ATM using Public Key Cryptography as part of the AKDS enhancement. Whilst the downloading of the working PIN and/or MAC keys will occur as the final step in the AKDS process, these keys will continue to be encrypted under the master key. They may also be downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Tidel AKDS enhancement is implemented, additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the BA60LOGM.ATTIDL event message documentation file.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Tidel AKDS.

- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases. All of the public key cryptography commands are defaulted to off in the security policy delivered from the factory due to security exposure. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application); nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in your security policy using the 108 and 109 commands:

- Generate RSA key pair (Command 120)
 - Generate AKB for root public key (Command 122)
 - Verify public key and generate AKB of public key (Command 123)
 - Generate digital signature (Command 124)
 - Verify digital signature (Command 125)
 - Generate message digest (Command 126)
 - Generate ATM master key and encrypt with public key (Command 12F)
- Thales e-Security version 6.02 firmware and the optional DSP plug-in (for RSA functions) provides public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The commands are as follows:
 - Generate RSA key set (Command EI)
 - Load a secret key (Command EK)
 - Generate a MAC on a public key (Command EO)
 - Validate a certificate and generate MAC on its public key (Command ES)
 - Generate a signature (Command EW)
 - Validate a signature (Command EY)
 - Export a DES key (Command GK)
 - Hash a Block of Data (GM)

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS. An EPP is a tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, and the cryptographic keys. All of the ATM's cryptographic functions are performed by the EPP.

Reference Documents

Atalla: *Atalla Key Block Banking Command Reference Manual* (524130-002, July 2002)

Thales: *Host Security Module RG7000 Programmer's Manual* (1270A514 Issue 4, November 2001)

Tidel ATM Communication Specification (Part Number 201-3653-017, revision Q, 10th August 2007)

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Tidel Mobile Top-Up

This section describes the implementation requirements associated with the Tidel Mobile Top-Up enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Tidel Mobile Top-Up enhancement.

The new BASE24-atm Tidel Mobile Top-Up Device Handler extension enables cardholders to replenish, or top up, their mobile telephone accounts at an ATM. The replenishment transaction is a non-currency dispense transaction. A telecommunications services provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account, and calculating any taxes applied to the transaction. The BASE24-atm product authorizes the funds portion of the transaction.

NOTE: The BASE changes in the BASE24-atm Mobile Top-Up Integration optional enhancement must be installed prior to implementation of the Tidel Mobile Top-Up enhancement. For more information refer to the *BASE24 Release 6.0 Implementation Guide (Version 5)*. A complete list of the files affected by the Release 6.0 Version 5 mobile top-up support can be found in the SCN file BA60UD08.SCNMTOP.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC

All BASE24-atm Tidel customers:

- Hooks in the main device handler module to support Mobile Top-Up (AT60TIDL.TIDELS)
- Inline changes to AT60TIDL.TIDELG, TIDELM
- Additional procedures in AT60TIDL.TIDELD
- Updated event (log) message manuals BA60LOGM.ATTIDL

All BASE24-atm Tidel customers taking the Mobile Top-Up enhancement:

- New subvolume, AT60TTDL, with files for the Tidel Device Handler Mobile Top-Up extension

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNTIMO.

Customers who take this enhancement should also install the enhancement Tidel General Enhancements. The large tokens used for Mobile Top-Up will need more space than is available in the Last Message Area of the TDF record. For more information refer to the *BASE24 Release 6.0 Implementation Guide (Version 7)*. A complete list of the files affected by the Release 6.0 Version 5 mobile top-up support can be found in the SCN file BA60UD0A.SCNTIDL.

CUSTOMACS Changes

Set the following flag to true to bind the Mobile Top-Up module to the Tidel Device Handler.

- atm_dh_tidel_mtu_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to or already exist in the LCONF to support this enhancement. Samples of the parameters are provided in the LCONFBA and LCONFAT files located on BA60MISC subvolume.

Different param records can be specified for each device handler process if desired.

MOBILE-PARAM-VERSION-NUM

A value used by the device handler to determine whether the mobile top-up information stored at the device must be replaced during a download request/response. This param value must be manually incremented when one or more of the following parameters are changed, otherwise the device handler will not send the updated information to the device:

- CARRIER-VT*identifier*
- CARRIER-AMTS-VT*identifier*
- MOBILE-OPERATOR-ID-VT*identifier*

This param is required by the BASE24-atm Tidel Mobile Top-Up add-on product. Valid values are 0001-9999. There is no default value.

CARRIER-AMTS-VTidentifier

This param is used by the Tidel Device Handler and should only be added for the definitions-based model of mobile top-up. It identifies the six amounts supported for this mobile operator.

The *identifier* is a two-digit numeric value, 00 – 08, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

Valid values must be formatted with eight amounts and each amount must be six digits separated by semicolons.

Example: 000030;000050;000075;000000;000000;000000;000000;000000

Represents \$30, \$50, and \$75 as valid amounts for this carrier.

CARRIER-VTidentifier

This param is used by the Tidel Device Handler. It identifies the four-digit mobile phone operator ID.

The *identifier* is a two-digit numeric value, 00 – 08, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

The value of the param links a mobile operator with an entry in the Mobile Operator File (MOF) and enables the mobile operator to be identified for the transaction.

MOBILE-NUM-MAX-LGTH

This param, used by the Tidel Device Handler, specifies the maximum length allowed for the mobile phone number entered by the cardholder. Valid values are 0 - 15. The default value is 11.

MOBILE-NUM-MIN-LGTH

This param, used by the Tidel Device Handler, specifies the minimum length allowed for the mobile phone number entered by the cardholder. Valid values are 0 - 15. The default value is 10.

MOBILE-OPERATOR-ID-VT*identifier*

This param is used by the Tidel Device Handler. It specifies the mobile phone operator IIN (7 digits) plus the mobile operator name (up to 20 characters) associated with the mobile phone operator.

The *identifier* is a two-digit numeric value, 00 – 08, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

Valid values must be formatted with a seven-digit operator IIN followed immediately by a mobile operator name.

Example: 1200991CARRIERNAME

Database Configuration

Mobile Operator File (MOF)

The Mobile Operator File (MOF) is a required file for the Mobile Top-Up for Tidel Device Handler Enhancement and contains one record for each telecommunications provider supplying mobile top-up services for its customers. It defines details of the mobile operators and the services provided. Information on use of the MOF file may be found in the *Base Files Maintenance Manual*.

Split Transaction Routing File (STRF)

The Split Transaction Routing File (STRF) is a required file for the Tidel Mobile Top-Up Enhancement. It contains one record for each Transaction Subtype in the BASE24 network that requires unique routing. The STRF enables the routing of transaction requests to two or more destinations based upon information obtained from a device for a single transaction. Information on use of the STRF file may be found in the *Base Files Maintenance Manual*.

Token file (TKN)

The following existing BASE24 tokens are utilized for this enhancement:

- The Non-Currency Dispense Token (token ID "A5")
- The Pre-Pay Receipt Token (token ID "BF")
- The Pre-Pay Top-Up Token (token ID "BI")
- The Transaction Subtype Token (token ID "BM")
- The Split Transaction Routing Token (token ID "BR")
- The Pre-Pay Original Data Token (token ID "BX")

If the tokens for this enhancement should be logged to the TLF or the ILF, the appropriate token records must be configured in the Token file.

If the tokens for this enhancement are to be extracted with the TLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Configure the AT60TTDL.TTDLNAMS table. This table controls mapping a Service Provider ID to a BASE24 Transaction Subtype. ALL IDs supported at the Tidel Terminal for Mobile Top-Up transactions must be entered in this table.

Device Configuration

To utilize this enhancement, terminals must run a version of firmware that supports Mobile Top-Up transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

In a standard BASE24-atm system, the device handlers always route transactions directly to the BASE24-atm Authorization module. In a mobile top-up environment, the device handler must accommodate the possibility of multiple authorization destinations. The BASE24-atm Tidel Device Handler will route mobile top-up transactions to the destination specified in the Split Transaction Routing File (STRF). This destination should be the process name of the BASE24-atm Transaction Context Manager (TCM).

The TCM will utilize the new Split Transaction Routing File (STRF) for purposes of routing. When the transaction is not a mobile top-up transaction, normal transaction processing and routing occurs.

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts:

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the BA60LOGM.ATTIDL event message documentation file.

If the contents of any of the following parameters change for a Tidel Device Handler process, an additional step must be performed.

CARRIER-VT*identifier*

CARRIER-AMTS-VT*identifier*

MOBILE-OPERATOR-ID-VT*identifier*

The contents of the LCONF param MOBILE-PARAM-VERSION-NUM for the same process(es) must be incremented. This param is a value used by the device handler to determine whether the mobile top-up information stored at the device must be replaced during a download request/response. If the param is not changed, the device handler will not send the updated information to the device.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the BASE24 Tokens Manual provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

Performance will be impacted in systems running mobile top-up transactions due to the increase in the transaction path length. The transaction path is longer because mobile top-up transactions are routed to two destinations (i.e., a funds authorizer and a mobile I/F). Additionally, there may be multiple interactions between the BASE24 device handler process, the BASE24-atm Transaction Context Manager process, and the Mobile Operator Interface process prior to the funds authorization and prior to the final top-up authorization. While each interaction with the device handler process resets the timer, BASE24-atm customers may need to examine the value of the LCONF parameter ATM-DH-AUTH-TIMER and determine if the increased length of the transaction path necessitates an increase in the value assigned to ATM-DH-AUTH-TIMER.

Vendor Impacts

Not applicable

Reference Documents

ATM COMMUNICATION SPECIFICATION, NCR EasyPoint LLC, Carrollton, Texas,
PART NUMBER: 201-3653-017, REVISION: Q, August 10, 2007

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Triton AKDS Support

This section describes the implementation requirements associated with Triton AKDS Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton AKDS Support enhancement.

The BASE24-atm Triton Device Handler has been enhanced to support the Triton Automated Key Distribution System (AKDS) enhancement. A new bindable extension, AT60ATRT.TRITAKDS, has been developed to support the Triton AKDS Support enhancement.

This enhancement enables customers to perform AKDS downloads from the Triton Device Handler to Triton ATMs configured to support certificate-based AKDS.

The Triton Device Handler has been modified to process AKDS messages for both dialup and leased line ATMs. Triton allows both the PIN and MAC Master keys to be separately downloaded to the ATM via AKDS.

AKDS commands can be performed via NCPCOM, or at the ATM, as one of the options presented in the Management Functions menu, under Key Management functions.

As chosen by Triton, the Certificate Authority (CA) to be used will be Verisign.

Prerequisite: The Transaction Security Services CSM for Triton AKDS on Version 07.4 must be installed prior to implementation of this enhancement. This CSM code will be included in product TSS Version 09.2.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS, FLGSDOC
- Changes to BA60DDL.DDLGTSS, DDLACNST
- Inline changes to BA60ASRC.SECAKDS
- Changes to BA60SRC.BASEMM, BASEFM, BASRCM, SECUTLLM, SECUTILS
- Updated event (log) message manual BA60LOGM.BASECUTL
- Changes to BA60MISC.BAMISCTD

All BASE24-atm customers:

- Inline changes to AT60DDL.DDLFTDF
- Changes to AT60SRC.ATDXG, ATMFM, ATMMM
- Inline changes to AT60AFT.RQATDS and SVATDTS
- Changes to BA60MISC.LCONFAT

All BASE24-atm Triton Device Handler customers:

- Inline changes to AT60TRIT.TRITDDL, TRITOND, TRITONG, TRITNAMS, TRITONM, TRITONS
- Updated event (log) message manual BA60LOGM.ATTRIT
- Changes to AT60TRIT.TRITMAN

All BASE24-atm Triton Device Handler customers requiring the new AKDS functionality:

- A new extension subvolume, AT60ATRT, with all files required to support the Triton AKDS functionality.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNTRAK.

CUSTOMACS Changes

The following Make flag has been added to the CUSTOMACS file.

Set the following flag to true to bind the Triton AKDS module to the Triton Device Handler.

atm_dh_triton_akd_on flag

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to support this enhancement. Note that the first of these params already exists for the NCR 5070 Device Handler, and will also be used by the Triton AKDS module. This param, therefore, already exists in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-AKDS-TSS-TIMER

This param is used by the Triton Device Handler. It specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

The following param is new for the Triton Device Handler. It has been added to the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-TRIT-AKDS-SERIAL-NUM-VRFY

This param is used by the Triton Device Handler. It specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

Terminal Definition File (TDF)

The field "Hardware" on screen 1 of the ATD previously only displayed data from the ATD, not the TDF. It now will display the data from the Hdwr field that has been added to the TDF. During AKDS processing, the Triton Device Handler will set this value to "6".

Previously, screen 22 was only displayed for ATD records. Now it is also displayed for Triton TDF records. Both the EPP Format and key-signer fields will display data from the fields that have now been added to the TDF.

The Key Signer field must be set to an appropriate value for Triton AKDS. This field is used to indicate the name to be used in the HOST field of the TSS Host Public Key Security Datasource (HSPKD). If multiple Host Public Keys are required for an individual vendor's ATMs, the Key Signer field can be used to assign a separate HOST name for each of those Public Keys.

This field allows up to 16 alphanumeric characters of free-format text to be entered. Embedded spaces are allowed. The default value is "HOST". The customer needs to ensure that the value entered in the Key Signer field matches the value used in the Host Name field in the KEYGEN Command. See the *BASE24 Transaction Security Services Processing Guide* for more information on the registration and certification process.

The EPP Format field is used to indicate whether the EPP, and therefore the ATM, supports Signature- or Certificate-based AKDS processing. During AKDS processing, the Triton Device Handler will set this value to indicate that certificate-based AKDS is being performed.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS. The only EPPs that currently support RKT are Sagem branded EPPs with firmware versions "414-0391 R1A2".

Terminal Software

The Triton ATM must be running Triton's Software, minimum version 2.3.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the TDF screen 22 must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services CSM for Triton AKDS on Version 07.4 must be installed prior to implementation of this enhancement. This CSM code is currently scheduled to be included in product TSS Version 09.2.

Device Definition

If the ATM is to be connected via a dialup connection, using the VISA protocol, changes will be necessary to the device definition in XPNET. If a TCP/IP connection is used, no changes are necessary. Changes for the VISA dialup protocol are noted here. The device defined in XPNET for the ATM's station, must be set up to use the ACISTD variant of the VISA protocol. This ensures that the connection to the ATM is not dropped after each step of the AKDS message sequence.

An example of such a definition follows:

> info device d74r^dial, detail

```
DEVICE:  \K9.P74R^NODE.D74R^DIAL
TYPE      255      ACLength      2
RECVLEN   2000     XMITLEN      2000
APPLID
PROFILE
TRADDRESS OFF      TTRANSPARENT OFF
DEFRSP    OFF      TRLEN        0
SELECTION NORMAL    NRSSCP       OFF
INPUTFORMAT
OUTPUTFORMAT
POLLFORMAT
SELECTFORMAT
DESTRROUTE
XNCONF
RPCONF
DEVICE:  \K9.P74R^NODE.D74R^DIAL
WILDCHAR  "?"      NUM DEST ENTRIES:  1

ENTRY      DESTINATION  OFFSET  MATCH CHARS
1          P74R^TRITON    4       "T1"
DEVICE:  \K9.P74R^NODE.D74R^DIAL
WILDCHAR  "?"      NUM PROT ENTRIES:  1

ENTRY      PROTOCOL    OFFSET  MATCH CHARS
1          ACISTD      0       "?"
```

File Conversions

BASE24-atm Terminal Definition File (TDF)

For users wishing to take the Triton AKDS enhancement, either the TDF conversion program must be run to convert the TDF, or the customer may modify the fields manually via PATHWAY for affected ATMS.

Details of the conversion program can be found in the General Changes earlier in Section 5.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-atm users that use OMF Extract to accept and process information from the TDF record.

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, the Enable PKI field on the Security Device Configuration window of the ACI Desktop must be checked. If this field is not checked for a security device, it is not used for PKI functions. This enables you to consolidate all PKI functions in just one or a few devices.

Initial Registration and Certification for the BASE24 System

Initial setup procedures must be carried out for Transaction Security Services before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain a certificate containing the EPP's public sign and verify key and the EPP's serial number, a certificate containing the EPP's public encrypt and decrypt key and the EPP's serial number, and a certificate containing the Certificate Authority's public sign and verify key and the Certificate Authority's unique identifier. It also must contain the EPP's private sign and verify key along with its private encrypt and decrypt key. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security File (HSPKD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair.

The public key must be sent to the Certificate Authority (Verisign) to be verified and signed. The Certificate Authority must return a certificate containing BASE24's public key and BASE24's unique identifier, as well as a certificate containing the Certificate Authority's own public key and its unique identifier. These values must be entered into the Transaction Security Services database.

Details of the registration and certification process for the BASE24 system can be found in the *BASE24 Transaction Security Services Processing Guide*.

Triton Device Handler – NCS/NCPCOM Commands

The following 9500 command is supported in the Triton Device Handler. It is entered at the BASE24 AFT NCS screen or through NCPCOM.

- **LOADALLKEYS**

This command potentially allows the authentication and public key exchange, the download of the Master Key, and a load of the working MAC and/or PIN key to be processed in a single command.

The first time that the Load All Keys command is issued, the Triton ATM will allow the authentication and public key exchange to proceed, followed by the download of the PIN Master key, and then the download of the MAC Master Key if MACing is supported. The load of the new MAC and/or PIN working key will be accomplished as the last step of the appropriate Master key download. Once the authentication and public key exchange has been successfully completed, further Load All Keys commands will allow just the download of the new PIN Master Key and/or the MAC Master Key, with the load of the new MAC and/or PIN working key accomplished as the last step of the appropriate Master key download. If another authentication and public key exchange is required, the customer must contact Triton support and perform a challenge and response sequence at the ATM before that step in the process is allowed to be performed again.

Once the device handler receives the 9500LOADALLKEYS message, it will set a flag in the TDF record indicating that an AKDS load is required. Nothing more happens until the ATM performs a financial transaction or a Host Totals admin transaction, at which time the device handler adds a new field to the response message to the ATM, indicating that an AKDS load must be initiated. It is then the ATM that will initiate the AKDS sequence. The device handler, EPP and Transaction Security Service (TSS) exchange a series of messages to authenticate each other and exchange the public keys. These public keys are used for encryption and mutual authentication in all future master key downloads. Then a series of messages follows in order to generate the Master Key and download it to the EPP.

The load of the new MAC and/or PIN working key will be accomplished as the last steps of the Master key download.

This command can be performed as often as is necessary to preserve the integrity of the ATM's keys.

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LOADALLKEYS <terminal ID>"

<process name>: The symbolic name assigned to the Triton Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the Terminal Data file (TDF).

Supported Keys

Only the PIN master key and/or the MAC Master key can be downloaded to an ATM using the AKDS enhancement. AKDS does not support the downloading of the working PIN and/or MAC keys. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Triton AKDS enhancement is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the BA60LOGM.ATTRIT event message documentation file.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Triton AKDS.

- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases. All of the public key cryptography commands are defaulted to off in the security policy delivered from the factory due to security exposure. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application); nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in your security policy using the 108 and 109 commands:

- Generate RSA key pair (Command 120)
 - Generate AKB for root public key (Command 122)
 - Verify public key and generate AKB of public key (Command 123)
 - Generate digital signature (Command 124)
 - Verify digital signature (Command 125)
 - Generate message digest (Command 126)
 - Generate ATM master key and encrypt with public key (Command 12F)
- Thales e-Security version 6.02 firmware and the optional DSP plug-in (for RSA functions) provides public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The commands are as follows:
 - Generate RSA key set (Command EI)
 - Load a secret key (Command EK)
 - Generate a MAC on a public key (Command EO)
 - Validate a certificate and generate MAC on its public key (Command ES)
 - Generate a signature (Command EW)
 - Validate a signature (Command EY)
 - Export a DES key (Command GK)
 - Hash a Block of Data (GM)

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS. An EPP is a tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, and the cryptographic keys. All of the ATM's cryptographic functions are performed by the EPP. The only EPPs that currently support RKT are Sagem branded EPPs with firmware versions "414-0391 R1A2".

Reference Documents

Atalla: *Atalla Key Block Banking Command Reference Manual* (524130-002, July 2002)

Thales: *Host Security Module RG7000 Programmer's Manual* (1270A514 Issue 4, November 2001)

Triton: *Comms Protocol and Message Format Spec* version 5.39 (Feb 5, 2008)

Triton: *TSRK Remote Key Extensions* version 1.1 (Jan 22, 2008)

Triton: *Remote Key Transport Host Developers Guide* version 1.6 (Jun 26, 2008)

Triton: *Remote Key Transport Technical Overview* version 0.2 (May 24, 2007)

Triton: *T5 RKT EPP Error Codes* v 0.2 (undated)

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6: BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from Release 6.0 Version 8.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to Release 6.0 Version 9.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to Release 6.0 Version 9.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Add Support for AMEX format is EMV IAD
- DUKPT Support with Message Data Encryption
- Hypercom Enhancements
- Institution Level Security with OLT for CRT Auth
- PCI – CVD2 Suppression in CRT Auth
- PCI – Track 2 Context Data Removal for APACS
- PCI – Track 2 Context Data Removal for HPDH
- PCI – Track 2 Context Data Removal for Visa II
- POS Limit Checking on Responses
- SPDH Support Additional EMV Data Elements
- SPDH Support for Network Mandates
- Support Additional Flexibility in EMV Processing
- Visa Contactless MSD with CVN 17
- Visa II Contactless Support

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes for all BASE24 Release 6.0 Version 9 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The POS Terminal Data file (PTD) has been modified to include an additional field on screen 19 for the Visa Contactless MSD with CVN 17 enhancement. The POS Terminal Data file screen, requester and server have been modified to process this new field. For further information, see Visa Contactless MSD with CVN 17, where the impacts of this enhancement are described later in Section 6.
- The POS Terminal Data file (PTD) has been modified in device-dependent data. This data is not displayed but the PTD conversion program must be run to correctly initialize the new fields. For further information, see PCI – Track 2 Context Data Removal for HPDH where the impacts of this enhancement are described later in Section 6.
- The POS Terminal Data File (PTDF) has been modified in device-dependent data. This data is not displayed but the PTDF conversion program must be run to correctly initialize the new fields. For further information, see PCI – Track 2 Context Data Removal for Visa II where the impacts of this enhancement are described later in Section 6.
- The POS Retailer Definition File (PRDF) has been modified to include an additional field on screen 4 for the Support Additional Flexibility in EMV Processing enhancement. The POS Retailer Definition File screen, requester and server have been modified to process these new fields. For further information, see Support Additional Flexibility in EMV Processing, where the impacts of this enhancement are described later in Section 6.

- Consolidate Variable in SPDH/RTAU Procedure. This enhancement modifies the SPDH to consolidate a number of variables into a single structure, to reduce the number of parameters in the argument list of procedure DCBCA^GET^REQUEST^FLDS. The effect is to reduce the total number of variables passed into the procedure, which provides capacity to define parameters required for CSMs. Equivalent changes have been applied to procedures DCBC^PROCESS^TRAN^REQ, DCBD^PROCESS^ADMIN^REQ and UTL_PASSTHRU_RVRS_CONCURRENT, all of which invoke procedure DCBCA^GET^REQUEST^FLDS. A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNSPCV.

Software Impacts

All BASE24-pos ACI Standard POS Device Handler customers:

- Inline changes to PS60SPDH.ASPDHS.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

POS Terminal Data files (PTD)

The PTD conversion program must be run to convert the PTD. Running the conversion program is required for any BASE24-pos customer utilizing the PTD in transaction processing. Users must create the new PTD file using the PSDDLUP file.

The PTD conversion program on the PS60UC04 subvolume can be used to convert the PTD records from the release 5.3 format to the Release 6.0 Version 9 format. The PTD conversion program on the PS60UC0C subvolume can be used to convert the PTD records from the Release 6.0 Version 8 format to the Release 6.0 Version 9 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPTDS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ptdcnv.t,name $ptdm,nowait/ cnvptdm –  
force –list –listreplace
```

Release 5.3 users should update the CNVPTDR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 Version 8 users should update the CNVPTDR obey file on the PS60UC0C subvolume with the appropriate information. The CNVPTDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0C on the BA60UC0C subvolume for Release 6.0 Version 8 customers.

POS Terminal Data File (PTDF)

The PTDF conversion program must be run to convert the PTDF. Running the conversion program is required for any BASE24-pos customer utilizing the PTDF in transaction processing. Users must create the new PTDF file using the PSDDLFP file.

The PTD conversion program on the PS60UC04 subvolume can be used to convert the PTDF records from the release 5.3 format to the Release 6.0 Version 9 format. The PTDF conversion program on the PS60UC0C subvolume can be used to convert the PTDF records from the Release 6.0 Version 8 format to the Release 6.0 Version 9 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPTDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ptdfcv.t,name $ptdfm,nowait/ cnvptdfm –  
force –list –listreplace
```

Release 5.3 users should update the CNVPTDR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 Version 8 users should update the CNVPTDFR obey file on the PS60UC0C subvolume with the appropriate information. The CNVPTDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0C on the BA60UC0C subvolume for Release 6.0 Version 8 customers.

POS Retailer Definition File (PRDF)

The PRDF conversion program must be run to convert the PRDF. Running the conversion program is required for any BASE24-pos customer utilizing the PRDF in transaction processing. Users must create the new PRDF file using the PSDDLUP file.

The PRDF conversion program on the PS60UC04 subvolume can be used to convert the PRDF records from the release 5.3 format to the Release 6.0 Version 9 format. The PRDF conversion program on the PS60UC0C subvolume can be used to convert the PRDF records from the Release 6.0 Version 8 format to the Release 6.0 Version 9 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPRDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#prdfcv.t,name $prdfm,nowait/ cnvprdfm –  
force –list –listreplace
```

Release 5.3 users should update the CNVPRDFR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 Version 8 users should update the CNVPRDFR obey file on the PS60UC0C subvolume with the appropriate information. The CNVPRDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0C on the BA60UC0C subvolume for Release 6.0 Version 8 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc

Add Support for AMEX format in EMV IAD

This section describes the implementation requirements associated with the Add Support for AMEX format in EMV IAD enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Add Support for AMEX format in EMV IAD enhancement.

This enhancement does not support AMEX offline PIN Change scripts. A TSS enhancement is needed to support that functionality.

Software Impacts

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24 EMV customers:

- Inline changes to BA60IEMV.CAMUTILS and EMVG.

All BASE24-pos EMV customers:

- Inline changes to PS60IATH.RTAUEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNEIAD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

A new value has been added to CPF screen 11 EMV IAD FORMAT and ALTERNATE IAD FORMAT. The value of 5 will represent the AMEX IAD format.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc

DUKPT Support with Message Data Encryption

This section describes the implementation requirements associated with the DUKPT Support with Message Data Encryption enhancement for SPDH.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the DUKPT Support with Message Data Encryption enhancement.

The ACI Standard POS Device Handler (SPDH) has been enhanced for Derived Unique Key Per Transaction (DUKPT) Support with Message Data Encryption.

This enhancement adds support for DUKPT as part of the full and configurable message data encryption functionality that was added to the ACI Standard POS Device Handler (SPDH) in BASE24-pos Release 6.0 Version 8.

The DUKPT Support with Message Data Encryption enhancement enables customers to protect data fields by employing either full message encryption or configurable message encryption. Message data is encrypted under the derived key, which supports double-length keys.

Using the DUKPT method, each terminal security module (TSM), or PIN pad, uses a derived unique key for each transaction. However, the TSM never contains any information that could lead to the determination of any past transaction keys used by the device, or of any past or future keys used by any other TSM. The originating TSM must derive the current transaction key from an initial key, loaded into the TSM on initialization. The receiving BASE24-pos security module (Atalla or Thales) then determines the current transaction key by using a key held on BASE24-pos, and non-secret information contained in the transaction message.

The POS Terminal Data File (PTD) Data Encryption Type field supports two new values of 05 (full message encryption - DUKPT) and 06 (configurable message encryption – DUKPT). The Derivation Key File (KEYD) stores the key locator in the derivation key field indicating the record in the Derivation Key Data Source file (DRVKD) used by TSS containing the derivation keys. The BASE24 AFT facility is used to maintain the data in the KEYD file and the ACI Desktop is used to maintain the data in the DRVKD file.

DUKPT full message encryption allows the customer to encrypt all optional data fields in an ACI Standard POS Message, with the exception of those listed in Table 1. DUKPT full message encryption is enabled by setting the Data Encryption Type field in the PTD to “05” (full message – DUKPT encryption).

FID	Description	Reason
G	Authentication Code	Created using FID H (Authentication Key)
H	Authentication Key	Already encrypted under the MAC terminal master key
I	Data Encryption Key	Cannot be used to encrypt itself.
M	PIN Communications Key	Already encrypted under the PIN terminal master key
b	PIN/Customer	Already encrypted under FID M (PIN Communications Key)
c	PIN/Supervisor	Already encrypted under FID M (PIN Communications Key)

Table 1 – Optional data fields that are never encrypted under the Data Encryption Key or the Derived Key

Configurable message encryption allows the customer to encrypt specific optional data fields in an ACI Standard POS Message, with the exception of those listed in Table 1. Configurable message encryption is enabled by setting the Data Encryption Type field in the PTD to “06” (configurable message – DUKPT encryption). The optional data fields to be encrypted are specified in the ACNF request and response field map records. If an optional data field is to be encrypted, an “E” is placed in the one-byte MSG field corresponding to the optional data field’s FID for the specific transaction type.

Although specific optional data fields can be encrypted, specific optional data subfields cannot. All subfields under FIDs 6, 7, 8 or 9 are encrypted, or none. The exception to this is FID 6, subfield T (Key Serial Number and Descriptor). FID 6, subfield T (Key Serial Number and Descriptor) must not be encrypted. The remaining subfields in FID 6 can be encrypted or not encrypted. For example, the first FID 6 would contain subFID T (KSN) and would reside in the unencrypted data area of the SPDH message. It could even contain other subFIDs that are unencrypted. The second FID 6 would contain other subFIDs and would reside in the encrypted data area of the SPDH message, after the encryption separator.

Whether using full message encryption or configurable message encryption, the format of the ACI Standard POS Message is the same. The message begins with the standard message header, which is never encrypted. Following the standard message header is the unencrypted data area, which contains all the optional data fields and subfields that are not encrypted under the data encryption key. Next comes the encryption separator (hexadecimal 1F), a separator character that is used to mark the beginning of the encrypted data area. The encrypted data area extends from the encryption separator to the end of the message, and contains all the optional data fields and subfields that are encrypted under the data encryption key. Field separator characters (hexadecimal 1C) and field identifiers (FIDs), which preface optional data fields, are included in the encrypted data. Likewise, record separator characters (hexadecimal 1E) and subfield identifiers (subFIDs), which preface optional data subfields, are included. The encryption separator itself is not encrypted, nor is the end-of-text character at the end of the message, if present.

Figure 1 illustrates the format of an encrypted ACI Standard POS Message.

Standard Message Header	Unencrypted Data Area	E S	Encrypted Data Area	E T X
-------------------------	-----------------------	--------	---------------------	-------------

Figure 1 – Format of an encrypted ACI Standard POS Message

By grouping optional data fields and subfields into the encrypted data area, the SPDH can encrypt or decrypt the data with a single call to the host security module (HSM).

A long-standing requirement of the ACI Standard POS Message is that all data must be in ASCII format, not binary. Furthermore, some HSMs cannot process binary data reliably if it contains certain special characters. For instance, the Atalla NSP cannot encrypt/decrypt binary data containing the special characters "<", "#", "^", ">", or CR. For these reasons, encrypted data in the ACI Standard POS Message is required to be in ASCII hexadecimal format when DUKPT full message encryption or DUKPT configurable message encryption is performed.

In Release 6.0 Version 3, available balance encryption was added to the SPDH as part of the POS Enhancements. The SPDH Message Data Element Encryption enhancement is integrated in a way that retains the original functionality of available balance encryption. If available balance encryption is enabled, the Available Balance field (FID J) is encrypted individually, as it was before. If full or configurable message data encryption is enabled, the Available Balance field is grouped with other optional data fields in the encrypted data area of the ACI Standard POS Message, rather than encrypted individually.

The Data Encryption Type field in the PTD takes on the new and redefined values shown in Table 2.

Value	New Definition	Previous Definition
00	No encryption	No encryption
01	Available balance encryption – ASCII	ASCII encryption
02	Available balance encryption – binary	Binary encryption
03	Full message encryption	Full message encryption
04	Configurable message encryption	Configurable message encryption
05	Full message encryption – DUKPT	
06	Configurable message encryption - DUKPT	

Table 2 – PTD Data Encryption Type

The configuration of a TSM can be converted to DUKPT encryption by adding the corresponding KEYD record, setting the PTD KEYD-GRP field, setting the LCONF param POS-DH-DUKPT-UPDATE-METHOD to a “Y” and changing out the TSM. If the KSN is present in the message (FID 6 sub-FID T), the PTD DATA-ENCRYPT-TYP field does not equal 05 (full message encryption - DUKPT) or 06 (configurable message encryption) and the LCONF param POS-DH-DUKPT-UPDATE-METHOD does not equal N, the device handler automatically updates the PTD DATA-ENCRYPT-TYP field and processes the transaction. If the DATA-ENCRYPT-TYP was 04 (configurable message encryption) then DATA-ENCRYPT-TYP is updated to 06 (configurable message encryption - DUKPT). Otherwise, if the DATA-ENCRYPT-TYP was 00 (no encryption) or 03 (full message encryption), the DATA-ENCRYPT-TYP is updated to 05 (full message encryption – DUKPT) as the default.

Message encryption is allowed for SPDH transactions of any message type, such as financial transactions (message type = F) and administrative transactions (message type = A). Message encryption is also allowed for transactions of any message subtype, such as online transactions (message subtype = O) and store-and-forward transactions (message subtype = S).

Message encryption is allowed for all SPDH transaction types except one: The download transaction message (transaction code 90) cannot be encrypted.

The SPDH Dynamic Key Management extension module was modified to exclude the PTD encrypt types for DUKPT (05 and) when determining whether new data keys should be generated.

Prerequisites: BASE24-pos Standard POS Device Handler (PS-DP140) and POS Data Encryption (PS-AE520) must be licensed.

SCN PS0C041 to PS60DSPD.SPDHDES must be installed. This corrects an issue found while testing this enhancement.

The Transaction Security Services (TSS) CSM for DUKPT Message Data Encryption on Version 07.4 must be installed prior to implementation of this enhancement.

An Atalla AKB HSM with updated firmware that supports command 388 (3DES UKPT Data Encrypt/Decrypt with OWF) is required for this enhancement. Thales does not yet support this functionality.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLPCNST
- Inline changes to BA60DDL.DDLGTSS
- Inline changes to BA60SRC.SECUTILS

All BASE24-pos customers:

- Inline changes to PS60AFT.SVPTDTBL and SVPTDTS
- Inline changes to PS60DDL.DDLFPTD

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHG and ASPDHS
- Updated event (log) message manual BA60LOGM.PSSPDH

All BASE24-pos ACI Standard POS Device Handler (SPDH) Data Encryption customers:

- Inline changes to PS60DSPD.SPDHDES

All BASE24-pos ACI Standard POS Device Handler (SPDH) Dynamic Key Management customers:

- Inline changes to PS60KSPD.SPDHDKMS

CUSTMACS Changes

Set the following Make flag to true to bind the Data Encryption module to the SPDH.

- pos_dh_spdh_de_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-pos Terminal Data file (PTD)

Enter a value in the Derivation Key Group field in the PTD. The Derivation Key File (KEYD) stores the key locator in the derivation key field indicating the record in the Derivation Key Data Source file (DRVKD) used by TSS containing the derivation keys. The BASE24 AFT facility is used to maintain the data in the KEYD file and the ACI Desktop is used to maintain the data in the DRVKD file. See the BASE24 Transaction Security Services Processing Guide for more information about entering the Derivation Key.

Depending on the type of data encryption supported by the device, the Data Encryption Type field in the PTD must be set to one of the following values.

- 00 = No encryption
- 01 = Available balance encryption – ASCII
- 02 = Available balance encryption – binary
- 03 = Full message encryption
- 04 = Configurable message encryption
- 05 = Full message encryption – DUKPT
- 06 = Configurable message encryption - DUKPT

ACI Standard Device Configuration File (ACNF)

No data modifications are needed in the ACNF if DUKPT full message encryption is used. If DUKPT configurable message encryption is used, data in the ACNF should be modified as follows.

For each optional data field that is to be encrypted in a device request or response message, place an “E” in the one-byte MSG field corresponding to the optional data field’s ID (FID) for the specific transaction type. The value “E” means the optional data field will be included in a message and it will be encrypted under the data encryption key. Following is an example.

[illegible]

In this example, FID B (Amount 1) is configured to be included, but not encrypted, in device request messages for normal purchase, pre-authorization, pre-authorization completion, mail/phone order, and merchandise return transactions (MSG field = "Y"). FID q (Track 2/Customer) is configured to be included and will also be encrypted under the data encryption key (MSG field = "E"). The remaining FIDs are not included in the request messages.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The SPDH compatible device must be configured for DUKPT support with message data encryption.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) CSM for DUKPT Message Data Encryption on Version 07.4 must be installed in order to use this enhancement.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files in BA60LOGM.PSSPHD.

Performance Impacts

DUKPT message data encryption may impact device performance due to the extra processing required to encrypt data in outgoing messages and to decrypt data in incoming messages.

DUKPT message data encryption may impact BASE24 system performance due to extra I/O with the HSM. Each message from the POS device that contains encrypted data will require one extra I/O with the HSM to decrypt the data. Each message to the POS device that contains encrypted data will require one extra I/O with the HSM to encrypt the data.

Vendor Impacts.

POS device software must be modified to encrypt and decrypt data in the ACI Standard POS Message using a derived unique key per transaction.

A separator character has been defined to delimit the encrypted portion of an ACI Standard POS Message:

- Encryption Separator – Hexadecimal 1F (Octal 37)

The encryption separator marks the beginning of encrypted data in the message. All data between the encryption separator and the end of the message (not including the end-of-text character) is considered to be encrypted under the data encryption key. The following rules must be observed when encrypting data in the ACI Standard POS Message:

- The standard message header must not be encrypted. Only optional data fields and subfields may be encrypted.
- All unencrypted optional data fields and subfields must immediately follow the standard message header.
- If there is encrypted data, the encryption separator must immediately follow the last unencrypted optional data field or subfield.
- All encrypted optional data fields and subfields must immediately follow the encryption separator.
- Field separator characters (hexadecimal 1C) and field identifiers (FIDs), which preface optional data fields, are included in the encrypted data.
- Record separator characters (hexadecimal 1E) and subfield identifiers (subFIDs), which preface optional data subfields, are included in the encrypted data.
- If an end-of-text (ETX) character is appended to the message, it must not be encrypted.

The following examples illustrate the placement of the encryption separator and encrypted data. Bold text represents data encrypted under the derived key. Plain text represents unencrypted data. The symbols used in the examples are defined below.

[std msg hdr]	=	standard message header
[es]	=	encryption separator character (hexadecimal 1F)
[fs]	=	field separator character (hexadecimal 1C)
[fid]	=	field identifier (A-Z, a-z, 0-9)
[field]	=	optional data field associated with the FID
[rs]	=	record separator character (hexadecimal 1E)
[sfid]	=	subfield identifier (A-Z, a-z, 0-9)
[subfield]	=	optional data subfield associated with the SFID
[etx]	=	End-of-text character (hexadecimal 03)

The following message is unencrypted.

[std msg hdr][fs][fid][field][fs][fid][field][etx]

All optional data fields are encrypted in the following message.

```
[std msg hdr][fs][fid 6][rs][sfid T][subfield][es][fs][fid][field]  
[fs][fid][field][etx]
```

All optional data fields are encrypted in the following message. There is no ETX.

```
[std msg hdr][fs][fid 6][rs][sfid T][subfield][es][fs][fid][field]  
[fs][fid][field]
```

All optional data fields and subfields are encrypted in the following message except for FID6, subFID T (Key Serial Number and Descriptor).

```
[std msg hdr][fs][fid 6][rs][sfid T][subfield][es][fs][fid][field]  
[fs][fid 6][rs][sfid][subfield][etx]
```

One optional data field is unencrypted in the following message. The rest are encrypted.

```
[std msg hdr][fs][fid][field][fs][fid 6][rs][sfid T][subfield][es]  
[fs][fid][field][fs][fid][field][etx]
```

Reference Documents

Atalla's *Command 388 for ACI 20080326*, dated March 26, 2008.

Hypercom Enhancements

This section describes the implementation requirements associated with the Hypercom Enhancements for the HPDH.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Hypercom Enhancements.

The Hypercom Device Handler (HPDH) has been enhanced to support the following:

- Populate the Point of Service Data token (token ID C4) with EMV terminal capability information stored in the POS Terminal Data File (PTD).
- Process the terminal Issuer Script Results data provided by the terminal from Private Use Field 63 in a 0400 reversal message and populate the EMV Issuer Script Results token (token ID BJ).

Software Impacts

All BASE24-pos Hypercom Device Handler (HPDH) customers:

- Inline changes to PS60HPDH.HPDHDDL
- Inline changes to PS60HPDH.HPDHG and HPDHS
- Inline changes to PS60HPDH.HPDHEMVD

All BASE24-pos Hypercom Device Handler (HPDH) EMV customers:

- Inline changes to PS60IHPD.HPDHEMVG and HPDHEMVS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

This enhancement uses the following tokens:

- Point of Service Data token (token ID C4)
- EMV Issuer Script Results token (token ID BJ)

If any of these tokens are to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If any of these tokens are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

This enhancement uses the following tokens:

- Point of Service Data token (token ID C4)
- EMV Issuer Script Results token (token ID BJ)

If any of these tokens are to be included in PTLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-pos users must consider this additional data in host programs that process this data.

User/Operational Impacts

Disk Impacts

The PTLF, and ILF will require additional disk space if any of the tokens used by this enhancement are included in the transaction records logged to these files.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the *BASE24 Tokens Manual* provides information on assessing the impacts of tokens on log file disk space requirements.

BASE24-pos
General Enhancements
Hypercom Enhancements

Vendor Impacts

Not applicable

Reference Documents

Hypercom's *Message Specification*, version 3.44, May 2, 2005

Institution Level Security with OLT for CRT Auth

This section describes the implementation requirements associated with the Institution Level Security with OLT for CRT Auth.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Institution Level Security with OLT for CRT Auth enhancement.

The BASE24 CRT Authorization sub-system has been enhanced to support the following functions:

- Ensure that access to the Online Transaction Screen (OLT) can be restricted by retailer financial institution identifier (FIID)
- Allow institution level security to the Multiple Referrals File (MRF) and POS Referrals File (PRF) to be configurable
- Ensure that reversals generated from the OLT screen update the card's offline accumulators, if the transaction has been authorized in stand-in by BASE24-pos.

In BASE24 Release 6.0 Version 7, the Institution Level Security enhancement was implemented to control which users had access to MRF and PRF screens, but not the OLT screen. The OLT Requester has been enhanced to include similar functionality as that added to the MRF Requester and PRF Requester in Release 6.0 Version 7. The BASE24 user will be allowed to enter transactions only for those retailers where the FIID defined in the POS Retailer Definition File (PRDF) matches one of the allowed FIIDs specified for the user on Security File (SEC) screen 2. When transaction details are entered on the OLT screen, a message is sent to the PRDF Server, using the MERCHANT value entered on the screen. Existing logic in the PRDF Server reads the PRDF and returns the appropriate record to the OLT Requester. A message is then sent to the SEC Server, using the FIID from the PRDF record. Existing logic in the SEC Server reads the SEC and determines whether the operator is allowed access for the institution. If a PRDF record is not read successfully, or institution access is not permitted, then the operator is not allowed to complete the transaction through the OLT screen.

The logic added to the MRF Requester and PRF Requester in Release 6.0 Version 7 has also been enhanced. The processing has been made configurable (via a new flag in COBNAMES), in case of performance concerns over the additional disk I/Os. Also, the institution security check has been inhibited if a PRDF record cannot be found, in order to allow operator access to referrals at not-on-us merchants. This

modification required a small change to the PRDF Server to ensure that the correct error code is returned when a PRDF record cannot be found. Note that the new logic added to the OLT Requester also includes the checking of the new flag in COBNAMES. However, the new logic added to the OLT Requester does NOT allow the institution security check to be inhibited if a PRDF record cannot be found, as OLT transactions can be performed for on-us retailers only.

This enhancement also resolves an issue caused by the failure of the CRT Authorization subsystem to store the COMPL-REQ field from the PSTM when a transaction is entered on the OLT screen. If the transaction is authorized in stand-in by BASE24 (because the request message could not be sent to the authorizing Host system), the Host Interface sets the COMPL-REQ field to a value of '2'. If the BASE24-pos Router/Authorization process approves the transaction in stand-in, this value ensures that the offline accumulators in the appropriate CAF or UAF record is updated. Because the COMPL-REQ field is not stored when the response message is returned to the CRT Authorization Device Handler, a subsequent PSTM reversal will contain a value of '0' in this field. This results in the offline accumulators in the appropriate CAF or UAF record not being updated when the reversal message is processed. The solution is to ensure the COMPL-REQ field in the response message is present in the record written to the Terminal Queue File (TQF), so that the value is available for inclusion in a subsequent reversal message.

Software Impacts

All BASE24 customers:

- Inline changes to BA60SRC.COBNAMEs

All BASE24-pos customers:

- Inline changes to PS60AFT.SVPRDFS

All BASE24-pos CRT Authorization customers:

- Inline changes to BP60CRTA.CRTAUDHS, DDLGCATH, RQMRFS, RQOLTS and RQPRFS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNCRTA.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

A new flag called INST-SECURITY-CHECK-FLG has been added to COBNAMES, and has been assigned a default value of "1" (perform institution security check). The flag indicates whether the institution security check should be performed when an operator attempts to process transactions through the OLT, MRF and PRF screens.

The value of the INST-SECURITY-CHECK-FLG in COBNAMES must be changed to "0" (do not perform institution security check) to inhibit the institution security check.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Screen Impacts

If the INST-SECURITY-CHECK-FLG in COBNAMES is set to "1" (perform institution security check), then the operator may find that access to the OLT screen is no longer permitted for certain retailers. This is the default value.

If the INST-SECURITY-CHECK-FLG in COBNAMES is set to "0" (do not perform institution security check), then the operator may find that access to the MRF and PRF screens is no longer inhibited for certain retailers.

Performance Impacts

The institution security check will have an impact on system performance. However, the security check is performed by the AFT sub-system, so there will be no direct impact on transaction processing.

The institution security check requires two additional disk reads (PRDF and SEC) whenever an operator attempts to process transactions through the OLT, MRF and PRF screens.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

PCI – CVD2 Suppression in CRT Auth

This section describes the implementation requirements associated with the PCI – CVD2 Suppression in CRT Auth enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI – CVD2 Suppression in CRT Auth enhancement.

The BASE24 CRT Authorization sub-system has been enhanced to suppress the storage of electronic card verification digits (within the Track 2 discretionary data), and the display of the manual card verification digits (CVD2). The following changes have been implemented:

- Authorization processing has been modified to ensure that the Track 2 discretionary data and CVD2 data is never included in a record written to the POS Referral File (PRF);
- Online Transaction (OLT) screen processing will be modified to ensure that CVD2 data is not logged to the Terminal Queue File (TQF), and is not included in the PSTM unless the message type is a request.

The effect of these changes is to ensure that CVD2 will not be viewable via the BASE24 CRT Authorization sub-system once an authorization request has been submitted. Also, Track 2 discretionary data will no longer be present in PRF records.

The suppression of the Track 2 discretionary in the PRF and the CVD2 data in the PRF and TQF ensures that the BASE24 CRT Authorization sub-system is compliant with Payment Card Industry: Data Security Standard (PCI:DSS).

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS

All BASE24-pos CRT Authorization customers:

- Inline changes to BP60CRTA.CRTAUDHS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNPCIC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Note Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Track 2 discretionary data included in a PRF record will be suppressed unconditionally. This means that a subsequent referral override will contain suppressed discretionary data, which may affect the data sent to Hosts in advice messages.

User/Operational Impacts

CVD2 data included in a PRF record and a TQF record will be suppressed unconditionally. The CVD2 data serves no purpose once a transaction has been authorized. This means that an operator will no longer be able to see the CVD2 data on the OLT screen once an authorization request has been submitted or on the PRF screen.

Vendor Impacts

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1.

ACI Worldwide Inc

PCI – Track 2 Context Data Removal for APACS

This section describes the implementation requirements associated with the PCI – Track 2 Context Data Removal for APACS enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI – Track 2 Context Data Removal for APACS enhancement.

The BASE24-pos APACS Device Handler has been enhanced to suppress the storage of Track 2 discretionary data and manual card verification digits (CVD2) in the device-dependent data area of the BASE24-pos Terminal Data File (PTDF). Unlike some other BASE24 modules, suppression is not configurable in the LCONF. The APACS Device Handler suppresses this data unconditionally. At the conclusion of a transaction, the APACS Device Handler retains the primary account number (PAN) and expiration date of Track 2 in the device-dependent data area of the PTDF, while replacing the rest of the Track 2 data with ASCII zeroes. Manual card verification digits are suppressed by replacing the CVD field in the BASE24-pos Release 5.1 Token with blanks before the token is stored in device-dependent data.

The suppression of the Track 2 discretionary data and manual card verification digits in the PTDF ensures that the APACS Device Handler is compliant with Payment Card Industry: Data Security Standard (PCI:DSS).

Software Impacts

All BASE24-pos APACS Device Handler customers:

- Inline changes to PS60APDH.APACSDHS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNPCIA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Note Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host systems may no longer receive full track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternatively, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35. Host systems must be capable of processing response, reversal, and advice messages without the discretionary data or by using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

ACI Worldwide Inc

PCI – Track 2 Context Data Removal for HPDH

This section describes the implementation requirements associated with the PCI – Track 2 Context Data Removal for HPDH enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI – Track 2 Context Data Removal for HPDH enhancement.

The BASE24-pos Hypercom Device Handler (HPDH) has been enhanced to suppress the storage of Track 2 discretionary data and manual card verification digits (CVD2) in the device-dependent data area of the BASE24-pos Terminal Data files (PTD). Unlike some other BASE24 modules, suppression is not configurable in the LCONF. The Hypercom Device Handler suppresses this data unconditionally. At the conclusion of a transaction, the Hypercom Device Handler retains the primary account number (PAN) and expiration date of Track 2 in the device-dependent data area of the PTD, while replacing the rest of the Track 2 data with ASCII zeroes. The member number is also retained and is available for reversal and advice processing. Manual card verification digits are suppressed by replacing the CVD field in the BASE24-pos Release 5.1 Token with blanks before the token is stored in device-dependent data.

The suppression of the Track 2 discretionary data and manual card verification digits in the PTD ensures that the Hypercom Device Handler is compliant with Payment Card Industry: Data Security Standard (PCI:DSS).

Software Impacts

All BASE24-pos Hypercom Device Handler (HPDH) customers:

- Inline changes to PS60HPDH.HPDHDDL and HPDHS.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNPCIH.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Note Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The PTD conversion program must be run to convert the POS Terminal Data files (PTD). Details of this conversion program can be found in the General Changes earlier in Section 6.

Host/Co-Network Impacts

Host systems may no longer receive full track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternatively, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35. Host systems must be capable of processing response, reversal, and advice messages without the discretionary data or by using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

ACI Worldwide Inc

PCI – Track 2 Context Data Removal for Visa II

This section describes the implementation requirements associated with the PCI – Track 2 Context Data Removal for Visa II enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PCI – Track 2 Context Data Removal for Visa II enhancement.

The BASE24-pos Visa II Device Handler (VVDH) has been enhanced to suppress the storage of Track 1 discretionary data, Track 2 discretionary data and manual card verification digits (CVV2) in the device-dependent data area of the BASE24-pos Terminal Data File (PTDF). Unlike some other BASE24 modules, suppression is not configurable in the LCONF. The Visa II Device Handler suppresses this data unconditionally. At the conclusion of a transaction, the Visa II Device Handler retains the primary account number (PAN) and expiration date of Track 1 and Track 2 in the device-dependent data area of the PTDF, while replacing the rest of the Track 1 and Track 2 data with ASCII zeroes. The member number is also retained and is available for reversal and advice processing. Manual card verification digits are suppressed by replacing the CVV2 field in device-dependent data with blanks.

The suppression of the Track 1 discretionary data, Track 2 discretionary data and manual card verification digits in the PTDF ensures that the Visa II Device Handler is compliant with Payment Card Industry: Data Security Standard (PCI:DSS).

Software Impacts

All BASE24-pos Visa II Device Handler (VVDH) customers:

- Inline changes to PS60VVDH.VVDHDDL, VVDHG and VVDHS.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNPCIV.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Note Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The PTDF conversion program must be run to convert the POS Terminal Data File (PTDF). Details of this conversion program can be found in the General Changes earlier in Section 6.

Host/Co-Network Impacts

Host systems may no longer receive full track data in Data Element 35 (Track 2 Data). The discretionary data within the track may be masked with zeroes. Alternatively, host systems may receive Data Element 2 (PAN) and Data Element 14 (Expiration Date) instead of Data Element 35. Host systems must be capable of processing response, reversal, and advice messages without the discretionary data or by using Data Element 2 and Data Element 14 instead of Data Element 35.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Payment Card Industry (PCI) Security Standards Council – Payment Card Industry (PCI) Data Security Standard, Version 1.1

ACI Worldwide Inc

POS Limit Checking on Responses

This section describes the implementation requirements associated with the POS Limit Checking on Responses enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the POS Limit Checking on Responses Enhancement.

The BASE24-pos Router/Authorization module has been amended to help prevent a situation that can happen when two transactions with the same PAN happen at exactly the same time.

Without these changes, the full withdrawal daily limit can be approved for each transaction, causing the Issuer Institution to debit the cardholder's account for up to double the daily limit.

This enhancement allows the Issuer Institution to configure a new LCONF param, POS-RESP-LMTS-CHK. If set, this allows the BASE24-pos Router/Authorization module to recheck the daily limits on a response from the Host. This will deny the second of the transactions to be processed by the Router/Authorization module. An EMS event message is raised, which allows the Issuer Institution to investigate the transaction, and manually post a reversal to the cardholder's account if necessary. This scenario only affects customers using an Authorization level of online/offline, and an authorization method of CAF or CAF/PBF.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60AUTH.AUTHLIBS, ROUTERS and RTAUG.
- Updated event (log) message manuals BA60LOGM.PSAUTH and PSROUTER

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support these changes. Details of the new parameter are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-RESP-LMTS-CHK

This param, used by the BASE24-pos Router/Authorization module, specifies whether the BASE24-pos Router/Authorization module should recheck the daily limits on the processing of any response from a Host, if the Authorization level is online/offline, and the Authorization method is CAF or CAF/PBF.

- Y = Recheck the daily limits on the receipt of a response from the Host. If the limits have now been exceeded, deny the transaction back to the terminal.
- N = Do not recheck the daily limits on the receipt of a response from the Host. Processing continues as it did before this enhancement was introduced. The default value is "N".

Database Configuration

Not Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not Applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not Applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files in BA60LOGM.PSAUTH and PSROUTER.

Reversal processing

If simultaneous transactions are detected that cause the cardholder to exceed their daily limit, an event (log) message will be raised. This will allow the Issuer Institution to investigate the transaction, and manually post a reversal to the cardholder's account if necessary.

Vendor Impacts

Not Applicable

Reference Documents

Not Applicable

SPDH Additional EMV Data Elements

This section describes the implementation requirements associated with the SPDH Additional EMV Data Elements enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH Additional EMV Data Elements enhancement.

The SPDH has been enhanced to support the following:

- Populate the EMV Status Token (token ID B4) with EMV terminal capability information from the terminal, if sent in FID 6 SFID P. Otherwise, continue to populate this field with information from the POS Terminal Data File (PTD).
- Add the EMV Issuer Script Results token (token ID BJ) if Issuer Script Results data is provided by the terminal a reversal message in FID 6 SFID R.

Software Impacts

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24-pos SPDH customers:

- Inline changes to PS60SPDH.ASPDHS and SPDHEMVD.

All BASE24-pos SPDH EMV customers:

- Inline changes to PS60ISPD.SPDHEMVG and SPDHEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNEIAD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

This enhancement uses the following tokens:

- EMV Status Token (token ID B4)
- EMV Issuer Script Results token (token ID BJ)

If any of these tokens are to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If any of these tokens are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The SPDH compatible device must be able to support sending the new version of FID 6 SFID P in request messages and FID 6 SFID R in reversal messages.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

This enhancement used the following tokens:

- EMV Status Token (token ID B4)
- EMV Issuer Script Results token (token ID BJ)

The EMV Terminal Capabilities field will go into the existing EMV Status token. No changes are needed to support this change.

Reversal messages may now contain the EMV Issuer Script Results token. Host changes may be required to access the data in the new token.

If any of these tokens are to be included in PTLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-pos users must consider this additional data in host programs that process this data.

User/Operational Impacts

Not applicable

BASE24-pos
General Enhancements
SPDH Additional EMV Data Elements

Vendor Impacts

Not applicable

Reference Documents

Not applicable

SPDH Support for Network Mandates

This section describes the implementation requirements associated with the SPDH Support for Network Mandates enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH Support for Network Mandates enhancement.

The BASE24 Standard POS Device Handler (SPDH) has been enhanced to support several 2007 and 2008 VisaNet and MasterCard mandates. Changes to the SPDH include the following.

- New Amount Types for Healthcare Auto-Substantiation Transactions
- New Card Level Result Codes
- IIAS Auto-Substantiation
- New Fields in Visa Fleet Card Data
- Reduced Cash Back Amount in Response Message to Device

These mandates are described as follows.

New Amount Types for Healthcare Auto-Substantiation Transactions

The *October 2007 VisaNet Business Enhancements* define four new, optional, amount types for healthcare auto-substantiation transactions. The SPDH supports the new amount types in FID 6 subFID 1 (Healthcare/Transit Data). They are in bold below.

- 3S = Amount co-payment
- 4S = Amount healthcare
- 4T = Amount transit
- 4U = Amount prescription/Rx
- 4V = Amount vision/optical
- 4W = Amount clinic/other qualified medical
- 4X = Amount dental
- 57 = Original amount

The *October 2007 VisaNet Business Enhancements* also assign additional meaning to the existing amount type 4S (amount healthcare).

Participating merchants will insert the total amount of qualified healthcare products as amount type 4S. This includes over-the-counter (OTC) products and other applicable types of qualified medical expenses such as prescriptions, vision/optical (non-OTC products), and clinic (non-OTC products). Some health plans only cover specific types of expenditures, such as prescription or vision. Thus, merchants may optionally include additional information in amount type fields for prescription/Rx (4U), vision/optical services (4V), clinic/other qualified medical services (4W), and dental (4X). The use of these amount types is shown in the following table.

Amount Type	Definition	Data Requirement
4S	Total of all qualified healthcare amounts	The amount in 4S can include - the OTC amount only. - the total of all amounts contained within each healthcare category 4U, 4V, 4W and 4X. - the OTC amount plus the total of all amounts contained within each healthcare category 4U, 4V, 4W and 4X.
4S	Over-the-counter (OTC) only	4S OTC amount
4U	Prescription/Rx	4U prescription/Rx amount
4V	Vision/optical	4V vision/optical amount
4W	Clinic/other qualified medical	4W clinic/other qualified medical amount
4X	Dental	4X dental amount

The following examples show how these amount types are used in FID 6 subFID 1.

Example 1	Processing
Purchase includes healthcare OTC of \$20.00.	FID 6 SubFID 1 contains one Additional Amount table entry: - Amount Type = 4S - Amount = \$20.00

Example 2	Processing
Purchase includes healthcare OTC of \$15.00 and prescription/Rx of \$30.00.	FID 6 SubFID 1 contains the following depending upon merchant support for the optional amount types. If the merchant does not support the optional amount types, the total amount is placed in the first entry of the Additional Amount table in FID 6 SubFID 1: - Amount Type = 4S - Amount = \$45.00 Or, if the merchant supports the optional amount types, the total amount is placed in the first entry of the Additional Amount table and the prescription/Rx amount is placed in the second entry: - Amount Type = 4S - Amount = \$45.00 - Amount Type = 4U - Amount = \$30.00

Example 3	Processing
Purchase includes healthcare OTC of \$15.00, prescription/Rx of \$30.00, and vision/optical amount of \$60.00.	FID 6 SubFID 1 contains the following depending upon merchant support for the optional amount types. If the merchant does not support the optional amount types, the total amount is placed in the first entry of the Additional Amount table in FID 6 SubFID 1: - Amount Type = 4S - Amount = \$105.00 Or, if the merchant supports the optional amount types, the total amount is placed in the first entry of the Additional Amount table, the prescription/Rx amount is placed in the second entry, and the vision/optical amount is placed in the third entry: - Amount Type = 4S - Amount = \$105.00 - Amount Type = 4U - Amount = \$30.00 - Amount Type = 4V - Amount = \$60.00

Example 4	Processing
Purchase includes prescription/Rx for \$50.00 and vision/optical for \$90.00.	FID 6 SubFID 1 contains the following depending upon merchant support for the optional amount types. If the merchant does not support the optional amount types, the total amount is placed in the first entry of the Additional Amount table in FID 6 SubFID 1: - Amount Type = 4S - Amount = \$140.00 Or, if the merchant supports the optional amount types, the total amount is placed in the first entry of the Additional Amount table, the prescription/Rx amount is placed in the second entry, and the vision/optical amount is placed in the third entry: - Amount Type = 4S - Amount = \$140.00 - Amount Type = 4U - Amount = \$50.00 - Amount Type = 4V - Amount = \$90.00

New Card Level Result Codes

Visa periodically implements changes to support additional transaction information in authorization messages according to MasterCard specifications. With the *October 2007 VisaNet Business Enhancements*, Visa allows acquirer processors that process MasterCard transactions through VisaNet to support Account Level Management (ALM) data elements in authorization messages.

MasterCard is introducing Account Level Management processing so MasterCard issuer processors may optionally define the card product for individual cardholders. Acquirer processors that process MasterCard transactions will receive an account category value in Field 38 – Authorization Identification Response to indicate that the cardholder account participates in ALM. This is similar to Visa's Card Level processing. With this enhancement, Card Level processing is supported for Visa-acquired MasterCard transactions.

The SPDH receives the MasterCard account category value in the Card Level Product ID Value field of the Switch Common Data Token (BY) in the 0210 response message, just as it does for Visa's Card Level value. The SPDH moves this value to FID 6 subFID k (Card Level Results) of the response message to the device.

The values for the Card Level Product ID Value field of the Switch Common Data Token (BY) and FID 6 subFID k (Card Level Results) are defined below. New and redefined values are in bold. Values that were added or redefined to support Visa-acquired MasterCard transactions are shaded.

A	= Visa Traditional Non-Rewards
AX	= American Express
B	= Visa Traditional Rewards / MasterCard Enhanced (Consumer)
C	= Visa Signature / MasterCard Consumer World
D	= Visa Signature Preferred / MasterCard Consumer World Elite
DI	= Discover
G	= Visa Business / MasterCard Business World
G1	= Visa Signature Business
G2	= Visa Business Check Card
H	= Visa Check Card / MasterCard Business World Elite
I	= Visa Commerce / MasterCard Corporate World
J	= MasterCard Corporate World Elite
J1	= Visa General Prepaid
J2	= Visa Prepaid Gift
J3	= Visa Prepaid Healthcare
J4	= Visa Prepaid Commercial
K	= Visa Corporate
K1	= Visa GSA Corporate T & E
M	= MasterCard/Euro Card and Diners
Q	= Private Label
Q1	= Private Label Prepaid
R	= Proprietary
S	= Visa Purchasing
S1	= Visa Purchasing with Fleet
S2	= Visa GSA Purchasing
S3	= Visa GSA Purchasing with Fleet
U	= Visa TravelMoney
Z	= Does not participate (MasterCard default value)

IIAS Auto-Substantiation

Inventory Information Approval System (IIAS) is a point-of-sale technology that provides the instant approval of flexible spending account (FSA) and health reimbursement arrangement (HRA) payment card purchases of qualified healthcare products. When an IIAS is used, the cardholder is not required to submit documentation to a plan administrator. The IRS has mandated that by January 1, 2008, when FSA and HRA payment cards are used at non-medical merchants, the merchants must be able to support the IIAS auto-substantiation approach. Non-medical merchants include grocery and discount stores, warehouse clubs, mail order, telephone order, Internet, and other merchants that sell over-the-counter qualified

healthcare products. By January 1, 2009, drug stores must also be able to support the IIAS auto-substantiation approach in order to accept FSA and HRA cards.

No changes were required to the SPDH to support the IIAS auto-substantiation approach defined in the *April 2008 VisaNet Business Enhancements*. For IIAS auto-substantiation transactions, Visa uses Field 54 (Additional Amounts) with the Amount Type subfield set to the value "4S" (amount healthcare). This value is currently supported by the SPDH in FID 6 subFID 1 (Healthcare/Transit Data). Visa uses Field 62.4 (Market-Specific Data Identifier) as the IIAS indicator, set to the value "M" (healthcare – medical). This value is currently supported by the SPDH in FID 1 (PS2000 Data). Visa also uses Field 62.20 (Merchant Verification Value or MVV) for IIAS auto-substantiation transactions. The MVV will be provided by the VisaNet interchange interface and is therefore not the responsibility of the POS device or the SPDH.

To support the IIAS auto-substantiation approach defined in the *8 June 2007 Update to the Banknet Release 07.2 Document* and the *28 March 2008 Update to the Banknet Release 08.2 Document*, the SPDH is enhanced to accept FID 6 subFID r (Auto-Substantiation Data) in transaction requests.

FID 6 subFID r – Auto-Substantiation Data		
This subFID contains information associated with auto-substantiation transactions. The format is as follows.		
IIAS Indicator	PIC X	
Assigned ID	PIC X(6)	
User Field ACI	PIC X(43)	
These fields are further described below.		
Position	Length	Description
01	1	IIAS Indicator An indicator used to specify that an Inventory Informational Approval System (IIAS) was used by the merchant to identify qualified medical items at the point of sale. Valid values are as follows: 0 = An IIAS was not used at the point of sale. 1 = An IIAS was used at the point of sale.
02-07	6	Assigned ID An identifier assigned to the merchant by MasterCard for IIAS validation. Identifies the merchant for real-time substantiation.
08-50	43	User Field ACI Reserved for future use.
Request:	Optional. Fixed length of 7 bytes.	
Response:	Not applicable.	

When the SPDH receives this new subFID in a transaction request, it moves the IIAS Indicator and Assigned ID from the subFID to the Auto-Substantiation Data Token (CY) and adds the token to the 0200 request PSTM. The Banknet interchange interface uses the token's IIAS Indicator to set DE 48 (Additional Data—Private Use), sub-element 61 (POS Data, Extended Condition Codes), subfield 3 (Real-time Substantiation Indicator) to indicate whether the merchant terminal verified the purchased items against an IIAS. The interface uses the token's Assigned ID to set DE 48 (Additional Data—Private Use), sub-element 32 (MasterCard Assigned ID).

New Fields in Visa Fleet Card Data

The *April 2008 VisaNet Business Enhancements* identify new subfields in Field 104, Dataset ID 5C. The SPDH is enhanced to support this new information in the Visa Fleet Card Data section of FID 6 subFID D (Purchasing Card or Fleet Card Data), shown below. New information and fields are shown in bold. For the purpose of this document, only the layout of the Visa Fleet Card Data (Card Type F) is shown. Refer to the *ACI Standard POS Device Message Specifications Manual* for the layout of other card types.

FID 6 subFID D – Purchasing Card or Fleet Card Data		
Request: Optional. Variable length of 30–876 bytes, depending on the Card Type.		
Response: Not applicable.		
The Purchasing Card or Fleet Card Data subfield contains purchasing or fleet card information. The format of this subfield is as follows:		
Card Type		PIC X
Customer Identification Number		PIC X(17)
Tax Amount		PIC 9(12)
Purchase Card or Fleet Card Data		PIC X(846)
These fields are further described below.		
<u>Position</u>	<u>Length</u>	<u>Description</u>
01	1	Card Type
		A code indicating the type of card data present. Valid values are as follows:
		C = MasterCard fleet card
		F = Visa fleet card
		M = MasterCard purchasing card level 3
		V = Visa purchasing card level 3
		c = Level 2 purchasing card (where c is a blank)
02-18	17	Customer Identification Number
19-30	12	Tax Amount
31-876	846	Visa Fleet Card Data
		The following fields contain Visa fleet card data that redefines the Purchase Card or Fleet Card Data field when the Card Type field is set to a value of F.
31	1	Purchase Type
32-33	2	Fuel Type

BASE24-pos
General Enhancements
SPDH Support for Network Mandates

34	1	Unit of Measure
		A code indicating the unit of measure. Valid values are as follows:
		G = Gallon
		I = Imperial gallon
		L = Liter
		K = Kilo
		P = Pound
35-46	12	Quantity
47-58	12	Unit Cost
59-70	12	Gross Fuel Price
71-82	12	Net Fuel Price
83-94	12	Net Nonfuel Price
95-111	17	Vehicle Number
112-128	17	Driver Identification Number
129-135	7	Odometer
136-139	4	Value-Added Tax Rate
140-151	12	Miscellaneous Fuel Tax
152-163	12	Other Tax
		The other tax amount.
164-175	12	Miscellaneous Nonfuel Tax
176	1	Service Type
177-192	2	Product Code (occurs 8 times)
193-204	12	Gross Nonfuel Price
205	1	Miscellaneous Fuel Tax Exemption Status
206	1	Miscellaneous Nonfuel Tax Exemption Status
207	1	Federal Nonfuel Excise Tax Exemption Status
208-219	12	Federal Nonfuel Excise Tax Amount
220	1	Federal Fuel Excise Tax Exemption Status
221-232	12	Federal Fuel Excise Tax
233	1	State Motor Fuel Tax Exemption Status
234-245	12	State Motor Fuel Tax Amount
246	1	County Fuel Sales Tax Exemption Status
247-258	12	County Fuel Sales Tax Amount
259	1	Nonfuel State and Local Sales Tax Exemption Status
260-271	12	Nonfuel State and Local Sales Tax Amount
272	1	County Motor Fuel Tax Exemption Status
273-284	12	County Motor Fuel Tax Amount
285	1	City Fuel Sales Tax Exemption Status
286-297	12	City Fuel Sales Tax Amount

298	1	City Motor Fuel Sales Tax Exemption Status
299-310	12	City Motor Fuel Sales Tax Amount or Local Tax Amount
311	1	Secondary State Fuel Tax Exemption Status
312-323	12	Secondary State Fuel Tax Amount
324	1	Federal Sales Tax Exemption Status
325-336	12	Federal Sales Tax Amount or National Tax Amount
		The federal sales tax amount or national tax amount. Two decimal places are implied.
337	1	National Tax Included
		The national tax included indicator. Valid values are as follows: 0 = Not subject to tax 1 = Subject to tax
338	1	Local Tax Included
		The local tax included indicator. Valid values are as follows: 0 = Tax not included 1 = State or provincial tax included 2 = Transaction is not subject to tax
339-358	20	Merchant VAT Registration / Single Business Reference Number
		The merchant value-added tax registration number or single business reference number.
359-371	13	Customer VAT Registration Number
		The customer value-added tax registration number.
372-388	17	Customer Code / Customer Reference Identifier
		The customer code or customer reference identifier.
389-403	15	Message Identifier
		The message identifier that is used to link the separate line item detail messages.
404	1	Additional Data Indicator
		The additional data indicator. Valid values are as follows: Y = Yes, additional data is provided separately in Visa BASEII Draft Data TC 50 text message transaction. N = No, additional data is not provided.
405-408	4	Summary Commodity Code
		The summary commodity code.
409-876	468	User Field 6

When the SPDH receives a transaction request containing Visa Fleet Card Data in FID 6 subFID D, it includes the new fields with the data that it moves from the subFID to the Purchase Card and Fleet Card Token (C2).

Reduced Cash Back Amount in Response Message to Device

Previously, when a purchase with cash back or a preauthorization purchase with cash back transaction was authorized for a lesser amount, Router/Auth reduced the total transaction amount (AMT-1) and the cash back amount (AMT-2) by the same amount. FID B (Amount 1) of the device response message contained the reduced total transaction amount, but FID C (Amount 2) did not contain the reduced cash back amount. The SPDH has been modified so FID C contains the reduced cash back amount.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLBATKN.

All BASE24-pos Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHG and ASPDHS.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0C.SCNSPDH.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Note Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

SPDH devices may require modifications to support reduced cash-back amounts in FID C (Amount 2) of transaction responses.

SPDH devices may require modifications to support new fields in the Visa Fleet Card Data section of FID 6 subFID D (Purchasing Card or Fleet Card Data).

SPDH devices may require modifications to support new values for FID 6 subFID k (Card Level Results).

SPDH devices may require modifications to support new amount types in FID 6 subFID l (Healthcare/Transit Data). The devices' handling of multiple amount types in a given transaction may need to be changed.

SPDH devices may require modifications to add FID 6 subFID r (Auto-Substantiation Data) to transaction requests.

SPDH devices may require modifications to interface with or support an Inventory Information Approval System (IIAS).

Reference Documents

Visa Healthcare Auto-Substantiation Transactions – Consolidated Technical Requirements, April 13, 2007

October 2007 VisaNet Business Enhancements, Technical Letter and Implementation Guide, Version 1.0, April 19, 2007

8 June 2007 Update to the Banknet Release 07.2 Document

April 2008 VisaNet Business Enhancements, Technical Letter and Implementation Guide, Version 2.0, December 19, 2007

28 March 2008 Update to the Banknet Release 08.2 Document

Support Additional Flexibility in EMV Processing

This section describes the implementation requirements associated with the Support Additional Flexibility in EMV Processing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Support Additional Flexibility in EMV Processing enhancement.

This enhancement includes the introduction of support for delayed authorizations, which is a function specific to POS processing. When an EMV card requests an online authorization, but the terminal cannot send the transaction to the acquirer, it is likely that the card will decline the transaction. In some cases, the merchant may want to approve the transaction, e.g. if the cardholder is known to the merchant. When the terminal is able to communicate with the acquirer again, the transaction may be sent online as a “delayed authorization”, to determine whether the issuer is prepared to accept the transaction.

BASE24-pos EMV has been enhanced to allow issuers to configure whether delayed authorizations are supported at the BIN level on the Card Prefix File (CPF). Valid values are ‘N’ and ‘Y’ (specifying ‘no’ and ‘yes’ respectively). BASE24-pos has also been enhanced to allow acquirers to configure whether delayed authorizations are supported at the merchant level on the POS Retailer Definition File (PRDF). Valid values are ‘N’ and ‘Y’ (specifying ‘no’ and ‘yes’ respectively). The “Delayed Authorization” support by cannot be configured by individual terminal, only at the retailer level or BIN level.

BASE24-pos EMV has been enhanced to use the configured values when processing EMV authorization requests. A delayed authorization is identified by the presence of an AAC as the Application Cryptogram (i.e. a transaction declined by the card).

In EMV issuer processing, Application Transaction Counter (ATC) checking is inhibited when processing a delayed authorization, because the transaction may have taken place before a subsequent transaction on the card that has already been processed by BASE24. This would cause ATC checking to fail, because the ATC in the delayed authorization would be less than the ATC held on the Cardholder Authorization File (CAF).

For additional detailed information on this enhancement see Section 4 (Base).

Software Impacts

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24-pos customers:

- Inline changes to PS60DDL.DDLFPRDF
- Inline changes to PS60AFT.SCRNPRDF, RQPRDFS
- Inline changes to PS60RTAU.ROUTERS, RTAUEMVD
- Inline changes to PS60EXT.PSOMFXS
- Inline changes to PS60UC04.CNVPRDFS
- New conversion program (PS60UC0C.CNVPRDFS) to convert the PRDF to the required format.

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVG, RTAUEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNFLEX.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Card Prefix File (CPF)

The following new field makes it possible to configure whether a “delayed authorization” is permitted for cards with this prefix:

Screen Name: DELAYED AUTH SUPPORT
Screen: CPF Screen 13
DDL Name: DELAYED-AUTH-SPPT
Valid Values are: N, Y (default is N)

The FALLBACK ACTION (POS) field has also been moved to Screen 13, having previously been on Screen 11.

A full description of this field can be found in the *BASE24-pos EMV Support Manual*.

POS Retailer Definition File (PRDF)

The following new field makes it possible to configure whether a “delayed authorization” is permitted from the retailer:

Screen Name: EMV DELAYED AUTH SUPPORT
Screen: PRDF Screen 4
DDL Name: EMV-DELAYED-AUTH-SPPT
Valid Values are: N, Y (default is N)

A full description of this field can be found in the *BASE24-pos Files Maintenance Manual*.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

In order for acquirers to decline “Delayed Authorization” transactions, it is essential that the Cryptogram Information Data (EMV tag ‘9F27’) is included in the authorization request message in FID 6 SFID O.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to initialize the new Delayed Auth Support field. Details of this conversion program can be found in the General Changes in Section 4.

POS Retailer Definition File (PRDF)

The PRDF conversion program must be run to initialize the new EMV Delayed Auth Support field. Details of this conversion program can be found in the General Changes earlier in Section 6.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process the new field in records containing a PRDF record image.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Visa Contactless MSD with CVN 17

This section describes the implementation requirements associated with the Visa Contactless MSD with CVN 17 enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Visa Contactless MSD with CVN 17 enhancement.

BASE24 Release 6.0 Version 8 supports the validation of the Dynamic Card Verification Value (dCVV) in Visa Magnetic Stripe Data (MSD/dCVV) transactions. Here, the terminal replaces the Track 2 discretionary data with the dCVV itself, and also other data elements used in the dCVV validation, e.g. the Application Transaction Counter (ATC). The dCVV is different for each transaction performed on the card, hence the term dynamic card verification. The advantage of using Track 2 to include the dCVV and associated data elements is that there is no impact on message formats passed between terminals, acquirers, interchanges and issuers.

BASE24 Release 6.0 Version 8 also supports the validation of an Application Cryptogram (AC) in Visa qVSDC transactions, where the Cryptogram Version Number (CVN) is either 10 or 17. CVN 10 uses the conventional set of EMV data elements in the AC calculation, whereas CVN 17 uses a subset of these. The advantage of using qVSDC for card authentication is that it provides the additional security inherent in EMV transactions.

Visa Contactless Magnetic Stripe with CVN 17 (MSD/CVN 17) is essentially a combination of the MSD and qVSDC implementations, so most of the logic required for the enhancement is already available in BASE24. In order to authorize MSD/CVN 17 transactions (POS Entry Mode = '91x'), the Application Cryptogram (AC) is validated, using the POS Router/Auth EMV extension module. The parameters required to perform the authentication are configured on the EMV segment in the appropriate Card Prefix File (CPF) record.

The philosophy of this enhancement, therefore, is to identify MSD/CVN 17 transactions, and process them essentially as qVSDC/CVN 17 transactions. For acquirer processing, this means ensuring that the required data (identical to qVSDC/CVN 17 transactions, except for the POS Entry Mode) is routed to the VisaNet Interface. For issuer processing (where the authorization is to be performed on BASE24), this means ensuring that the EMV authentication routines are invoked (to allow the cryptogram to be validated in the same way as a qVSDC/CVN 17 transaction).

The modules that can receive MSD/CVN 17 transaction requests from outside BASE24 must ensure that the PSTM and token data is set correctly to allow other BASE24 modules to identify MSD/CVN 17 transactions. This means ensuring the POS Entry Mode in the PSTM is set to '91x', and the EMV Status Token and EMV Request Token are present.

According to the Visa specifications, supplementary EMV data must be included in DE 55 in some types of contactless transactions. To support this requirement, a new subFID ('6q') has been added to the SPDH message, and a new token ('S4') has been added to BASE24. When the SPDH receives the new subFID in a message, the data is moved from the subFID to the new token. This allows the VisaNet Interchange Interface to extract the data from the token and include the data in DE 55. Similarly, if the supplementary EMV data is present in DE 55 in an incoming message from VisaNet, the data is moved to the new token to allow the data to be logged.

Device Handler Processing

The ACI Standard POS Device Handler (SPDH) is the only BASE24-pos module enhanced to acquire MSD/CVN 17 transactions. The additional data required to support MSD/CVN 17 transactions is carried in the existing EMV subFIDs and the new EMV supplementary data subFID ('6q') in the SPDH message. The SPDH module moves this data to the existing EMV tokens and the new EMV Supplementary Data Token ('S4').

In the SPDH, the POS Entry Mode is set from the value sent in subFID '6E', so MSD/CVN 17 transactions must contain a value of '91x' in this subFID. The EMV Status Token is added if Processing Flag 2 in the incoming message header is set to '5' (indicating an EMV-capable terminal). Terminals capable of processing MSD/CVN 17 transactions may not be EMV-capable, so may not set Processing Flag 2 to '5'. The SPDH has been enhanced to ensure that the EMV Status Token is added for MSD/CVN 17 transactions. The logic that determines whether to add the EMV Status Token has been enhanced to check specifically for an MSD/CVN 17 transaction (subFID '6E' = '91x' and subFID '6O' is present), in addition to checking for a Processing Flag 2 value of '5'. Note that the presence of subFID '6O' (EMV Request Data) already results in the addition of the EMV Request Token (as required for MSD/CVN 17 transactions).

Interface Processing

The VisaNet Interchange Interface has been enhanced to recognize MSD/CVN 17 transactions received from VisaNet. The additional data required to support MSD/CVN 17 transactions is extracted from DE 55, and moved to the existing EMV tokens and the new EMV Supplementary Data Token.

In the VisaNet Interchange Interface, the POS Entry Mode is set from the value sent in DE 22, which is '91x' for MSD/CVN 17 transactions. The EMV Status Token is

added if DE 60.2 in the incoming message is set to '5' (indicating an EMV-capable terminal). For MSD/CVN 17 transactions, DE 60.2 in the incoming message may be set to '8' (indicating a contactless terminal). So, the logic that determines whether to add the EMV Status Token has been enhanced to check specifically for a contactless transaction containing a cryptogram (DE 60.2 = '8' and DE 55 or the tertiary bit map is present), in addition to checking for a DE 60.2 value of '5'. Note that the presence of DE 55 or the tertiary bit map (EMV Request Data) already results in the addition of the EMV Request Token (as required for MSD/CVN 17 transactions).

Visa supports an "issuer-selection option" that allows issuers to specify whether DE 55 should be sent by VisaNet when the cryptogram is verified by Visa as a "pre-screen" or "stand-in" function. For issuers that perform dCVV validation on BASE24 (or on a backend Host), it is assumed that DE 55 will be sent by VisaNet, when Visa performs the cryptogram validation in MSD/CVN 17 transactions. This option will allow the ATC in MSD/CVN 17 transactions to be updated on the issuer's system, which is important for facilitating ATC derivation in MSD/dCVV transactions. Assuming that the same ATC is used for both MSD/dCVV and MSD/CVN 17 transactions, it is vital that the ATC in MSD/CVN 17 transactions is updated on the issuer's database so that the ATC in MSD/dCVV transactions can be derived correctly. For issuers who validate neither the dCVV nor the cryptogram, there is no need to maintain the ATC on the issuer system at all, and for issuers who validate both the dCVV and the cryptogram, the ATC will always be available in messages received from Visa (in either Track 2 or DE 55). Note that the VisaNet Interface EMV extension does not have to be implemented if the sole purpose of DE 55 in MSD/CVN 17 transactions is to update the ATC on the issuer's system.

The VisaNet Interchange Interface has also been enhanced to recognize MSD/CVN 17 transactions received from BASE24-pos. The interface extracts the data carried in the existing EMV tokens and the new EMV Supplementary Data Token to ensure that the required information is included in the external message sent to VisaNet.

Authorization Processing

Within the BASE24-pos Router/Authorization modules, the method selected for authorizing contactless transactions depends on the value of the POS Entry Mode in the PSTM:

- If the POS Entry Mode = '91x', then the dCVV is validated;
- If the POS Entry Mode = '07x', then the AC is validated.

For MSD/CVN 17 transactions, the requirement is to validate the AC, even though the POS Entry Mode is '91x'. As with contactless EMV transactions, the AC and the data used to calculate the value is included in the EMV data in the incoming message received by BASE24.

The easiest way to ensure that the AC is validated is to set the POS Entry Mode in the PSTM to '07x' within the BASE24-pos Router/Authorization modules for MSD/CVN 17 transactions. This ensures the following:

- No changes are required to invoke the EMV processing, which is required to validate the cryptogram;
- The Dynamic Card Verification (DCV) processing will not be invoked, because the POS Entry Mode value will not be '91x'.

When the BASE24-pos Router/Authorization modules receive an incoming MSD/CVN 17 transaction (identified by a value of '91x' in the PSTM POS Entry Mode field and the presence of the EMV Status Token and the EMV Request Token), the PSTM POS Entry Mode field is changed to '07x'. Note that the POS Entry Mode field in the EMV Status Token is still set to '91x', so that the PSTM POS Entry Mode value can be "reset" to '91x' when the transaction leaves the Router/Authorization modules. This means that MSD/CVN 17 transactions contain a PSTM POS Entry Mode value of '07x' inside the Router/Authorization modules, but a PSTM POS Entry Mode value of '91x' outside the Router/Authorization modules, including when the transaction is written to the POS Transaction Log File (PTLF).

EMV authorization processing for MSD/CVN 17 transactions must be modified slightly to ensure that the EMV Response Token is not added to the PSTM, because Issuer Authentication Data is not returned in MSD/CVN 17 transactions. In addition, a recent case (#467543) has highlighted a MasterCard requirement for issuers not to generate Issuer Authentication Data in "PayPass – M/Chip" transactions. Therefore, this new logic will apply to the authorization of any contactless transaction containing an EMV cryptogram. Visa also supports contactless VSDC (in addition to qVSDC and MSD) as a proximity solution, and it is valid for the issuer to generate Issuer Authentication Data in contactless VSDC transactions. However, it is unlikely that contactless VSDC will be used as a proximity solution, and so the response message will not contain Issuer Authentication Data when a contactless VSDC transaction is authorized by BASE24-pos.

EMV authorization processing has also been enhanced to allow ATC limit checking to be performed even when the EMV segment is not present in the appropriate CAF record. A new value is supported in the ATC CHECK TYPE field in the EMV segment of the CPF to allow the ATC to be checked against the ATC LIMIT field in the Base segment of the CPF. This enhanced logic applies to both MSD/CVN 17 transactions and EMV transactions (including CAP and ATM EMV transactions).

Terminal Capabilities

The card data input capability indicator on the PTD includes values for contactless chip and contactless magnetic stripe, but does not allow these to be configured in conjunction with contact capabilities, e.g. EMV chip, magnetic stripe.

A new field has been added to the POS Terminal Data file (PTD) to specify the contactless capabilities of the terminal, allowing the existing field to be used to specify the contact capabilities of the terminal. The new field may be updated either by an operator via the AFT screen, or dynamically during transaction processing when a contactless transaction is received.

In BASE24 Release 6.0 Version 8, the terminal input capability field in the EMV Status Token (TERM_ENTRY_CAP) always contains a value of '5', indicating an EMV-capable terminal. The SPDH sets the field directly from Processing Flag 2 in the incoming message header and the VisaNet Interchange Interface sets the field directly from DE 60.2. The modifications to the SPDH and the VisaNet Interchange Interface mean that these fields may not always be set to '5' when the EMV Status Token is added.

This enhancement has introduced a new value of '8' in the TERM_ENTRY_CAP field in the EMV Status Token, to identify terminals that do not have a contact EMV reader, but are capable of providing an EMV cryptogram in contactless transactions. A value of '8' matches the DE 60.2 value used by Visa to identify a contactless terminal without an EMV contact reader, and the logic in the VisaNet Interchange Interface naturally moves this value to the TERM_ENTRY_CAP field in the EMV Status Token. The SPDH has been enhanced to set the TERM_ENTRY_CAP field in the EMV Status Token to '8', if the terminal input capability information on the PTD indicates that the terminal supports contactless transactions, but does not have an EMV contact reader.

The importance of this new value is that it allows the VisaNet Interchange Interface to set the correct value in DE 60.2 in outgoing messages. Before this enhancement, the VisaNet Interchange Interface could not differentiate between a terminal with a contact EMV reader and a terminal with only a contactless EMV reader, unless the Point of Service Data Token is present. The Point of Service Data Token contains additional information on the terminal's card data input capabilities, but may not be present in transactions received via some acquiring endpoints, e.g. POS HISO.

Software Architecture

In order to support MSD/CVN 17 processing as an issuer (where the cryptogram verification is performed on BASE24), the customer must license (as a minimum) the following EMV extension subvolumes:

- BA60IEMV;
- PS60IRTA;
- SW60IVIS.

In order to support MSD/CVN 17 processing as an acquirer, the customer must license (as a minimum) the following EMV extension subvolumes:

- BA60IEMV
- PS60ISPD
- SW60IVIS

Note that the POS Router/Authorization EMV extension subvolume is not required to support the acquiring of MSD/CVN 17 transactions. Note that the VisaNet Interface EMV extension subvolume is not required for issuer systems where the cryptogram is verified by Visa as a “pre-screen” function.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCPF, DDLBATKN, DDLPSTKN
- Inline changes to BA60AFT.SCRNCPF, RQCPFS, SVCDFS, COBTKN
- Inline changes to BA60SRC.BATKNID, BATKNCVS, PSTKNCVS

All BASE24-pos customers:

- Inline changes to PS60DDL.DDLFPTD, DDLGPTD
- Inline changes to PS60AFT.SCRNPTD, RQPTDS, SVPTDTS, SVPTDTBL
- Inline changes to PS60RTAU.RTAUG, AUTHLIBS, RTAUEMVD
- Inline changes to PS60UC04.CNVPTDS
- New conversion program (PS60UC0C.CNVPTDS) to convert the PTD to the required format

All BASE24-pos ACI Standard POS Device Handler customers:

- Inline changes to PS60SPDH.ASPDHS, SPDHEMVG

All BASE24 VisaNet Interchange Interface customers:

- Inline changes to SW60VISA.VISADDLS, VISAFMTS

All BASE24 EMV customers:

- Inline changes to BA60IEMV.EMVG

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVS

All BASE24-pos EMV CAP customers:

- Inline changes to PS60JRTA.RTAUCAPS

All BASE24-pos EMV ACI Standard POS Device Handler customers:

- Inline changes to PS60ISPD.SPDHEMVG, SPDHEMVS

All BASE24 VisaNet Interchange Interface EMV customers:

- Inline changes to SW60IVIS.VISAEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNCV17.

CUSTMACS Changes

In order to build the required system components to support MSD/CVN 17 processing as an issuer (where the cryptogram verification is performed on BASE24), the customer must set the following Make flags to “true”:

- emv_on
- pos_emv_on
- visa_emv_on

In order to build the required system components to support MSD/CVN 17 processing as an acquirer, the customer must set the following Make flags to “true”:

- emv_on
- spdh_emv_on
- visa_emv_on

Note that the pos_emv_on Make flag does not have to be set to “true” to support the acquiring of MSD/CVN 17 transactions, as the POS Router/Authorization EMV extension subvolume is not required. Note that the visa_emv_on Make flag does not have to be set to “true” for issuer systems where the cryptogram is verified by Visa as a “pre-screen” function.

LCONF Assign Changes

Not applicable

LCONF Param Changes

In order to send MSD/CVN 17 transactions to VisaNet, the LCONF parameter USE-FIELD-55-FOR-EMV-DATA must be set to “Y”.

Database Configuration

Card Prefix File (CPF)

The CPF includes configuration data that determines the authorization processing performed on MSD and EMV (qVSDC and VSDC) contactless transactions.

Issuers of MSD cards that are able to generate Application Cryptograms (ACs) must add the EMV segment to the appropriate CPF record, if the cryptogram is to be verified by BASE24. If the cryptogram is verified on a backend Host, or by Visa as a “pre-screen” function (for an issuer-only system), then there is no requirement to add the EMV segment to the appropriate CPF record.

The EMV segment is added to CPF records by turning the segment on in the SEG-TABLE of the PITABLE file.

To perform ATC limit checking when the EMV segment is not present in the appropriate CAF record (see below), the ATC CHECK TYPE field on Screen 11 of the CPF must be set to '8'. A full description of this field can be found in the *BASE24-pos EMV Support Manual*.

Cardholder Authorization File (CAF)

BASE24 authorization processing for MSD/CVN 17 transactions does not require the EMV segment to be present in the appropriate CAF record, unless there is a requirement to hold the ATC for the card on the EMV segment instead of the Base segment (configured through the ATC CHECK TYPE field on Screen 11 of the CPF).

The EMV segment is added to CAF records by turning the segment on in the appropriate IDF record (screen 5).

POS Terminal Data files (PTD)

The following new field makes it possible to configure the contactless capabilities of a terminal:

Screen Name:	CONTACTLESS INPUT CAPABILITIES
Screen:	PTD Screen 19
DDL Name:	CTLS-INPUT-CAP-IND
Valid Values are:	Space = not used
	0 = unknown or unspecified (default)
	1 = none (no terminal used or contact only)
	2 = contactless magnetic stripe
	3 = contactless EMV
	4 = contactless magnetic stripe and contactless EMV

A full description of this field can be found in the *BASE24-pos Files Maintenance Manual*.

Token File (TKN)

This enhancement uses the following new token:

- EMV Supplementary Data Token (Token ID 'S4').

This enhancement uses the following existing tokens:

- EMV Request Token (Token ID 'B2').
- EMV Status Token (Token ID 'B4').
- POS Data 2 Token (Token ID 'CZ').

If any of these tokens are to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If any of these tokens are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

SPDH devices must provide BASE24 with the data required to authenticate an Application Cryptogram generated using CVN 17.

A new SPDH variable-length subFID ('6q') is being introduced to carry the supplementary data required by Visa in DE 55 in a contactless magnetic stripe transaction. The definition of the subFID comprises a 2-byte Dataset Id field (numeric) and a (maximum) 160-byte buffer (ASCII hexadecimal) for holding the EMV data (TLV-encoded).

A Dataset Id of '01' is used to identify the supplementary data required by Visa in DE 55 in a contactless magnetic stripe transaction. The data elements that may be present in the subFID are:

- Form Factor Indicator (tag '9F6E');
- Customer Exclusive Data (tag '9F7C');
- Terminal Transaction Qualifiers (tag '9F66').

The 'L' part of the TLV-encoded data element(s) contains the length of the associated value when represented in binary. So, for example, the sub-FID may contain the following data (spaces are inserted only to aid viewing):

01 9F6E0400000000 9F7C060102FFFFFFFF 9F660400000000

Note that the SPDH message supports display-format data only, so each character in the string above is transmitted as a single ASCII byte (ASCII hexadecimal). This contrasts with the Visa definition of these data elements, where a single ASCII byte contains two hexadecimal characters (binary-coded hexadecimal).

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

The CPF needs to be updated if a customer decides to add the EMV segment to the CPF. No conversion program is provided to do this, so the user must add the required EMV data via the AFT screen, using the CPF requester and server. To accomplish this, the EMV segment must be turned on in the Segment Table (SEG-TABLE in the PITABLE file). RQMEGA must be recompiled using this Segment Table. Next, each CPF record must be read and updated to add the EMV segment. Consideration should be given to the various fields on CPF Screens 11, 12 and 13 when adding this new segment. The segment will be added with default values unless changes are made on these screens.

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Strictly speaking, the Transaction Security Services (TSS) do not have to be implemented for this enhancement, because an Application Cryptogram (AC) generated with CVN 17 can be authenticated via a direct connection to an HSM. However, the current Visa requirement is that “the MSD Path shall be capable of supporting Dynamic Card Verification Value (dCVV) until the migration to Cryptogram 17 is completed”. The Transaction Security Services (TSS) are required to validate the dCVV, so TSS must be configured to authorize MSD transactions.

File Conversions

POS Terminal Data files (PTD)

The PTD conversion program must be run to initialize the new Contactless Input Capabilities field. Details of this conversion program can be found in the General Changes earlier in Section 6.

Host/Co-Network Impacts

This enhancement uses the following new token:

- EMV Supplementary Data Token (Token ID 'S4').

This enhancement uses the following existing tokens:

- EMV Request Token (Token ID 'B2').
- EMV Status Token (Token ID 'B4').
- POS Data 2 Token (Token ID 'CZ').

If any of these tokens are to be included in PTLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-pos users must consider this additional data in host programs that process this data.

User/Operational Impacts

Disk Impacts

The PTLF and ILF may require additional disk space if any of the tokens used by this enhancement are included in the transaction records logged to these files.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-pos users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the *BASE24 Tokens Manual* provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Contactless POS terminals must provide the Host with data compliant with Version 2.0.2 of the Visa Contactless Payment Specification.

Reference Documents

Visa's *Contactless Payment Specifications*, v2.0.2, dated August 2007.

Visa II Contactless Support

This section describes the implementation requirements associated with the Visa II Contactless Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Visa II Contactless Support enhancement.

The Visa II Device Handler (VVDH) has been enhanced to support contactless magnetic stripe transactions, Visa's Contactless MSD with CVN 17 (MSD/CVN 17) initiative and the addition of the Point of Service Data token (C4):

- For both contactless magnetic stripe and MSD/CVN 17 transactions, the POS Entry Mode is set to '91x' by the VVDH to indicate a contactless transaction.
- If MSD/CVN 17 data is present in the transaction (Group 3 version 42 data), then that data is stored in the EMV Request token (B2) and the new EMV Supplementary Data token (S4). The EMV Status token (B4) is also added for an MSD/CVN 17 transaction, to ensure that the POS Router/Auth process performs EMV processing.
- If the POS Data Group is present in the transaction (Group 3 version 27 data), then the data is mapped to the Point of Service Data token (C4).

Contactless Magnetic Stripe Transactions

If the POS Data Group is present in the transaction request (Group 3 version 27 data), and subfield 7: card data – input mode = A (contactless magnetic stripe), then the Visa II Device Handler sets the POS Entry Mode in the PSTM to "91x".

Contactless MSD/CVN 17 Transactions

If the Visa Contactless group is present in the transaction request (Group 3 version 42 data), then the Visa II Device Handler sets the POS Entry Mode in the PSTM to "91x". In addition, the following tokens are added to the PSTM, using the data specified:

EMV Status Token:

Field Name	Value
PT-SRV-ENTRY-MDE	910
TERM-ENTRY-CAP	8
LAST-EMV-STAT	0
DATA-SUSPECT	0
APPL-PAN-SEQ-NUM	Spaces
DEV-INFO	Spaces
RSN-ONL-CDE	Spaces
ARQC-VRFY	0
ISO-RC-IND	Space

EMV Request Token:

- Cryptogram amount
- Application cryptogram
- Application transaction counter
- Issuer application data
- Unpredictable number

EMV Supplementary Data Token:

- Customer exclusive data
- Form factor indicator

A Visa II Device Handler EMV extension has not been created to support contactless MSD/CVN 17 transactions. The code is in the main file.

For additional detailed information on supporting MSD/CVN 17 transactions in BASE24-pos, see Visa Contactless MSD with CVN 17 earlier in Section 6.

Point of Service Data Token Support

If the POS Data Group is present in the transaction request (Group 3 version 27 data), then the Visa II Device Handler adds the Point of Service Data token to the PSTM, using the mappings specified:

POS Data Group		Point of Service Data Token Buffer		
Sub-field	Value	Pos.	Field Name	Value
1	0	11	TERM-INPUT-CAP-IND	0
1	1	11	TERM-INPUT-CAP-IND	1
1	2	11	TERM-INPUT-CAP-IND	2
1	4	11	TERM-INPUT-CAP-IND	0
1	5	11	TERM-INPUT-CAP-IND	9
1	6	11	TERM-INPUT-CAP-IND	6
1	A	11	TERM-INPUT-CAP-IND	4
1	B	11	TERM-INPUT-CAP-IND	7
1	C	11	TERM-INPUT-CAP-IND	8
1	D	11	TERM-INPUT-CAP-IND	5
1	E	11	TERM-INPUT-CAP-IND	9
1	M	11	TERM-INPUT-CAP-IND	3
1	V, X (or any other value)	11	TERM-INPUT-CAP-IND	0
3	0	6	CRD-CAPTR-IND	0
3	1	6	CRD-CAPTR-IND	1
3	9 (or any other value)	6	CRD-CAPTR-IND	0
4	0	1 3 10	TERM-ATTEND-IND TERM-LOC-IND CRDHLDLDR-ACTVT-TERM-IND	2 3 0
4	1	1 3 10	TERM-ATTEND-IND TERM-LOC-IND CRDHLDLDR-ACTVT-TERM-IND	0 0 0
4	2	1 3 10	TERM-ATTEND-IND TERM-LOC-IND CRDHLDLDR-ACTVT-TERM-IND	1 0 1 (if subfield 8 = '1') 2 (otherwise)
4	3	1 3 10	TERM-ATTEND-IND TERM-LOC-IND CRDHLDLDR-ACTVT-TERM-IND	0 1 0

BASE24-pos
General Enhancements
Visa II Contactless Support

POS Data Group		Point of Service Data Token Buffer		
Sub-field	Value	Pos.	Field Name	Value
4	4	1	TERM-ATTEND-IND	1
		3	TERM-LOC-IND	1
		10	CRDHLD-RTACTVT-TERM-IND	1 (if subfield 8 = '1') 2 (otherwise)
4	5	1	TERM-ATTEND-IND	1
		3	TERM-LOC-IND	2
		10	CRDHLD-RTACTVT-TERM-IND	6
4	6	1	TERM-ATTEND-IND	1
		3	TERM-LOC-IND	1
		10	CRDHLD-RTACTVT-TERM-IND	1 (if subfield 8 = '1') 2 (otherwise)
4	9, S, T (or any other value)	1	TERM-ATTEND-IND	0
		3	TERM-LOC-IND	0
		10	CRDHLD-RTACTVT-TERM-IND	0
5	0	4	CRDHLD-RTPRESENT-IND	0
5	1	4	CRDHLD-RTPRESENT-IND	1
5	2	4	CRDHLD-RTPRESENT-IND	2
5	3	4	CRDHLD-RTPRESENT-IND	3
5	4	4	CRDHLD-RTPRESENT-IND	4
5	5	4	CRDHLD-RTPRESENT-IND	5
5	8	4	CRDHLD-RTPRESENT-IND	4
5	9 (or any other value)	4	CRDHLD-RTPRESENT-IND	0
6	0	5	CRD-PRESENT-IND	1
6	1	5	CRD-PRESENT-IND	0
6	9 (or any other value)	5	CRD-PRESENT-IND	0
6	W	10	CRDHLD-RTACTVT-TERM-IND	7
8	0	12	CRDHLD-ID-METHOD	3 (if position 4 = '0') 4 (otherwise)
8	1	12	CRDHLD-ID-METHOD	2
8	2	12	CRDHLD-ID-METHOD	6
8	5	12	CRDHLD-ID-METHOD	1
8	6	12	CRDHLD-ID-METHOD	9
8	9	12	CRDHLD-ID-METHOD	0
8	S, T (or any other value)	12	CRDHLD-ID-METHOD	9

In addition, the Visa II Device Handler sets the third digit of the POS Entry Mode in the PSTM, based on the value of subfield 7: cardholder authentication method in the POS Data Group:

- 2 (if subfield 7 contains '0');
- 0 (if subfield 7 contains '1')
- 1 (otherwise).

Software Impacts

All BASE24-pos Visa II Device Handler customers:

- Inline changes to PS60VVDH.VVDHG, VVDHS, VVDHM.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNVIL.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

This enhancement uses the following new token:

- EMV Supplementary Data Token (Token ID 'S4').

This enhancement uses the following existing tokens:

- EMV Request Token (Token ID 'B2').
- EMV Status Token (Token ID 'B4').
- POS Data 2 Token (Token ID 'CZ').

If any of these tokens are to be included in PTLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-pos users must consider this additional data in host programs that process this data.

If any of these tokens are to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If any of these tokens are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Visa II devices must be able to provide BASE24 with Group 3 Version 27 data to allow BASE24 to identify contactless magnetic stripe transactions, and to add the Point of Service Data token.

Visa II devices must be able to provide BASE24 with Group 3 Version 42 data to allow BASE24 to process MSD/CVN 17 transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

This enhancement uses the following new token:

- EMV Supplementary Data Token (Token ID 'S4').

This enhancement uses the following existing tokens:

- EMV Request Token (Token ID 'B2').
- EMV Status Token (Token ID 'B4').
- POS Data 2 Token (Token ID 'CZ').

If any of these tokens are to be included in PTLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-pos users must consider this additional data in host programs that process this data.

If any of these tokens are to be included in PTLF or ILF Extracts, sent to the host, or passed through the BIC ISO Interchange Interface, BASE24-pos users must consider this additional data in host programs that process this data.

User/Operational Impacts

Disk Impacts

The PTLF and ILF may require additional disk space if any of the tokens used by this enhancement are included in the transaction records logged to these files.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-pos users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these

limitations. Appendix A of the *BASE24 Tokens Manual* provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

POS terminals must provide the Host with data compliant with Version 7.7.1 of the TSYS External Interface Specification Authorization Record Formats (EIS 1080).

Reference Documents

TSYS's *EIS 1080 External Interface Specification Authorization Record Formats*, v7.7.1, dated March 2008.

Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with Release 6.0 Version 9, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-pos.

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7: BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from Release 6.0 Version 8.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to Release 6.0 Version 9.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to Release 6.0 Version 9.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with Release 6.0 Version 9, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-teller.

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8: BASE24-remote banking

Section 8 discusses the changes and enhancements made to BASE24-remote banking in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing BASE24-remote banking users must consider when upgrading from Release 6.0 Version 8. This section covers changes made to the following products within the BASE24 family:

- BASE24-telebanking
- BASE24-billpay
- BASE24 Kernel/IEP

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-remote banking users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24- remote banking users must address.
- Optional enhancements describe the changes that affect only those BASE24-remote banking users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-remote banking users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24-remote banking users need to consider when upgrading to Release 6.0 Version 9.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-remote banking users must address in order to upgrade to Release 6.0 Version 9.

There are no general enhancements inherent to BASE24-remote banking.

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Optional Enhancements

This section identifies the BASE24-remote banking enhancements that can be enabled or disabled with Release 6.0 Version 9, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-remote banking.

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9: BASE24 Interchange Interfaces

Section 9 discusses the changes and enhancements made to BASE24 Interchange Interfaces in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from Release 6.0 Version 8.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 9 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to Release 6.0 Version 9.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to Release 6.0 Version 9.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Visa Contactless MSD with CVN 17

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- None

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Visa Contactless MSD with CVN 17

This section describes the implementation requirements associated with the Visa Contactless MSD with CVN 17 enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Visa Contactless MSD with CVN 17 enhancement.

The VisaNet Interchange Interface has been enhanced to recognize MSD/CVN 17 transactions received from VisaNet. The additional data required to support MSD/CVN 17 transactions is extracted from DE 55, and moved to the existing EMV tokens and the new EMV Supplementary Data Token.

The VisaNet Interchange Interface has also been enhanced to recognize MSD/CVN 17 transactions received from BASE24-pos. The interface extracts the data carried in the existing EMV tokens and the new EMV Supplementary Data Token to ensure that the required information is included in the external message sent to VisaNet.

For additional detailed information on this enhancement see Section 6 (POS).

Software Impacts

All BASE24 VISA Interface customers:

- Inline changes to SW60VISA.VISADDLS, VISAFMTS.

All BASE24 VISA Interface EMV customers:

- Inline changes to SW60IVIS.VISAEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0C.SCNCV17.

Note that the software modifications were applied as part of the “VisaNet October 2008 Business Enhancements” development project, available in August 2008.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

In order to send MSD/CVN 17 transactions to VisaNet, the LCONF parameter USE-FIELD-55-FOR-EMV-DATA must be set to "Y".

Database Configuration

Note Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Visa's "*Contactless Payment Specifications*", v2.0.2, dated August 2007.

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Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with Release 6.0 Version 9, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

There are no optional enhancements for the BASE24 Interchange Interface modules.

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10: BASE24 Value-Added Products

Section 10 discusses the changes and enhancements made to the BASE24 value-added products in Release 6.0 Version 9, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from Release 6.0 Version 8.

The value-added products discussed in this section include the following:

- BASE24-card
- BASE24-customer service
- BASE24-infoBase
- BASE24-inventory Voucher Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 10 must be used in conjunction with Section 4 to identify the changes that BASE24 value-added product users need to consider when upgrading to Release 6.0 Version 9.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all BASE24 Value-added product users must address in order to upgrade to Release 6.0 Version 9.

There are no general enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-infoBase
- BASE24-inventory Voucher Manager

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with Release 6.0 Version 9 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-infoBase
- BASE24-inventory Voucher Manager

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A: Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with Release 6.0 Version 9 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the *BASE24 Logical Network Configuration File (LCONF) Manual* for instructions on adding them to the LCONF.

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ASSIGNS

There were no new LCONF assigns added as part of BASE24 Release 6.0 Version 9.

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PARAMS

The following params were added or modified for Release 6.0 Version 9. Refer to the specified page numbers for explanations of how these params can or must be set.

PCI – Event and Trace Messages

PCI-DISPOSITION4-19

POS Limit Checking on Responses

POS-RESP-LMTS-CHK2-25, 6-49, 6-50

Tidel AKDS Support

ATM-DH-TIDL-AKDS-HOST-LOAD-TIMER5-79

ATM-DH-TIDL-AKDS-SERIAL-NUM-VRFY5-79

Triton AKDS Support

ATM-DH-TRIT-AKDS-SERIAL-NUM-VRFY5-97

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B: Host/CO-Network Impacts

Appendix B contains a list of Release 6.0 Version 9 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

Hypercom Enhancements.....	6-27
PCI - CVD2 Suppression in CRT Auth.....	6-35
PCI - Track 2 Context Data Removal for Diebold CSP	5-41
PCI - Track 2 Context Data Removal for Tidel.....	5-45
PCI - Track 2 Context Data Removal for Triton	5-49
PCI - Track 2 Context Data Removal for APACS	6-39
PCI - Track 2 Context Data Removal for HPDH	6-43
PCI - Track 2 Context Data Removal for VISA II	6-47
SPDH Additional EMV Data Elements	6-55
Support Additional Flexibility in EMV Processing (BASE).....	5-53
Support Additional Flexibility in EMV Processing (POS).....	5-53
Tidel AKDS Support.....	5-82
Triton AKDS Support	5-101
Triton EMV Offline PIN Synchronization	5-62
Triton Mini-Statement Support	5-66
Visa Contactless MSD with CVN 17 Support.....	6-83
VISA II Contactless Support	6-91

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C: CUSTMACS File Changes

Appendix C contains a list of Release 6.0 Version 9 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 9 code.

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Enhancements

The following enhancements have CUSTMACS impacts:

DUKPT Support with Message Data Encryption

pos_dh_spdh_de_on..... 6-19

NCR Standard APTRA Edge NDC Support

atm_dh_n50_edge_on5-74

Tidel AKDS Support

atm_dh_tidel_akd_on flag5-78

Tidel Mobile Top-Up

atm_dh_tidel_mtu_on.....5-88

Triton AKDS Support

atm_dh_triton_akd_on flag5-96

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D: Re-Sequenced Files and New Files

Appendix D consists of the following information:

- Information regarding the resequencing of the VISAFMTS file for the VISA Interface.
- A table illustrating the new files added to BASE24 for the Release 6.0 Version 9 enhancements.

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Re-Sequencing of the VISAFMTS File

The following information is in reference to the re-sequencing of the SW60VISA. VISAFMTS file for the VISA Interface. This affects all customers with a license for the VISA Interface (SW DS500).

ACI re-evaluated BASE24 Release 6.0 and determined that it was necessary to re-sequence the VISAFMTS file due to change activity for fixes and enhancements.

Below are detailed instructions for working with the newly re-sequenced source file. Additionally, a special report has been provided that maps the new sequence number to the corresponding former sequence number of each line. This will aid in the insertion of custom code into the newly re-sequenced code.

Working with the Re-sequenced VISAFMTS File

Customers who license the VISA Interface fall into three general categories regarding the re-sequencing of the VISAFMTS file. Regardless of the method used, any fixes added to the VISAFMTS file since the last time fixes have been applied should be reviewed. All dependencies should be checked to verify that other VISA and Non-VISA Interface files are in sync with the VISAFMTS file.

Listed below these categories are the general guidelines for use when working with the re-sequenced VISAFMTS file:

1. No CSM Code
 - Replace the old copy of VISAFMTS with the newly re-sequenced version of VISAFMTS file.
2. CSM code using the proc replacement methodology
 - Create a new VISAFMTQS by copying the history section of the old VISAFMTQS and the corresponding PROCs from the newly re-sequenced VISAFMTS file.
 - Choose one of the methods noted later in this section for moving CSM code into the new VISAFMTQS file.
3. Inline CSM code
 - Choose one of the methods noted later in this section for moving CSM code into the new VISAFMTQS file.

There are three general methods for moving CSM changes into the newly re-sequenced VISAFMTS file, with many variations on each method depending upon your environment and personal preferences.

1. Using EDIT or TEDIT - A special mapping file (SEQMAP08) has been provided that will aid in the proper placement of CSM code. This file is a copy of the re-sequenced VISAFMTS file and will include, on each line of code, both the new sequence number and the corresponding old sequence number for that line of code. This file will not compile and is to be used only as a reference tool.
 - In the CSM source file, find the sequence number immediately preceding the beginning of each CSM code section
 - Search for this number in the special mapping file to locate the corresponding sequence number in the VISAFMTS file
 - Insert your CSM code at the appropriate new sequence number
2. Using CodeWright - During re-sequencing, all redundant blank lines were removed from the source code. The CodeWright difference function does not do a good job of displaying changes side-by-side when multiple blank lines are removed from the source code. Using the CodeWright difference function is probably the least effective method. CodeWright can, however, be used quite successfully by cutting and pasting between screens.
 - Open the CSM code on one screen
 - Open the newly re-sequenced VISAFMTS file on a second screen
 - Open the SEQMAP08 on a third screen to locate corresponding sequence numbers
3. Using BeyondCompare – This is a tool that is very reasonable in cost and makes the migration of CSM code into the newly re-sequenced VISAFMTS file very easy. If your code is heavily customized, ACI suggests using the BeyondCompare tool available from Scooter Software (<http://www.scootersoftware.com/>). Scooter offers a 30-day trial version of BeyondCompare that can be used to evaluate the benefits of the tool.

ACI uses BeyondCompare for text file and folder comparisons on a frequent basis. The side-by-side comparison of files that have blank lines removed is superior to that of CodeWright. Using BeyondCompare is by far the quickest method in testing different ways of reinserting custom code into the newly re-sequenced VISAFMTS source code.

Following are suggestions for setting up the BeyondCompare environment:

- Strip off the sequence numbers in the CSM'd code
- Open the CSM code in the left window and the re-sequenced code in the right window
- In the menu bar, choose Tools/Pick Rules: select "Everything Else"
- In the menu bar, choose Tools/Edit Current Rules
- In the File Comparison Rules dialog box, choose Importance
- Check the Blank Lines box
- Click the first NEW button
- In the Unimportant Text dialog box, select Category/ Columns. In the Start column box, enter 71 and check the stopping at end of line box
- In the Toolbar, you have options to view all text, just changes, changes with context, and just matching text
- CSM code will appear in RED in the left window. Select lines and move to the re-sequenced code using the toolbar arrow
- Save modified source file

Note: Regardless of the method used, the customer should review any fixes that were added to VISAFMTS since that last time they applied fixes to this file. All dependencies should be checked to verify that other VISA and non-VISA Interface files are in sync with the VISAFMTS file.

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New Files Added to BASE24 with Version 9

The following table lists the files added by the Release 6.0 Version 9 enhancements:

Subvolume	Filename	Language	Enhancement Name
AT60ATDL	ATATDLFM	TACL	Tidel AKDS Support
	ATATDLMM	TACL	Tidel AKDS Support
	TIDLAKDG	TAL	Tidel AKDS Support
	TIDLAKDM	TACL	Tidel AKDS Support
	TIDLAKDS	TAL	Tidel AKDS Support
AT60ATRT	ATATRTFM	TACL	Triton AKDS Support
	ATATRTM	TACL	Triton AKDS Support
	ATATRTMM	TACL	Triton AKDS Support
	TRITAKDG	TAL	Triton AKDS Support
	TRITAKDM	TACL	Triton AKDS Support
	TRITAKDS	TAL	Triton AKDS Support
AT60DDLD	TDF0C	N/A	Tidel AKDS Support Triton AKDS Support
	TIDL0C	N/A	Tidel AKDS Support Tidel Display Balances on Screen Support Tidel Mobile Top-Up
	TRIT0A	N/A	Triton AKDS Support Triton Mini-Statement Support
	TRIT0C	N/A	Triton AKDS Support Triton Mini-Statement Support
AT60EN50	ATEN50FM	TACL	NCR Standard APTRA Edge NDC Support
	ATEN50MM	TACL	NCR Standard APTRA Edge NDC Support
	N50EDGE G	TAL	NCR Standard APTRA Edge NDC Support
	N50EDGEM	TACL	NCR Standard APTRA Edge NDC Support
	N50EDGES	TAL	NCR Standard APTRA Edge NDC Support
AT60TTDL	ATTTDLFM	TACL	Tidel Mobile Top-Up
	ATTTDLMM	TACL	Tidel Mobile Top-Up
	TIDLMTUG	TAL	Tidel Mobile Top-Up
	TIDLMTUM	TACL	Tidel Mobile Top-Up
	TIDLMTUS	TAL	Tidel Mobile Top-Up
	TTDLNAMS	TAL	Tidel Mobile Top-Up
AT60U100	U100NAMS	TAL	Multi-Currency with Like Denominations

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 9

Subvolume	Filename	Language	Enhancement Name
AT60UC0C	AAREAD0C	N/A	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
	ATUC0CFM	TACL	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
	ATUC0CM	TACL	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
	ATUC0CMM	TACL	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
	CNVTD FM	TACL	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
	CNVTD FR	TACL	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
	CNVTD FS	TAL	Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support
BA10LMCF	AAREADME	N/A	PCI - Event and Trace Messages
	BALMCFFM	TACL	PCI - Event and Trace Messages
	BALMCFM	TACL	PCI - Event and Trace Messages
	BALMCFMM	TACL	PCI - Event and Trace Messages
	CNVLMCFM	TACL	PCI - Event and Trace Messages
	CNVLMCFR	TACL	PCI - Event and Trace Messages
	CNVLMCFS	TAL	PCI - Event and Trace Messages
	CONTENTS	N/A	PCI - Event and Trace Messages
	LMCFFUP	N/A	PCI - Event and Trace Messages
	LMCFHIST	N/A	PCI - Event and Trace Messages
	LMCFPCI	N/A	PCI - Event and Trace Messages
BA60DDLD	CPF0C	N/A	Add support for AMEX format in EMV IAD Support Additional Flexibility in EMV Processing Visa Contactless MSD with CVN 17

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 9

Subvolume	Filename	Language	Enhancement Name
BA60UC0C	AAREAD0C	N/A	PCI - Track2 Context Data Removal for HPDH PCI - Track2 Context Data Removal for Visa II Support Additional Flexibility in EMV Processing Tidel AKDS Support Tidel Display Balances on Screen Support Triton AKDS Support Triton Mini-Statement Support Visa Contactless MSD with CVN 17
	BAUC0CFM	TACL	Support Additional Flexibility in EMV Processing
	BAUC0CM	TACL	Support Additional Flexibility in EMV Processing
	BAUC0CMM	TACL	Support Additional Flexibility in EMV Processing
	CNVCPFM	TACL	Support Additional Flexibility in EMV Processing
	CNVCPFR	TACL	Support Additional Flexibility in EMV Processing
	CNVCPFS	TAL	Support Additional Flexibility in EMV Processing
PS60DDL	HPDH04	N/A	PCI - Track2 Context Data Removal for HPDH
	HPDH0B	N/A	PCI - Track2 Context Data Removal for HPDH
	HPDH0C	N/A	PCI - Track2 Context Data Removal for HPDH
	PRDF0C	N/A	Support Additional Flexibility in EMV Processing
	PTD0C	N/A	DUKPT Support with Message Data Encryption Visa Contactless MSD with CVN 17
	VVDH04	N/A	PCI - Track2 Context Data Removal for Visa II
	VVDH0B	N/A	PCI - Track2 Context Data Removal for Visa II
	VVDH0C	N/A	PCI - Track2 Context Data Removal for Visa II

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 9

Subvolume	Filename	Language	Enhancement Name
PS60UC0C	AAREAD0C	N/A	PCI - Track2 Context Data Removal for HPDH PCI - Track2 Context Data Removal for Visa II Support Additional Flexibility in EMV Processing Visa Contactless MSD with CVN 17
	CNVPRDFM	TACL	Support Additional Flexibility in EMV Processing
	CNVPRDFR	TACL	Support Additional Flexibility in EMV Processing
	CNVPRDFS	TAL	Support Additional Flexibility in EMV Processing
	CNVPTDFM	TACL	PCI - Track2 Context Data Removal for Visa II
	CNVPTDFR	TACL	PCI - Track2 Context Data Removal for Visa II
	CNVPTDFS	TAL	PCI - Track2 Context Data Removal for Visa II
	CNVPTDM	TACL	PCI - Track2 Context Data Removal for HPDH Visa Contactless MSD with CVN 17
	CNVPTDR	TACL	PCI - Track2 Context Data Removal for HPDH Visa Contactless MSD with CVN 17
	CNVPTDS	TAL	PCI - Track2 Context Data Removal for HPDH Visa Contactless MSD with CVN 17
	PSUC0CFM	TACL	PCI - Track2 Context Data Removal for HPDH PCI - Track2 Context Data Removal for Visa II Support Additional Flexibility in EMV Processing Visa Contactless MSD with CVN 17
	PSUC0CM	TACL	PCI - Track2 Context Data Removal for HPDH PCI - Track2 Context Data Removal for Visa II Support Additional Flexibility in EMV Processing Visa Contactless MSD with CVN 17
	PSUC0CMM	TACL	PCI - Track2 Context Data Removal for HPDH PCI - Track2 Context Data Removal for Visa II Support Additional Flexibility in EMV Processing Visa Contactless MSD with CVN 17

The following table lists the same files but by Version 9 enhancement name:

Enhancement Name	Subvolume	Filename	Language
Add support for AMEX format in EMV IAD	BA60DDLD	CPF0C	N/A
DUKPT Support with Message Data Encryption	PS60DDLD	PTD0C	N/A
Multi-Currency with Like Denominations	AT60U100	U100NAMS	TAL
NCR Standard APTRA Edge NDC Support	AT60EN50	ATEN50FM	TACL
		ATEN50MM	TACL
		N50EDGE G	TAL
		N50EDGE M	TACL
		N50EDGE S	TAL
PCI - Event and Trace Messages	BA10LMCF	AAREADME	N/A
		BALMCF FM	TACL
		BALMCF M	TACL
		BALMCF MM	TACL
		CNVLMCF M	TACL
		CNVLMCF R	TACL
		CNVLMCF S	TAL
		CONTENTS	N/A
		LMCF FUP	N/A
		LMCF HIST	N/A
		LMCF PCI	N/A
PCI - Track2 Context Data Removal for HPDH	BA60UC0C	AAREAD0C	N/A
	PS60DDLD	HPDH04	N/A
		HPDH0B	N/A
		HPDH0C	N/A
	PS60UC0C	AAREAD0C	N/A
		CNVPTDM	TACL
		CNVPTDR	TACL
		CNVPTDS	TAL
		PSUC0CFM	TACL
		PSUC0CM	TACL
		PSUC0CMM	TACL
PCI - Track2 Context Data Removal for Visa II	BA60UC0C	AAREAD0C	N/A
	PS60DDLD	VVDH04	N/A
		VVDH0B	N/A
		VVDH0C	N/A
	PS60UC0C	AAREAD0C	N/A
		CNVPTDFM	TACL
		CNVPTDFR	TACL
		CNVPTDFS	TAL
		PSUC0CFM	TACL
		PSUC0CM	TACL
		PSUC0CMM	TACL

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 9

Enhancement Name	Subvolume	Filename	Language
Support Additional Flexibility in EMV Processing	BA60DDLD	CPF0C	N/A
	BA60UC0C	AAREAD0C	N/A
		BAUC0CFM	TACL
		BAUC0CM	TACL
		BAUC0CMM	TACL
		CNVCPFM	TACL
		CNVCPFR	TACL
		CNVCPFS	TAL
	PS60DDLD	PRDF0C	N/A
	PS60UC0C	AAREAD0C	N/A
		CNVPRDFM	TACL
		CNVPRDFR	TACL
		CNVPRDFS	TAL
		PSUC0CFM	TACL
		PSUC0CM	TACL
		PSUC0CMM	TACL
Tidel AKDS Support	AT60ATDL	ATATDLFM	TACL
		ATATDLMM	TACL
		TIDLAKDG	TAL
		TIDLAKDM	TACL
		TIDLAKDS	TAL
	AT60DDLD	TDF0C	N/A
		TIDL0C	N/A
	AT60UC0C	AAREAD0C	N/A
		ATUC0CFM	TACL
		ATUC0CM	TACL
		ATUC0CMM	TACL
		CNVTDFFM	TACL
		CNVTDFFR	TACL
		CNVTDFFS	TAL
	BA60UC0C	AAREAD0C	N/A
Tidel Display Balances on Screen Support	AT60DDLD	TIDL0C	N/A
	AT60UC0C	AAREAD0C	N/A
		ATUC0CFM	TACL
		ATUC0CM	TACL
		ATUC0CMM	TACL
		CNVTDFFM	TACL
		CNVTDFFR	TACL
		CNVTDFFS	TAL
	BA60UC0C	AAREAD0C	N/A

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 9

Enhancement Name	Subvolume	Filename	Language
Tidel Mobile Top-Up	AT60DDLD	TIDL0C	N/A
	AT60TTDL	ATTTDLFM	TACL
		ATTTDLMM	TACL
		TIDLMTUG	TAL
		TIDLMTUM	TACL
		TIDLMTUS	TAL
		TTDLNAMS	TAL
Triton AKDS Support	AT60ATRT	ATATRTFM	TACL
		ATATRTM	TACL
		ATATRTMM	TACL
		TRITAKDG	TAL
		TRITAKDM	TACL
		TRITAKDS	TAL
	AT60DDLD	TDF0C	DDL
		TRIT0A	DDL
		TRIT0C	DDL
	AT60UC0C	AAREAD0C	N/A
		ATUC0CFM	TACL
		ATUC0CM	TACL
		ATUC0CMM	TACL
		CNVTD FM	TACL
		CNVTD FR	TACL
		CNVTD FS	TAL
	BA60UC0C	AAREAD0C	N/A
Triton Mini-Statement Support	AT60DDLD	TRIT0A	DDL
		TRIT0C	DDL
	AT60UC0C	AAREAD0C	N/A
		ATUC0CFM	TACL
		ATUC0CM	TACL
		ATUC0CMM	TACL
		CNVTD FM	TACL
		CNVTD FR	TACL
		CNVTD FS	TAL
	BA60UC0C	AAREAD0C	N/A
Visa Contactless MSD with CVN 17	BA60DDLD	CPF0C	N/A
	BA60UC0C	AAREAD0C	N/A
	PS60DDLD	PTD0C	N/A
	PS60UC0C	AAREAD0C	N/A
		CNVPTDM	TACL
		CNVPTDR	TACL
		CNVPTDS	TAL
		PSUC0CFM	TACL
		PSUC0CM	TACL
		PSUC0CMM	TACL

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E: Moved, Renamed, Deleted Files

Appendix E contains a list of files that were moved, renamed, or deleted in BASE24 Release 6.0 Version 9.

No files were moved, renamed or deleted in Release 6.0 Version 9.

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F: Regional Specific Information

Appendix F contains the information that is required to implement the changes for Release 6.0 Version 9 that are specific to each region.

There were no Regional Specific changes made as part of Release 6.0 Version 9.

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